

Filter Summary Report: CG,TIA,Automated,CROSS,COUPLED,some,parasitic,Z1,Z5,ZL

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Contents

1 Examined $H(z)$ for CG TIA Automated CROSS COUPLED some parasitic Z1 Z5 ZL: $\frac{Z_1 Z_5 Z_L g_m r_o + Z_1 Z_5 Z_L - Z_1 Z_L r_o}{Z_1 Z_5 g_m r_o + Z_1 Z_5 + 2Z_1 Z_L g_m r_o + 4Z_1 Z_L + Z_1 r_o + Z_5 Z_L + Z_5 r_o + Z_L r_o}$

$$H(z) = \frac{Z_1 Z_5 Z_L g_m r_o + Z_1 Z_5 Z_L - Z_1 Z_L r_o}{Z_1 Z_5 g_m r_o + Z_1 Z_5 + 2Z_1 Z_L g_m r_o + 4Z_1 Z_L + Z_1 r_o + Z_5 Z_L + Z_5 r_o + Z_L r_o}$$

2 AP

3 BP

3.1 BP-1 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s(L_L R_1 R_5 g_m r_o + L_L R_1 R_5 - L_L R_1 r_o)}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o + s^2(C_L L_L R_1 R_5 g_m r_o + C_L L_L R_1 R_5 + C_L L_L R_1 r_o + C_L L_L R_5 r_o) + s(2L_L R_1 g_m r_o + 4L_L R_1 + L_L R_5 + L_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_L} R_1 R_5 g_m r_o + \sqrt{C_L} R_1 R_5 + \sqrt{C_L} R_1 r_o + \sqrt{C_L} R_5 r_o}{2\sqrt{L_L} R_1 g_m r_o + 4\sqrt{L_L} R_1 + \sqrt{L_L} R_5 + \sqrt{L_L} r_o}$
 wo: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$
 bandwidth: $\frac{2\sqrt{L_L} R_1 g_m r_o + 4\sqrt{L_L} R_1 + \sqrt{L_L} R_5 + \sqrt{L_L} r_o}{\sqrt{C_L} \sqrt{L_L} (\sqrt{C_L} R_1 R_5 g_m r_o + \sqrt{C_L} R_1 R_5 + \sqrt{C_L} R_1 r_o + \sqrt{C_L} R_5 r_o)}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{2R_1 g_m r_o + 4R_1 + R_5 + r_o}$
 Qz: None
 Wz: None

3.2 BP-2 $Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, R_5, R_L \right)$

$$H(s) = \frac{s(L_1 R_5 R_L g_m r_o + L_1 R_5 R_L - L_1 R_L r_o)}{R_5 R_L + R_5 r_o + R_L r_o + s^2(C_1 L_1 R_5 R_L + C_1 L_1 R_5 r_o + C_1 L_1 R_L r_o) + s(L_1 R_5 g_m r_o + L_1 R_5 + 2L_1 R_L g_m r_o + 4L_1 R_L + L_1 r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} R_5 R_L + \sqrt{C_1} R_5 r_o + \sqrt{C_1} R_L r_o}{\sqrt{L_1} R_5 g_m r_o + \sqrt{L_1} R_5 + 2\sqrt{L_1} R_L g_m r_o + 4\sqrt{L_1} R_L + \sqrt{L_1} r_o}$
 wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
 bandwidth: $\frac{\sqrt{L_1} R_5 g_m r_o + \sqrt{L_1} R_5 + 2\sqrt{L_1} R_L g_m r_o + 4\sqrt{L_1} R_L + \sqrt{L_1} r_o}{\sqrt{C_1} \sqrt{L_1} (\sqrt{C_1} R_5 R_L + \sqrt{C_1} R_5 r_o + \sqrt{C_1} R_L r_o)}$
 K-LP: 0
 K-HP: 0
 K-BP: $\frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}$
 Qz: None
 Wz: None

4 BS

4.1 BS-1 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o + s^2(C_L L_L R_1 R_5 g_m r_o + C_L L_L R_1 R_5 - C_L L_L R_1 r_o)}{2R_1 g_m r_o + 4R_1 + R_5 + r_o + s^2(2C_L L_L R_1 g_m r_o + 4C_L L_L R_1 + C_L L_L R_5 + C_L L_L r_o) + s(C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o)}$$

Parameters:

Q: $\frac{2\sqrt{L_L} R_1 g_m r_o + 4\sqrt{L_L} R_1 + \sqrt{L_L} R_5 + \sqrt{L_L} r_o}{\sqrt{C_L} R_1 R_5 g_m r_o + \sqrt{C_L} R_1 R_5 + \sqrt{C_L} R_1 r_o + \sqrt{C_L} R_5 r_o}$
 wo: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$
 bandwidth: $\frac{\sqrt{C_L} R_1 R_5 g_m r_o + \sqrt{C_L} R_1 R_5 + \sqrt{C_L} R_1 r_o + \sqrt{C_L} R_5 r_o}{\sqrt{C_L} \sqrt{L_L} (2\sqrt{L_L} R_1 g_m r_o + 4\sqrt{L_L} R_1 + \sqrt{L_L} R_5 + \sqrt{L_L} r_o)}$
 K-LP: $\frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{2R_1 g_m r_o + 4R_1 + R_5 + r_o}$

K-HP: $\frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{2 R_1 g_m r_o + 4 R_1 + R_5 + r_o}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_L} \sqrt{L_L}}$

4.2 BS-2 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, R_L \right)$

$$H(s) = \frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o + s^2 (C_1 L_1 R_5 R_L g_m r_o + C_1 L_1 R_5 R_L - C_1 L_1 R_L r_o)}{R_5 g_m r_o + R_5 + 2 R_L g_m r_o + 4 R_L + r_o + s^2 (C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 + 2 C_1 L_1 R_L g_m r_o + 4 C_1 L_1 R_L + C_1 L_1 r_o) + s (C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{L_1} R_5 g_m r_o + \sqrt{L_1} R_5 + 2 \sqrt{L_1} R_L g_m r_o + 4 \sqrt{L_1} R_L + \sqrt{L_1} r_o}{\sqrt{C_1} R_5 R_L + \sqrt{C_1} R_5 r_o + \sqrt{C_1} R_L r_o}$
wo: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$
bandwidth: $\frac{\sqrt{C_1} R_5 R_L + \sqrt{C_1} R_5 r_o + \sqrt{C_1} R_L r_o}{\sqrt{C_1} \sqrt{L_1} (\sqrt{L_1} R_5 g_m r_o + \sqrt{L_1} R_5 + 2 \sqrt{L_1} R_L g_m r_o + 4 \sqrt{L_1} R_L + \sqrt{L_1} r_o)}$
K-LP: $\frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{R_5 g_m r_o + R_5 + 2 R_L g_m r_o + 4 R_L + r_o}$
K-HP: $\frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{R_5 g_m r_o + R_5 + 2 R_L g_m r_o + 4 R_L + r_o}$
K-BP: 0
Qz: None
Wz: $\frac{1}{\sqrt{C_1} \sqrt{L_1}}$

5 GE

5.1 GE-1 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, R_L \right)$

$$H(s) = \frac{-C_5 L_5 R_1 R_L r_o s^2 - R_1 R_L r_o + s (L_5 R_1 R_L g_m r_o + L_5 R_1 R_L)}{2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_L r_o + s^2 (2 C_5 L_5 R_1 R_L g_m r_o + 4 C_5 L_5 R_1 R_L + C_5 L_5 R_1 r_o + C_5 L_5 R_L r_o) + s (L_5 R_1 g_m r_o + L_5 R_1 + L_5 R_L + L_5 r_o)}$$

Parameters:

Q: $\frac{2 \sqrt{C_5} R_1 R_L g_m r_o + 4 \sqrt{C_5} R_1 R_L + \sqrt{C_5} R_1 r_o + \sqrt{C_5} R_L r_o}{\sqrt{L_5} R_1 g_m r_o + \sqrt{L_5} R_1 + \sqrt{L_5} R_L + \sqrt{L_5} r_o}$
wo: $\frac{1}{\sqrt{C_5} \sqrt{L_5}}$
bandwidth: $\frac{\sqrt{L_5} R_1 g_m r_o + \sqrt{L_5} R_1 + \sqrt{L_5} R_L + \sqrt{L_5} r_o}{\sqrt{C_5} \sqrt{L_5} (2 \sqrt{C_5} R_1 R_L g_m r_o + 4 \sqrt{C_5} R_1 R_L + \sqrt{C_5} R_1 r_o + \sqrt{C_5} R_L r_o)}$
K-LP: $-\frac{R_1 R_L r_o}{2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_L r_o}$
K-HP: $-\frac{R_1 R_L r_o}{2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_L r_o}$
K-BP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$
Qz: $-\frac{\sqrt{C_5} r_o}{\sqrt{L_5 g_m r_o + L_5}}$
Wz: $\frac{1}{\sqrt{C_5} \sqrt{L_5}}$

5.2 GE-2 $Z(s) = \left(R_1, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_5 R_1 R_L r_o s + R_1 R_L g_m r_o + R_1 R_L + s^2 (C_5 L_5 R_1 R_L g_m r_o + C_5 L_5 R_1 R_L)}{R_1 g_m r_o + R_1 + R_L + r_o + s^2 (C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1 + C_5 L_5 R_L + C_5 L_5 r_o) + s (2 C_5 R_1 R_L g_m r_o + 4 C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{L_5} R_1 g_m r_o + \sqrt{L_5} R_1 + \sqrt{L_5} R_L + \sqrt{L_5} r_o}{2 \sqrt{C_5} R_1 R_L g_m r_o + 4 \sqrt{C_5} R_1 R_L + \sqrt{C_5} R_1 r_o + \sqrt{C_5} R_L r_o}$
wo: $\frac{1}{\sqrt{C_5} \sqrt{L_5}}$
bandwidth: $\frac{2 \sqrt{C_5} R_1 R_L g_m r_o + 4 \sqrt{C_5} R_1 R_L + \sqrt{C_5} R_1 r_o + \sqrt{C_5} R_L r_o}{\sqrt{C_5} \sqrt{L_5} (\sqrt{L_5} R_1 g_m r_o + \sqrt{L_5} R_1 + \sqrt{L_5} R_L + \sqrt{L_5} r_o)}$
K-LP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$
K-HP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$
K-BP: $-\frac{R_1 R_L r_o}{2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_L r_o}$
Qz: $\frac{-\sqrt{L_5} g_m r_o - \sqrt{L_5}}{\sqrt{C_5} r_o}$

$$\text{Wz: } \frac{1}{\sqrt{C_5} \sqrt{L_5}}$$

6 HP

7 LP

$$\mathbf{7.1 \quad LP-1} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_5 g_m r_o + R_5 - r_o}{C_1 C_L R_5 r_o s^2 + 2g_m r_o + s(C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2} \sqrt{C_1} \sqrt{C_L} \sqrt{R_5} r_o \sqrt{g_m + \frac{2}{r_o}}}{C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o} \\ \text{wo: } & \frac{\sqrt{\frac{2g_m r_o + 4}{r_o}}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_5}} \\ \text{bandwidth: } & \frac{\sqrt{2} \sqrt{\frac{2g_m r_o + 4}{r_o}} (C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o)}{2C_1 C_L R_5 r_o \sqrt{g_m + \frac{2}{r_o}}} \\ \text{K-LP: } & \frac{R_5 g_m r_o + R_5 - r_o}{2g_m r_o + 4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & 0 \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$\mathbf{7.2 \quad LP-2} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{C_1 C_L R_5 R_L r_o s^2 + R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o + s(C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_5} \sqrt{R_L} r_o \sqrt{R_5 g_m + \frac{R_5}{r_o} + 2R_L g_m + \frac{4R_L}{r_o} + 1}}{C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o} \\ \text{wo: } & \frac{\sqrt{\frac{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}{r_o}}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_5} \sqrt{R_L}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}{r_o}} (C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o)}{C_1 C_L R_5 R_L r_o \sqrt{R_5 g_m + \frac{R_5}{r_o} + 2R_L g_m + \frac{4R_L}{r_o} + 1}} \\ \text{K-LP: } & \frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & 0 \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

$$\mathbf{7.3 \quad LP-3} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{C_1 C_L R_1 R_5 r_o s^2 + 2R_1 g_m r_o + 4R_1 + R_5 + r_o + s(C_1 R_1 R_5 + C_1 R_1 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5} r_o \sqrt{2R_1 g_m + \frac{4R_1}{r_o} + \frac{R_5}{r_o} + 1}}{C_1 R_1 R_5 + C_1 R_1 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o} \\ \text{wo: } & \frac{\sqrt{\frac{2R_1 g_m r_o + 4R_1 + R_5 + r_o}{r_o}}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2R_1 g_m r_o + 4R_1 + R_5 + r_o}{r_o}} (C_1 R_1 R_5 + C_1 R_1 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o)}{C_1 C_L R_1 R_5 r_o \sqrt{2R_1 g_m + \frac{4R_1}{r_o} + \frac{R_5}{r_o} + 1}} \\ \text{K-LP: } & \frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{2R_1 g_m r_o + 4R_1 + R_5 + r_o} \end{aligned}$$

K-HP: 0
K-BP: 0
Qz: None
Wz: None

7.4 LP-4 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{C_1 C_L R_1 R_5 R_L r_o s^2 + R_1 R_5 g_m r_o + R_1 R_5 + 2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o + s (C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L} r_o \sqrt{\frac{R_1 R_5 g_m + \frac{R_1 R_5}{r_o} + 2 R_1 R_L g_m + \frac{4 R_1 R_L}{r_o} + R_1 + \frac{R_5 R_L}{r_o} + R_5 + R_L}}{C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o}$

wo: $\sqrt{\frac{\frac{R_1 R_5 g_m r_o + R_1 R_5 + 2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}{r_o}}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L}}}$

bandwidth: $\sqrt{\frac{\sqrt{\frac{R_1 R_5 g_m r_o + R_1 R_5 + 2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}{r_o}} (C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o)}{C_1 C_L R_1 R_5 R_L r_o \sqrt{R_1 R_5 g_m + \frac{R_1 R_5}{r_o} + 2 R_1 R_L g_m + \frac{4 R_1 R_L}{r_o} + R_1 + \frac{R_5 R_L}{r_o} + R_5 + R_L}}}$

K-LP: $\frac{R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{R_1 R_5 g_m r_o + R_1 R_5 + 2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}$

K-HP: 0

K-BP: 0

Qz: None

Wz: None

8 X-INVALID-NUMER

8.1 X-INVALID-NUMER-1 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_1 r_o s + R_1 g_m r_o + R_1}{C_5 C_L R_1 r_o s^2 + s (2 C_5 R_1 g_m r_o + 4 C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1} r_o \sqrt{\frac{1}{r_o}}}{2 C_5 R_1 g_m r_o + 4 C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

wo: $\frac{\sqrt{\frac{1}{r_o}}}{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1}}$

bandwidth: $\frac{2 C_5 R_1 g_m r_o + 4 C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}{C_5 C_L R_1 r_o}$

K-LP: $R_1 g_m r_o + R_1$

K-HP: 0

K-BP: $-\frac{C_5 R_1 r_o}{2 C_5 R_1 g_m r_o + 4 C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

Qz: None

Wz: None

8.2 X-INVALID-NUMER-2 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_1 R_L r_o s + R_1 R_L g_m r_o + R_1 R_L}{C_5 C_L R_1 R_L r_o s^2 + R_1 g_m r_o + R_1 + R_L + r_o + s (2 C_5 R_1 R_L g_m r_o + 4 C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1} \sqrt{R_L} r_o \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}{2 C_5 R_1 R_L g_m r_o + 4 C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}$

wo: $\sqrt{\frac{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{r_o}}{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1} \sqrt{R_L}}}$

bandwidth: $\sqrt{\frac{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{r_o} (2 C_5 R_1 R_L g_m r_o + 4 C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}{C_5 C_L R_1 R_L r_o \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}}$

K-LP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$

K-HP: 0

K-BP: $-\frac{C_5 R_1 R_L r_o}{2 C_5 R_1 R_L g_m r_o + 4 C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}$

Qz: None
Wz: None

8.3 X-INVALID-NUMER-3 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_1 R_5 r_o s + R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{C_5 C_L R_1 R_5 r_o s^2 + 2R_1 g_m r_o + 4R_1 + R_5 + r_o + s(2C_5 R_1 R_5 g_m r_o + 4C_5 R_1 R_5 + C_5 R_5 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5} r_o \sqrt{2R_1 g_m + \frac{4R_1}{r_o} + \frac{R_5}{r_o} + 1}}{2C_5 R_1 R_5 g_m r_o + 4C_5 R_1 R_5 + C_5 R_5 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o}$

wo: $\frac{\sqrt{\frac{2R_1 g_m r_o + 4R_1 + R_5 + r_o}{r_o}}}{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5}}$

bandwidth: $\frac{\sqrt{\frac{2R_1 g_m r_o + 4R_1 + R_5 + r_o}{r_o}} (2C_5 R_1 R_5 g_m r_o + 4C_5 R_1 R_5 + C_5 R_5 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o)}{C_5 C_L R_1 R_5 r_o \sqrt{2R_1 g_m + \frac{4R_1}{r_o} + \frac{R_5}{r_o} + 1}}$

K-LP: $\frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{2R_1 g_m r_o + 4R_1 + R_5 + r_o}$

K-HP: 0

K-BP: $-\frac{C_5 R_1 R_5 r_o}{2C_5 R_1 R_5 g_m r_o + 4C_5 R_1 R_5 + C_5 R_5 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o}$

Qz: None

Wz: None

8.4 X-INVALID-NUMER-4 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_1 R_5 R_L r_o s + R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{C_5 C_L R_1 R_5 R_L r_o s^2 + R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o + s(2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L} r_o \sqrt{R_1 R_5 g_m + \frac{R_1 R_5}{r_o} + 2R_1 R_L g_m + \frac{4R_1 R_L}{r_o} + R_1 + \frac{R_5 R_L}{r_o} + R_5 + R_L}}{2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o}$

wo: $\frac{\sqrt{\frac{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}{r_o}}}{\sqrt{C_5} \sqrt{C_L} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}{r_o}} (2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o)}{C_5 C_L R_1 R_5 R_L r_o \sqrt{R_1 R_5 g_m + \frac{R_1 R_5}{r_o} + 2R_1 R_L g_m + \frac{4R_1 R_L}{r_o} + R_1 + \frac{R_5 R_L}{r_o} + R_5 + R_L}}$

K-LP: $\frac{R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}$

K-HP: 0

K-BP: $-\frac{C_5 R_1 R_5 R_L r_o}{2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o}$

Qz: None

Wz: None

8.5 X-INVALID-NUMER-5 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s(C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 - C_5 R_1 r_o)}{s^2(C_5 C_L R_1 R_5 g_m r_o + C_5 C_L R_1 R_5 + C_5 C_L R_1 r_o + C_5 C_L R_5 r_o) + s(2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 R_5 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_5} \sqrt{C_L} R_1 R_5 g_m r_o \sqrt{\frac{1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_1 R_5 \sqrt{\frac{1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_1 r_o \sqrt{\frac{1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_5 r_o \sqrt{\frac{1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}}}{2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 R_5 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

wo: $\frac{1}{C_5 C_L R_1 R_5 g_m r_o + C_5 C_L R_1 R_5 + C_5 C_L R_1 r_o + C_5 C_L R_5 r_o}$

bandwidth: $\frac{(2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 R_5 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) \sqrt{\frac{1}{C_5 C_L R_1 R_5 g_m r_o + C_5 C_L R_1 R_5 + C_5 C_L R_1 r_o + C_5 C_L R_5 r_o}}}{\sqrt{C_5} \sqrt{C_L} R_1 R_5 g_m r_o \sqrt{\frac{1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_1 R_5 \sqrt{\frac{1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_1 r_o \sqrt{\frac{1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_5 r_o \sqrt{\frac{1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}}}$

K-LP: $R_1 g_m r_o + R_1$

K-HP: 0

K-BP: $\frac{C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 - C_5 R_1 r_o}{2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 R_5 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

Qz: None

Wz: None

8.6 X-INVALID-NUMER-6 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s (C_5 R_1 R_5 R_L g_m r_o + C_5 R_1 R_5 R_L - C_5 R_1 R_L r_o)}{R_1 g_m r_o + R_1 + R_L + r_o + s^2 (C_5 C_L R_1 R_5 R_L g_m r_o + C_5 C_L R_1 R_5 R_L + C_5 C_L R_1 R_L r_o + C_5 C_L R_5 R_L r_o) + s (C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_5 R_L + C_5 R_5 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_5} \sqrt{C_L} R_1 R_5 \sqrt{R_L} g_m r_o \sqrt{\frac{R_1 g_m r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_L}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_1 R_5 \sqrt{R_L} \sqrt{\frac{R_1 g_m r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_L}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_1 \sqrt{R_L} r_o}{C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_5 R_L + C_5 R_5 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o} \\ \text{wo: } & \sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_5 C_L R_1 R_5 R_L g_m r_o + C_5 C_L R_1 R_5 R_L + C_5 C_L R_1 R_L r_o + C_5 C_L R_5 R_L r_o}} \\ \text{bandwidth: } & \frac{\sqrt{C_5} \sqrt{C_L} R_1 R_5 \sqrt{R_L} g_m r_o \sqrt{\frac{R_1 g_m r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_L}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_1 R_5 \sqrt{R_L} \sqrt{\frac{R_1 g_m r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_1}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_L}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} + \sqrt{C_5} \sqrt{C_L} R_1 \sqrt{R_L} r_o}{\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_5 C_L R_1 R_5 R_L g_m r_o + C_5 C_L R_1 R_5 R_L + C_5 C_L R_1 R_L r_o + C_5 C_L R_5 R_L r_o}} (C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_5 R_L + C_5 R_5 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)} \\ \text{K-LP: } & \frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_5 R_1 R_5 R_L g_m r_o + C_5 R_1 R_5 R_L - C_5 R_1 R_L r_o}{C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_5 R_L + C_5 R_5 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.7 X-INVALID-NUMER-7 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_5 g_m r_o + R_5 - r_o + s (C_L R_5 R_L g_m r_o + C_L R_5 R_L - C_L R_L r_o)}{2g_m r_o + s^2 (C_1 C_L R_5 R_L + C_1 C_L R_5 r_o + C_1 C_L R_L r_o) + s (C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + 2C_L R_L g_m r_o + 4C_L R_L + C_L r_o) + 4}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2} \sqrt{C_1} \sqrt{C_L} R_5 R_L \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{2}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_5 r_o \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{2}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_L r_o \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{2}{R_5 R_L + R_5 r_o + R_L r_o}}}{C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + 2C_L R_L g_m r_o + 4C_L R_L + C_L r_o} \\ \text{wo: } & \sqrt{\frac{2g_m r_o + 4}{C_1 C_L R_5 R_L + C_1 C_L R_5 r_o + C_1 C_L R_L r_o}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2g_m r_o + 4}{C_1 C_L R_5 R_L + C_1 C_L R_5 r_o + C_1 C_L R_L r_o}} (C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + 2C_L R_L g_m r_o + 4C_L R_L + C_L r_o)}{\sqrt{2} \sqrt{C_1} \sqrt{C_L} R_5 R_L \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{2}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_5 r_o \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{2}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{2} \sqrt{C_1} \sqrt{C_L} R_L r_o \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{2}{R_5 R_L + R_5 r_o + R_L r_o}}} \\ \text{K-LP: } & \frac{R_5 g_m r_o + R_5 - r_o}{2g_m r_o + 4} \\ \text{K-HP: } & 0 \\ \text{K-BP: } & \frac{C_L R_5 R_L g_m r_o + C_L R_5 R_L - C_L R_L r_o}{C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + 2C_L R_L g_m r_o + 4C_L R_L + C_L r_o} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.8 X-INVALID-NUMER-8 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_5 R_L r_o s + R_L g_m r_o + R_L}{C_1 C_5 R_L r_o s^2 + g_m r_o + s (C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_L} r_o \sqrt{g_m + \frac{1}{r_o}}}{C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o} \\ \text{wo: } & \frac{\sqrt{\frac{g_m r_o + 1}{r_o}}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_L}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{g_m r_o + 1}{r_o}} (C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o)}{C_1 C_5 R_L r_o \sqrt{g_m + \frac{1}{r_o}}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & 0 \\ \text{K-BP: } & -\frac{C_5 R_L r_o}{C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \text{None} \end{aligned}$$

8.9 X-INVALID-NUMER-9 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_L r_o s + R_L g_m r_o + R_L}{g_m r_o + s^2 (C_1 C_5 R_L r_o + C_1 C_L R_L r_o + C_5 C_L R_L r_o) + s (C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_L} r_o \sqrt{g_m + \frac{1}{r_o}} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L}}{C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o + C_L R_L g_m r_o + C_L R_L}$
wo: $\sqrt{\frac{g_m r_o + 1}{C_1 C_5 R_L r_o + C_1 C_L R_L r_o + C_5 C_L R_L r_o}}$
bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{C_1 C_5 R_L r_o + C_1 C_L R_L r_o + C_5 C_L R_L r_o}} (C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o + C_L R_L g_m r_o + C_L R_L)}{\sqrt{R_L} r_o \sqrt{g_m + \frac{1}{r_o}} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L}}$
K-LP: R_L
K-HP: 0
K-BP: $-\frac{C_5 R_L r_o}{C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o + C_L R_L g_m r_o + C_L R_L}$
Qz: None
Wz: None

8.10 X-INVALID-NUMER-10 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$

$$H(s) = \frac{-C_5 R_5 R_L r_o s + R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{C_1 C_5 R_5 R_L r_o s^2 + R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o + s (C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2C_5 R_5 R_L g_m r_o + 4C_5 R_5 R_L + C_5 R_5 r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_5} \sqrt{R_L} r_o \sqrt{R_5 g_m + \frac{R_5}{r_o} + 2R_L g_m + \frac{4R_L}{r_o} + 1}}{C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2C_5 R_5 R_L g_m r_o + 4C_5 R_5 R_L + C_5 R_5 r_o}$
wo: $\sqrt{\frac{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}{r_o}}$
bandwidth: $\frac{\sqrt{\frac{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}{r_o}} (C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2C_5 R_5 R_L g_m r_o + 4C_5 R_5 R_L + C_5 R_5 r_o)}{C_1 C_5 R_5 R_L r_o \sqrt{R_5 g_m + \frac{R_5}{r_o} + 2R_L g_m + \frac{4R_L}{r_o} + 1}}$
K-LP: $\frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}$
K-HP: 0
K-BP: $-\frac{C_5 R_5 R_L r_o}{C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2C_5 R_5 R_L g_m r_o + 4C_5 R_5 R_L + C_5 R_5 r_o}$
Qz: None
Wz: None

8.11 X-INVALID-NUMER-11 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_5 r_o s + R_5 g_m r_o + R_5 - r_o}{2g_m r_o + s^2 (C_1 C_5 R_5 r_o + C_1 C_L R_5 r_o + C_5 C_L R_5 r_o) + s (C_1 R_5 + C_1 r_o + 2C_5 R_5 g_m r_o + 4C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

Parameters:

Q: $\frac{\sqrt{2} \sqrt{R_5} r_o \sqrt{g_m + \frac{2}{r_o}} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L}}{C_1 R_5 + C_1 r_o + 2C_5 R_5 g_m r_o + 4C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o}$
wo: $\sqrt{\frac{2g_m r_o + 4}{C_1 C_5 R_5 r_o + C_1 C_L R_5 r_o + C_5 C_L R_5 r_o}}$
bandwidth: $\frac{\sqrt{2} \sqrt{\frac{2g_m r_o + 4}{C_1 C_5 R_5 r_o + C_1 C_L R_5 r_o + C_5 C_L R_5 r_o}} (C_1 R_5 + C_1 r_o + 2C_5 R_5 g_m r_o + 4C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o)}{2\sqrt{R_5} r_o \sqrt{g_m + \frac{2}{r_o}} \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L}}$
K-LP: $\frac{R_5 g_m r_o + R_5 - r_o}{2g_m r_o + 4}$
K-HP: 0
K-BP: $-\frac{C_5 R_5 r_o}{C_1 R_5 + C_1 r_o + 2C_5 R_5 g_m r_o + 4C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o}$
Qz: None
Wz: None

8.12 X-INVALID-NUMER-12 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_5 R_L r_o s + R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o + s^2 (C_1 C_5 R_5 R_L r_o + C_1 C_L R_5 R_L r_o + C_5 C_L R_5 R_L r_o) + s (C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2C_5 R_5 R_L g_m r_o + 4C_5 R_5 R_L + C_5 R_5 r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{R_5} \sqrt{R_L} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{R_5 g_m + \frac{R_5}{r_o} + 2R_L g_m + \frac{4R_L}{r_o} + 1}}{C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2C_5 R_5 R_L g_m r_o + 4C_5 R_5 R_L + C_5 R_5 r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o}$

wo: $\sqrt{\frac{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}{C_1 C_5 R_5 R_L r_o + C_1 C_L R_5 R_L r_o + C_5 C_L R_5 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}{C_1 C_5 R_5 R_L r_o + C_1 C_L R_5 R_L r_o + C_5 C_L R_5 R_L r_o}} (C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2C_5 R_5 R_L g_m r_o + 4C_5 R_5 R_L + C_5 R_5 r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o)}{\sqrt{R_5} \sqrt{R_L} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{R_5 g_m + \frac{R_5}{r_o} + 2R_L g_m + \frac{4R_L}{r_o} + 1}}$

K-LP: $\frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}$

K-HP: 0

K-BP: $-\frac{C_5 R_5 R_L r_o}{C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2C_5 R_5 R_L g_m r_o + 4C_5 R_5 R_L + C_5 R_5 r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o}$

Qz: None

Wz: None

8.13 X-INVALID-NUMER-13 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s (C_5 R_5 R_L g_m r_o + C_5 R_5 R_L - C_5 R_L r_o)}{g_m r_o + s^2 (C_1 C_5 R_5 R_L + C_1 C_5 R_5 r_o + C_1 C_5 R_L r_o) + s (C_1 R_L + C_1 r_o + C_5 R_5 g_m r_o + C_5 R_5 + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o) + 1}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} R_5 R_L \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{1}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_5} R_5 r_o \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{1}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_5} R_L r_o \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{1}{R_5 R_L + R_5 r_o + R_L r_o}}}{C_1 R_L + C_1 r_o + C_5 R_5 g_m r_o + C_5 R_5 + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o}$

wo: $\sqrt{\frac{g_m r_o + 1}{C_1 C_5 R_5 R_L + C_1 C_5 R_5 r_o + C_1 C_5 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{g_m r_o + 1}{C_1 C_5 R_5 R_L + C_1 C_5 R_5 r_o + C_1 C_5 R_L r_o}} (C_1 R_L + C_1 r_o + C_5 R_5 g_m r_o + C_5 R_5 + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o)}{\sqrt{C_1} \sqrt{C_5} R_5 R_L \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{1}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_5} R_5 r_o \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{1}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_5} R_L r_o \sqrt{\frac{g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{1}{R_5 R_L + R_5 r_o + R_L r_o}}}$

K-LP: R_L

K-HP: 0

K-BP: $\frac{C_5 R_5 R_L g_m r_o + C_5 R_5 R_L - C_5 R_L r_o}{C_1 R_L + C_1 r_o + C_5 R_5 g_m r_o + C_5 R_5 + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o}$

Qz: None

Wz: None

8.14 X-INVALID-NUMER-14 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o + s (C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L - C_L R_1 R_L r_o)}{2R_1 g_m r_o + 4R_1 + R_5 + r_o + s^2 (C_1 C_L R_1 R_5 R_L + C_1 C_L R_1 R_5 r_o + C_1 C_L R_1 R_L r_o) + s (C_1 R_1 R_5 + C_1 R_1 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + 2C_L R_1 R_L g_m r_o + 4C_L R_1 R_L + C_L R_1 r_o + C_L R_5 R_L + C_L R_5 r_o + C_L R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_5 R_L \sqrt{\frac{2R_1 g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{4R_1}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_5}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{r_o}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_5 r_o \sqrt{\frac{2R_1 g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{4R_1}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_5}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{r_o}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_L r_o \sqrt{\frac{2R_1 g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{4R_1}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_5}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{r_o}{R_5 R_L + R_5 r_o + R_L r_o}}}{C_1 R_1 R_5 + C_1 R_1 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + 2C_L R_1 R_L g_m r_o + 4C_L R_1 R_L + C_L R_1 r_o + C_L R_5 R_L + C_L R_5 r_o + C_L R_L r_o}$

wo: $\sqrt{\frac{2R_1 g_m r_o + 4R_1 + R_5 + r_o}{C_1 C_L R_1 R_5 R_L + C_1 C_L R_1 R_5 r_o + C_1 C_L R_1 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{2R_1 g_m r_o + 4R_1 + R_5 + r_o}{C_1 C_L R_1 R_5 R_L + C_1 C_L R_1 R_5 r_o + C_1 C_L R_1 R_L r_o}} (C_1 R_1 R_5 + C_1 R_1 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + 2C_L R_1 R_L g_m r_o + 4C_L R_1 R_L + C_L R_1 r_o + C_L R_5 R_L + C_L R_5 r_o + C_L R_L r_o)}{\sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_5 R_L \sqrt{\frac{2R_1 g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{4R_1}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_5}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{r_o}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_5 r_o \sqrt{\frac{2R_1 g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{4R_1}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_5}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{r_o}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_L} \sqrt{R_1} R_L r_o \sqrt{\frac{2R_1 g_m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{4R_1}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_5}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{r_o}{R_5 R_L + R_5 r_o + R_L r_o}}}$

K-LP: $\frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{2R_1 g_m r_o + 4R_1 + R_5 + r_o}$

K-HP: 0

K-BP: $\frac{C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L - C_L R_1 R_L r_o}{C_1 R_1 R_5 + C_1 R_1 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + 2C_L R_1 R_L g_m r_o + 4C_L R_1 R_L + C_L R_1 r_o + C_L R_5 R_L + C_L R_5 r_o + C_L R_L r_o}$

Qz: None

Wz: None

8.15 X-INVALID-NUMER-15 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_5 R_1 R_L r_o s + R_1 R_L g_m r_o + R_1 R_L}{C_1 C_5 R_1 R_L r_o s^2 + R_1 g_m r_o + R_1 + R_L + r_o + s(C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_L} r_o \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}{C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o}$

wo: $\frac{\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{r_o}}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{r_o}} (C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o)}{C_1 C_5 R_1 R_L r_o \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}$

K-LP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$

K-HP: 0

K-BP: $-\frac{C_5 R_1 R_L r_o}{C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o}$

Qz: None

Wz: None

8.16 X-INVALID-NUMER-16 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_1 r_o s + R_1 g_m r_o + R_1}{s^2 (C_1 C_5 R_1 r_o + C_1 C_L R_1 r_o + C_5 C_L R_1 r_o) + s(C_1 R_1 + 2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

Parameters:

Q: $\frac{\sqrt{R_1} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{\frac{1}{r_o}}}{C_1 R_1 + 2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

wo: $\sqrt{\frac{1}{C_1 C_5 R_1 r_o + C_1 C_L R_1 r_o + C_5 C_L R_1 r_o}}$

bandwidth: $\frac{(C_1 R_1 + 2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) \sqrt{\frac{1}{C_1 C_5 R_1 r_o + C_1 C_L R_1 r_o + C_5 C_L R_1 r_o}}}{\sqrt{R_1} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{\frac{1}{r_o}}}$

K-LP: $R_1 g_m r_o + R_1$

K-HP: 0

K-BP: $-\frac{C_5 R_1 r_o}{C_1 R_1 + 2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o}$

Qz: None

Wz: None

8.17 X-INVALID-NUMER-17 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_1 R_L r_o s + R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o + s^2 (C_1 C_5 R_1 R_L r_o + C_1 C_L R_1 R_L r_o + C_5 C_L R_1 R_L r_o) + s(C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_L} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}{C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}$

wo: $\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_1 C_5 R_1 R_L r_o + C_1 C_L R_1 R_L r_o + C_5 C_L R_1 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_1 C_5 R_1 R_L r_o + C_1 C_L R_1 R_L r_o + C_5 C_L R_1 R_L r_o}} (C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}{\sqrt{R_1} \sqrt{R_L} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{R_1 g_m + \frac{R_1}{r_o} + \frac{R_L}{r_o} + 1}}$

K-LP: $\frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$

K-HP: 0

K-BP: $-\frac{C_5 R_1 R_L r_o}{C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o}$

Qz: None

Wz: None

8.18 X-INVALID-NUMER-18 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$

$$H(s) = \frac{-C_5 R_1 R_5 R_L r_o s + R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{C_1 C_5 R_1 R_5 R_L r_o s^2 + R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o + s(C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o + 2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L} r_o \sqrt{R_1 R_5 g_m + \frac{R_1 R_5}{r_o} + 2R_1 R_L g_m + \frac{4R_1 R_L}{r_o} + R_1 + \frac{R_5 R_L}{r_o} + R_5 + R_L}}{C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o + 2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o}$

wo: $\frac{\sqrt{\frac{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}{r_o}}}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1} \sqrt{R_5} \sqrt{R_L}}$

bandwidth: $\frac{\sqrt{\frac{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}{r_o}}}{C_1 C_5 R_1 R_5 R_L r_o \sqrt{R_1 R_5 g_m + \frac{R_1 R_5}{r_o} + 2R_1 R_L g_m + \frac{4R_1 R_L}{r_o} + R_1 + \frac{R_5 R_L}{r_o} + R_5 + R_L}}$

K-LP: $\frac{R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}$

K-HP: 0

K-BP: $-\frac{C_5 R_1 R_5 R_L r_o}{C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o + 2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o}$

Qz: None

Wz: None

8.19 X-INVALID-NUMER-19 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 R_1 R_5 r_o s + R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{2R_1 g_m r_o + 4R_1 + R_5 + r_o + s^2(C_1 C_5 R_1 R_5 r_o + C_1 C_L R_1 R_5 r_o + C_5 C_L R_1 R_5 r_o) + s(C_1 R_1 R_5 + C_1 R_1 r_o + 2C_5 R_1 R_5 g_m r_o + 4C_5 R_1 R_5 + C_5 R_5 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_5} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{2R_1 g_m + \frac{4R_1}{r_o} + \frac{R_5}{r_o} + 1}}{C_1 R_1 R_5 + C_1 R_1 r_o + 2C_5 R_1 R_5 g_m r_o + 4C_5 R_1 R_5 + C_5 R_5 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o}$

wo: $\sqrt{\frac{2R_1 g_m r_o + 4R_1 + R_5 + r_o}{C_1 C_5 R_1 R_5 r_o + C_1 C_L R_1 R_5 r_o + C_5 C_L R_1 R_5 r_o}}$

bandwidth: $\frac{\sqrt{\frac{2R_1 g_m r_o + 4R_1 + R_5 + r_o}{C_1 C_5 R_1 R_5 r_o + C_1 C_L R_1 R_5 r_o + C_5 C_L R_1 R_5 r_o}}}{\sqrt{R_1} \sqrt{R_5} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{2R_1 g_m + \frac{4R_1}{r_o} + \frac{R_5}{r_o} + 1}}$

K-LP: $\frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{2R_1 g_m r_o + 4R_1 + R_5 + r_o}$

K-HP: 0

K-BP: $-\frac{C_5 R_1 R_5 r_o}{C_1 R_1 R_5 + C_1 R_1 r_o + 2C_5 R_1 R_5 g_m r_o + 4C_5 R_1 R_5 + C_5 R_5 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o}$

Qz: None

Wz: None

8.20 X-INVALID-NUMER-20 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_1 R_5 R_L r_o s + R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o + s^2(C_1 C_5 R_1 R_5 R_L r_o + C_1 C_L R_1 R_5 R_L r_o + C_5 C_L R_1 R_5 R_L r_o) + s(C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o + 2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o)}$$

Parameters:

Q: $\frac{\sqrt{R_1} \sqrt{R_5} \sqrt{R_L} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{R_1 R_5 g_m + \frac{R_1 R_5}{r_o} + 2R_1 R_L g_m + \frac{4R_1 R_L}{r_o} + R_1 + \frac{R_5 R_L}{r_o} + R_5 + R_L}}{C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o + 2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o}$

wo: $\sqrt{\frac{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}{C_1 C_5 R_1 R_5 R_L r_o + C_1 C_L R_1 R_5 R_L r_o + C_5 C_L R_1 R_5 R_L r_o}}$

bandwidth: $\frac{\sqrt{\frac{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}{C_1 C_5 R_1 R_5 R_L r_o + C_1 C_L R_1 R_5 R_L r_o + C_5 C_L R_1 R_5 R_L r_o}}}{\sqrt{R_1} \sqrt{R_5} \sqrt{R_L} r_o \sqrt{C_1 C_5 + C_1 C_L + C_5 C_L} \sqrt{R_1 R_5 g_m + \frac{R_1 R_5}{r_o} + 2R_1 R_L g_m + \frac{4R_1 R_L}{r_o} + R_1 + \frac{R_5 R_L}{r_o} + R_5 + R_L}}$

K-LP: $\frac{R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}$

K-HP: 0

K-BP: $-\frac{C_5 R_1 R_5 R_L r_o}{C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o + 2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o}$

Qz: None

Wz: None

8.21 X-INVALID-NUMER-21 $Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s(C_5 R_1 R_5 R_L g_m r_o + C_5 R_1 R_5 R_L - C_5 R_1 R_L r_o)}{R_1 g_m r_o + R_1 + R_L + r_o + s^2(C_1 C_5 R_1 R_5 R_L + C_1 C_5 R_1 R_5 r_o + C_1 C_5 R_1 R_L r_o) + s(C_1 R_1 R_L + C_1 R_1 r_o + C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_5 R_L + C_5 R_5 r_o + C_5 R_L r_o)}$$

Parameters:

$$Q: \frac{\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}R_5RL\sqrt{\frac{R_1gm\tau_0}{R_5R_L+R_5\tau_0+R_L\tau_0}+\frac{R_1}{R_5R_L+R_5\tau_0+R_L\tau_0}+\frac{R_L}{R_5R_L+R_5\tau_0+R_L\tau_0}+\frac{r_0}{R_5R_L+R_5\tau_0+R_L\tau_0}}+\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}R_5r_0\sqrt{\frac{R_1gm\tau_0}{R_5R_L+R_5\tau_0+R_L\tau_0}+\frac{R_1}{R_5R_L+R_5\tau_0+R_L\tau_0}+\frac{R_L}{R_5R_L+R_5\tau_0+R_L\tau_0}+\frac{r_0}{R_5R_L+R_5\tau_0+R_L\tau_0}}+\sqrt{C_1}\sqrt{C_5}\sqrt{R_1}RLr_0\sqrt{\frac{R_1gm\tau_0}{R_5R_L+R_5\tau_0+R_L\tau_0}+\frac{R_1}{R_5R_L+R_5\tau_0+R_L\tau_0}+\frac{R_L}{R_5R_L+R_5\tau_0+R_L\tau_0}+\frac{r_0}{R_5R_L+R_5\tau_0+R_L\tau_0}}}{C_1R_1R_L+C_1R_1r_0+C_5R_1R_5gm\tau_0+C_5R_1R_5+2C_5R_1R_1gm\tau_0+4C_5R_1R_L+C_5R_1r_0+C_5R_5R_L+C_5R_5\tau_0+C_5R_5R_Lr_0}$$

$$\text{WO: } \sqrt{\frac{R_1 g_m r_o + R_1 + R_L + r_o}{C_1 C_5 R_1 R_5 R_L + C_1 C_5 R_1 R_5 r_o + C_1 C_5 R_1 R_L r_o}}$$

$$\text{bandwidth: } \frac{\sqrt{\frac{R_1 g m r_o + R_1 + R_L + r_o}{C_1 C_5 R_1 R_5 R_L + C_1 C_5 R_1 R_5 r_o + C_1 C_5 R_1 R_L r_o}} (C_1 R_1 R_L + C_1 R_1 r_o + C_5 R_1 R_5 g m r_o + C_5 R_1 R_5 + 2 C_5 R_1 R_L g m r_o + 4 C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_5 R_L + C_5 R_5 r_o + C_5 R_L r_o)}{\sqrt{C_1} \sqrt{C_5} \sqrt{R_1 R_5 R_L} \sqrt{\frac{R_1 g m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_1}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_L}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{r_o}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_5} \sqrt{R_1 R_5 r_o} \sqrt{\frac{R_1 g m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_1}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_L}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{r_o}{R_5 R_L + R_5 r_o + R_L r_o}} + \sqrt{C_1} \sqrt{C_5} \sqrt{R_1 R_L r_o} \sqrt{\frac{R_1 g m r_o}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_1}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{R_L}{R_5 R_L + R_5 r_o + R_L r_o} + \frac{r_o}{R_5 R_L + R_5 r_o + R_L r_o}}$$

$$\text{K-LP: } \frac{R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o}$$

K-HP: 0

$$\text{K-BP: } \frac{C_5 R_1 R_5 R_L g_m r_o + C_5 R_1 R_5 R_L - C_5 R_1 R_L r_o}{C_1 R_1 R_L + C_1 R_1 r_o + C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_5 R_L + C_5 R_5 r_o + C_5 R_L r_o}$$

Qz: None

Wz: None

8.22 X-INVALID-NUMER-22 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_5 g_m r_o + R_5 - r_o + s(C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 - C_1 R_1 r_o)}{2g_m r_o + s^2(C_1 C_L R_1 R_5 g_m r_o + C_1 C_L R_1 R_5 + C_1 C_L R_1 r_o + C_1 C_L R_5 r_o) + s(2C_1 R_1 g_m r_o + 4C_1 R_1 + C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

Parameters:

$$\text{Q: } \sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_5g_m r_o \sqrt{\frac{g_m r_o}{R_1R_5g_m r_o + R_1R_5 + R_1r_o + R_5r_o} + \frac{2}{R_1R_5g_m r_o + R_1R_5 + R_1r_o + R_5r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_5 \sqrt{\frac{g_m r_o}{R_1R_5g_m r_o + R_1R_5 + R_1r_o + R_5r_o} + \frac{2}{R_1R_5g_m r_o + R_1R_5 + R_1r_o + R_5r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1r_o \sqrt{\frac{g_m r_o}{R_1R_5g_m r_o + R_1R_5 + R_1r_o + R_5r_o} + \frac{2}{R_1R_5g_m r_o + R_1R_5 + R_1r_o + R_5r_o}} + \sqrt{2}\sqrt{C_1}\sqrt{C_L}R_5r_o \sqrt{\frac{g_m r_o}{R_1R_5g_m r_o + R_1R_5 + R_1r_o + R_5r_o} + \frac{2}{R_1R_5g_m r_o + R_1R_5 + R_1r_o + R_5r_o}}$$

$$\text{WO: } \sqrt{\frac{2g_m r_o + 4}{C_1 C_L R_1 R_5 g_m r_o + C_1 C_L R_1 R_5 + C_1 C_L R_1 r_o + C_1 C_L R_5 r_o}}$$

bandwidth: $\frac{\sqrt{2g_m r_o + 4}}{\sqrt{C_1 C_L R_1 R_5 g_m r_o + C_1 C_L R_1 R_5 + C_1 C_L R_1 r_o + C_1 C_L R_5 r_o}} (2C_1 R_1 g_m r_o + 4C_1 R_1 + C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o)$

$$\text{K-LP: } \frac{R_5 g_m r_o + R_5 - r_o}{2 g_m r_o + 4}$$

K-HP: 0

$$\text{K-BP: } \frac{C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 - C_1 R_1 r_o}{2C_1 R_1 g_m r_o + 4C_1 R_1 + C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o}$$

Qz: None

Wz: None

8.23 X-INVALID-NUMER-23 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o + s(C_1 R_1 R_5 R_L g_m r_o + C_1 R_1 R_5 R_L - C_1 R_1 R_L r_o)}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o + s^2(C_1 C_L R_1 R_5 R_L g_m r_o + C_1 C_L R_1 R_5 R_L + C_1 C_L R_1 R_L r_o + C_1 C_L R_5 R_L r_o) + s(C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 + 2C_1 R_1 R_L g_m r_o + 4C_1 R_1 R_L + C_1 R_1 r_o + C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o)}$$

Parameters:

$$Q: \frac{\sqrt{C_1} \sqrt{C_L} R_1 R_5 \sqrt{R_L} g m r_o}{\sqrt{\frac{R_5 g m r_o}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_5}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{2 R_L g m r_o}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{4 R_L}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{r_o}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}}} + \sqrt{C_1} \sqrt{C_L} R_1 R_5 \sqrt{R_L} \sqrt{\frac{R_5 g m r_o}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_5}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{2 R_L g m r_o}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{4 R_L}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{r_o}{R_1 R_5 g m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}}$$

$$\text{WO: } \sqrt{\frac{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}{C_1 C_L R_1 R_5 R_L g_m r_o + C_1 C_L R_1 R_5 R_L + C_1 C_L R_1 R_L r_o + C_1 C_L R_5 R_L r_o}}$$

bandwidth: $\frac{\sqrt{C_1} \sqrt{C_L} R_1 R_5 \sqrt{R_L} g_m r_o \sqrt{\frac{R_5 g_m r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_5}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{2 R_L g_m r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{4 R_L}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}}}{\sqrt{C_1} \sqrt{C_L} R_1 R_5 \sqrt{R_L}} \sqrt{\frac{R_5 g_m r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{R_5}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{2 R_L g_m r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{4 R_L}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o} + \frac{r_o}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o}} \sqrt{\frac{R_5 g_m r_o + R_5 + 2 R_L g_m r_o + 4 R_L}{C_1 C_L R_1 R_5 R_L g_m r_o + C_1 C_L R_1 R_5 R_L + C_1 C_L R_1 R_5}}$

$$\text{K-LP: } \frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o}$$

K-HP: 0

$$\text{K-BP: } \frac{C_1 R_1 R_5 R_L g_m r_o + C_1 R_1 R_5 R_L - C_1 R_1 R_L r_o}{C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 + 2C_1 R_1 R_L g_m r_o + 4C_1 R_1 R_L + C_1 R_1 r_o + C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o}$$

Qz: None

Wz: None

9 X-INVALID-ORDER

9.1 X-INVALID-ORDER-1 $Z(s) = (R_1, \infty, \infty, \infty, R_5, R_L)$

$$H(s) = \frac{R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o}$$

9.2 X-INVALID-ORDER-2 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5, \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o}{2R_1 g_m r_o + 4R_1 + R_5 + r_o + s(C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o)}$$

9.3 X-INVALID-ORDER-3 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5, \frac{R_L}{C_L R_L s + 1}\right)$

$$H(s) = \frac{R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o + s(C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L + C_L R_1 R_L r_o + C_L R_5 R_L r_o)}$$

9.4 X-INVALID-ORDER-4 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5, R_L + \frac{1}{C_L s}\right)$

$$H(s) = \frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o + s(C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L - C_L R_1 R_L r_o)}{2R_1 g_m r_o + 4R_1 + R_5 + r_o + s(C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + 2C_L R_1 R_L g_m r_o + 4C_L R_1 R_L + C_L R_1 r_o + C_L R_5 R_L + C_L R_5 r_o + C_L R_L r_o)}$$

9.5 X-INVALID-ORDER-5 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L\right)$

$$H(s) = \frac{-C_5 R_1 R_L r_o s + R_1 R_L g_m r_o + R_1 R_L}{R_1 g_m r_o + R_1 + R_L + r_o + s(2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o)}$$

9.6 X-INVALID-ORDER-6 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{-C_5 L_L R_1 r_o s^2 + s(L_L R_1 g_m r_o + L_L R_1)}{C_5 C_L L_L R_1 r_o s^3 + R_1 g_m r_o + R_1 + r_o + s^2(2C_5 L_L R_1 g_m r_o + 4C_5 L_L R_1 + C_5 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o) + s(C_5 R_1 r_o + L_L)}$$

9.7 X-INVALID-ORDER-7 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5 s}, L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{-C_5 C_L L_L R_1 r_o s^3 - C_5 R_1 r_o s + R_1 g_m r_o + R_1 + s^2(C_L L_L R_1 g_m r_o + C_L L_L R_1)}{s^3(2C_5 C_L L_L R_1 g_m r_o + 4C_5 C_L L_L R_1 + C_5 C_L L_L r_o) + s^2(C_5 C_L R_1 r_o + C_L L_L) + s(2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

9.8 X-INVALID-ORDER-8 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L\right)$

$$H(s) = \frac{-C_5 R_1 R_5 R_L r_o s + R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{R_1 R_5 g_m r_o + R_1 R_5 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o + s(2C_5 R_1 R_5 R_L g_m r_o + 4C_5 R_1 R_5 R_L + C_5 R_1 R_5 r_o + C_5 R_5 R_L r_o)}$$

9.9 X-INVALID-ORDER-9 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{L_L s}{C_L L_L s^2 + 1}\right)$

$$H(s) = \frac{-C_5 L_L R_1 R_5 r_o s^2 + s(L_L R_1 R_5 g_m r_o + L_L R_1 R_5 - L_L R_1 r_o)}{C_5 C_L L_L R_1 R_5 r_o s^3 + R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o + s^2(2C_5 L_L R_1 R_5 g_m r_o + 4C_5 L_L R_1 R_5 + C_5 L_L R_5 r_o + C_L L_L R_1 R_5 g_m r_o + C_L L_L R_1 R_5 + C_L L_L R_1 r_o + C_L L_L R_5 r_o) + s(C_5 R_1 R_5 r_o + 2L_L R_1 g_m r_o + 4L_L R_1 + L_L R_5 + L_L r_o)}$$

9.10 X-INVALID-ORDER-10 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, L_L s + \frac{1}{C_L s}\right)$

$$H(s) = \frac{-C_5 C_L L_L R_1 R_5 r_o s^3 - C_5 R_1 R_5 r_o s + R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o + s^2(C_L L_L R_1 R_5 g_m r_o + C_L L_L R_1 R_5 - C_L L_L R_1 r_o)}{2R_1 g_m r_o + 4R_1 + R_5 + r_o + s^3(2C_5 C_L L_L R_1 R_5 g_m r_o + 4C_5 C_L L_L R_1 R_5 + C_5 C_L L_L R_5 r_o) + s^2(C_5 C_L R_1 R_5 r_o + 2C_L L_L R_1 g_m r_o + 4C_L L_L R_1 + C_L L_L R_5 + C_L L_L r_o) + s(2C_5 R_1 R_5 g_m r_o + 4C_5 R_1 R_5 + C_5 R_5 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o)}$$

$$\mathbf{9.11 \quad X-INVALID-ORDER-11} \quad Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L \right)$$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s (C_5 R_1 R_5 R_L g_m r_o + C_5 R_1 R_5 R_L - C_5 R_1 R_L r_o)}{R_1 g_m r_o + R_1 + R_L + r_o + s (C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_5 R_L + C_5 R_5 r_o + C_5 R_L r_o)}$$

$$\mathbf{9.12 \quad X-INVALID-ORDER-12} \quad Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (C_5 L_L R_1 R_5 g_m r_o + C_5 L_L R_1 R_5 - C_5 L_L R_1 r_o) + s (L_L R_1 g_m r_o + L_L R_1)}{R_1 g_m r_o + R_1 + r_o + s^3 (C_5 C_L L_L R_1 R_5 g_m r_o + C_5 C_L L_L R_1 R_5 + C_5 C_L L_L R_1 r_o + C_5 C_L L_L R_5 r_o) + s^2 (2C_5 L_L R_1 g_m r_o + 4C_5 L_L R_1 + C_5 L_L R_5 + C_5 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o) + s (C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 + C_5 R_1 r_o + C_5 R_5 r_o + L_L)}$$

$$\mathbf{9.13 \quad X-INVALID-ORDER-13} \quad Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 (C_5 C_L L_L R_1 R_5 g_m r_o + C_5 C_L L_L R_1 R_5 - C_5 C_L L_L R_1 r_o) + s^2 (C_L L_L R_1 g_m r_o + C_L L_L R_1) + s (C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 - C_5 R_1 r_o)}{s^3 (2C_5 C_L L_L R_1 g_m r_o + 4C_5 C_L L_L R_1 + C_5 C_L L_L R_5 + C_5 C_L L_L r_o) + s^2 (C_5 C_L R_1 R_5 g_m r_o + C_5 C_L R_1 R_5 + C_5 C_L R_1 r_o + C_5 C_L R_5 r_o + C_L L_L) + s (2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 R_5 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.14 \quad X-INVALID-ORDER-14} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 r_o s^2 - R_1 r_o + s (L_5 R_1 g_m r_o + L_5 R_1)}{C_5 C_L L_5 R_1 r_o s^3 + 2R_1 g_m r_o + 4R_1 + r_o + s^2 (2C_5 L_5 R_1 g_m r_o + 4C_5 L_5 R_1 + C_5 L_5 r_o + C_L L_5 R_1 g_m r_o + C_L L_5 R_1 + C_L L_5 r_o) + s (C_L R_1 r_o + L_5)}$$

$$\mathbf{9.15 \quad X-INVALID-ORDER-15} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_L r_o s^2 - R_1 R_L r_o + s (L_5 R_1 R_L g_m r_o + L_5 R_1 R_L)}{C_5 C_L L_5 R_1 R_L r_o s^3 + 2R_1 R_L g_m r_o + 4R_1 R_L + R_1 r_o + R_L r_o + s^2 (2C_5 L_5 R_1 R_L g_m r_o + 4C_5 L_5 R_1 R_L + C_5 L_5 R_1 r_o + C_5 L_5 R_L r_o + C_L L_5 R_1 R_L g_m r_o + C_L L_5 R_1 R_L + C_L L_5 R_L r_o) + s (C_L R_1 R_L r_o + L_5 R_1 g_m r_o + L_5 R_1 + L_5 R_L + L_5 r_o)}$$

$$\mathbf{9.16 \quad X-INVALID-ORDER-16} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_5 R_1 R_L r_o s^3 - R_1 r_o + s^2 (-C_5 L_5 R_1 r_o + C_L L_5 R_1 R_L g_m r_o + C_L L_5 R_1 R_L) + s (-C_L R_1 R_L r_o + L_5 R_1 g_m r_o + L_5 R_1)}{2R_1 g_m r_o + 4R_1 + r_o + s^3 (2C_5 C_L L_5 R_1 R_L g_m r_o + 4C_5 C_L L_5 R_1 R_L + C_5 C_L L_5 R_1 r_o + C_5 C_L L_5 R_L r_o) + s^2 (2C_5 L_5 R_1 g_m r_o + 4C_5 L_5 R_1 + C_5 L_5 r_o + C_L L_5 R_1 g_m r_o + C_L L_5 R_1 + C_L L_5 R_L + C_L L_5 r_o) + s (2C_L R_1 R_L g_m r_o + 4C_L R_1 R_L + C_L R_1 r_o + C_L R_L r_o + L_5)}$$

$$\mathbf{9.17 \quad X-INVALID-ORDER-17} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_5 L_L R_1 r_o s^3 - L_L R_1 r_o s + s^2 (L_5 L_L R_1 g_m r_o + L_5 L_L R_1)}{C_5 C_L L_5 L_L R_1 r_o s^4 + R_1 r_o + s^3 (2C_5 L_5 L_L R_1 g_m r_o + 4C_5 L_5 L_L R_1 + C_5 L_5 L_L r_o + C_L L_5 L_L R_1 g_m r_o + C_L L_5 L_L R_1 + C_L L_5 L_L r_o) + s^2 (C_5 L_5 R_1 r_o + C_L L_L R_1 r_o + L_5 L_L) + s (L_5 R_1 g_m r_o + L_5 R_1 + L_5 r_o + 2L_L R_1 g_m r_o + 4L_L R_1 + L_L r_o)}$$

$$\mathbf{9.18 \quad X-INVALID-ORDER-18} \quad Z(s) = \left(R_1, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_5 L_L R_1 r_o s^4 - R_1 r_o + s^3 (C_L L_5 L_L R_1 g_m r_o + C_L L_5 L_L R_1) + s^2 (-C_5 L_5 R_1 r_o - C_L L_L R_1 r_o) + s (L_5 R_1 g_m r_o + L_5 R_1)}{2R_1 g_m r_o + 4R_1 + r_o + s^4 (2C_5 C_L L_5 L_L R_1 g_m r_o + 4C_5 C_L L_5 L_L R_1 + C_5 C_L L_5 L_L r_o) + s^3 (C_5 C_L L_5 R_1 r_o + C_L L_5 L_L) + s^2 (2C_5 L_5 R_1 g_m r_o + 4C_5 L_5 R_1 + C_5 L_5 r_o + C_L L_5 R_1 g_m r_o + C_L L_5 R_1 + C_L L_5 r_o + 2C_L L_L R_1 g_m r_o + 4C_L L_L R_1 + C_L L_L r_o) + s (C_L R_1 r_o + L_5)}$$

$$\mathbf{9.19 \quad X-INVALID-ORDER-19} \quad Z(s) = \left(R_1, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 R_1 r_o s + R_1 g_m r_o + R_1 + s^2 (C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1)}{s^3 (C_5 C_L L_5 R_1 g_m r_o + C_5 C_L L_5 R_1 + C_5 C_L L_5 r_o) + s^2 (C_5 C_L R_1 r_o + C_5 L_5) + s (2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.20 \quad X-INVALID-ORDER-20} \quad Z(s) = \left(R_1, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_5 R_1 R_L r_o s + R_1 R_L g_m r_o + R_1 R_L + s^2 (C_5 L_5 R_1 R_L g_m r_o + C_5 L_5 R_1 R_L)}{R_1 g_m r_o + R_1 + R_L + r_o + s^3 (C_5 C_L L_5 R_1 R_L g_m r_o + C_5 C_L L_5 R_1 R_L + C_5 C_L L_5 R_L r_o) + s^2 (C_5 C_L R_1 R_L r_o + C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1 + C_5 L_5 R_L + C_5 L_5 r_o) + s (2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L + C_L R_L r_o)}$$

$$\mathbf{9.21 \quad X-INVALID-ORDER-21} \quad Z(s) = \left(R_1, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 (C_5 C_L L_5 R_1 R_L g_m r_o + C_5 C_L L_5 R_1 R_L) + s^2 (-C_5 C_L R_1 R_L r_o + C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1) + s (-C_5 R_1 r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^3 (C_5 C_L L_5 R_1 g_m r_o + C_5 C_L L_5 R_1 + C_5 C_L L_5 R_L + C_5 C_L L_5 r_o) + s^2 (2C_5 C_L R_1 R_L g_m r_o + 4C_5 C_L R_1 R_L + C_5 C_L R_1 r_o + C_5 C_L R_L r_o + C_5 L_5) + s (2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o) + 1}$$

$$\mathbf{9.22 \quad X-INVALID-ORDER-22} \quad Z(s) = \left(R_1, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_L R_1 r_o s^2 + s^3 (C_5 L_5 L_L R_1 g_m r_o + C_5 L_5 L_L R_1) + s (L_L R_1 g_m r_o + L_L R_1)}{R_1 g_m r_o + R_1 + r_o + s^4 (C_5 C_L L_5 L_L R_1 g_m r_o + C_5 C_L L_5 L_L R_1 + C_5 C_L L_5 L_L r_o) + s^3 (C_5 C_L L_L R_1 r_o + C_5 L_5 L_L) + s^2 (C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1 + C_5 L_5 r_o + 2C_5 L_L R_1 g_m r_o + 4C_5 L_L R_1 + C_5 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o) + s (C_5 R_1 r_o + L_L)}$$

$$\mathbf{9.23 \quad X-INVALID-ORDER-23} \quad Z(s) = \left(R_1, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_L R_1 r_o s^3 - C_5 R_1 r_o s + R_1 g_m r_o + R_1 + s^4 (C_5 C_L L_5 L_L R_1 g_m r_o + C_5 C_L L_5 L_L R_1) + s^2 (C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1 + C_L L_L R_1 g_m r_o + C_L L_L R_1)}{C_5 C_L L_5 L_L s^4 + s^3 (C_5 C_L L_5 R_1 g_m r_o + C_5 C_L L_5 R_1 + C_5 C_L L_5 r_o + 2C_5 C_L L_L R_1 g_m r_o + 4C_5 C_L L_L R_1 + C_5 C_L L_L r_o) + s^2 (C_5 C_L R_1 r_o + C_5 L_5 + C_L L_L) + s (2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.24 \quad X-INVALID-ORDER-24} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5, R_L \right)$$

$$H(s) = \frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o + s (C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o)}$$

$$\mathbf{9.25 \quad X-INVALID-ORDER-25} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s (L_L R_5 g_m r_o + L_L R_5 - L_L r_o)}{C_1 C_L L_L R_5 r_o s^3 + R_5 g_m r_o + R_5 + r_o + s^2 (C_1 L_L R_5 + C_1 L_L r_o + C_L L_L R_5 g_m r_o + C_L L_L R_5 + C_L L_L r_o) + s (C_1 R_5 r_o + 2L_L g_m r_o + 4L_L)}$$

$$\mathbf{9.26 \quad X-INVALID-ORDER-26} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, R_5, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_5 g_m r_o + R_5 - r_o + s^2 (C_L L_L R_5 g_m r_o + C_L L_L R_5 - C_L L_L r_o)}{2g_m r_o + s^3 (C_1 C_L L_L R_5 + C_1 C_L L_L r_o) + s^2 (C_1 C_L R_5 r_o + 2C_L L_L g_m r_o + 4C_L L_L) + s (C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

$$\mathbf{9.27 \quad X-INVALID-ORDER-27} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 r_o s + g_m r_o + 1}{s^2 (C_1 C_5 r_o + C_1 C_L r_o + C_5 C_L r_o) + s (C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.28 \quad X-INVALID-ORDER-28} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_L r_o s^2 + g_m r_o + s (-C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}{C_1 C_5 C_L R_L r_o s^3 + s^2 (C_1 C_5 r_o + C_1 C_L R_L + C_1 C_L r_o + 2C_5 C_L R_L g_m r_o + 4C_5 C_L R_L + C_5 C_L r_o) + s (C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.29 \quad X-INVALID-ORDER-29} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_L r_o s^2 + s (L_L g_m r_o + L_L)}{g_m r_o + s^3 (C_1 C_5 L_L r_o + C_1 C_L L_L r_o + C_5 C_L L_L r_o) + s^2 (C_1 L_L + 2C_5 L_L g_m r_o + 4C_5 L_L + C_L L_L g_m r_o + C_L L_L) + s (C_1 r_o + C_5 r_o) + 1}$$

$$\mathbf{9.30 \quad X-INVALID-ORDER-30} \quad Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_L r_o s^3 - C_5 r_o s + g_m r_o + s^2 (C_L L_L g_m r_o + C_L L_L) + 1}{C_1 C_5 C_L L_L r_o s^4 + s^3 (C_1 C_L L_L + 2C_5 C_L L_L g_m r_o + 4C_5 C_L L_L) + s^2 (C_1 C_5 r_o + C_1 C_L r_o + C_5 C_L r_o) + s (C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.31 \quad X-INVALID-ORDER-31} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_5 R_L r_o s^2 + R_5 g_m r_o + R_5 - r_o + s(-C_5 R_5 r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L - C_L R_L r_o)}{C_1 C_5 C_L R_5 R_L r_o s^3 + 2g_m r_o + s^2(C_1 C_5 R_5 r_o + C_1 C_L R_5 R_L + C_1 C_L R_5 r_o + C_1 C_L R_L r_o + 2C_5 C_L R_5 R_L g_m r_o + 4C_5 C_L R_5 R_L + C_5 C_L R_5 r_o) + s(C_1 R_5 + C_1 r_o + 2C_5 R_5 g_m r_o + 4C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + 2C_L R_L g_m r_o + 4C_L R_L + C_L r_o) + 4}$$

$$\mathbf{9.32 \quad X-INVALID-ORDER-32} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_L R_5 r_o s^2 + s(L_L R_5 g_m r_o + L_L R_5 - L_L r_o)}{R_5 g_m r_o + R_5 + r_o + s^3(C_1 C_5 L_L R_5 r_o + C_1 C_L L_L R_5 r_o + C_5 C_L L_L R_5 r_o) + s^2(C_1 L_L R_5 + C_1 L_L r_o + 2C_5 L_L R_5 g_m r_o + 4C_5 L_L R_5 + C_L L_L R_5 g_m r_o + C_L L_L R_5 + C_L L_L r_o) + s(C_1 R_5 r_o + C_5 R_5 r_o + 2L_L g_m r_o + 4L_L)}$$

$$\mathbf{9.33 \quad X-INVALID-ORDER-33} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_L R_5 r_o s^3 - C_5 R_5 r_o s + R_5 g_m r_o + R_5 - r_o + s^2(C_L L_L R_5 g_m r_o + C_L L_L R_5 - C_L L_L r_o)}{C_1 C_5 C_L L_L R_5 r_o s^4 + 2g_m r_o + s^3(C_1 C_L L_L R_5 + C_1 C_L L_L r_o + 2C_5 C_L L_L R_5 g_m r_o + 4C_5 C_L L_L R_5) + s^2(C_1 C_5 R_5 r_o + C_1 C_L R_5 r_o + C_5 C_L R_5 r_o + 2C_L L_L g_m r_o + 4C_L L_L) + s(C_1 R_5 + C_1 r_o + 2C_5 R_5 g_m r_o + 4C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

$$\mathbf{9.34 \quad X-INVALID-ORDER-34} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s(C_5 R_5 g_m r_o + C_5 R_5 - C_5 r_o) + 1}{C_1 C_5 C_L R_5 r_o s^3 + s^2(C_1 C_5 R_5 + C_1 C_5 r_o + C_1 C_L r_o + C_5 C_L R_5 g_m r_o + C_5 C_L R_5 + C_5 C_L r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.35 \quad X-INVALID-ORDER-35} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s(C_5 R_5 R_L g_m r_o + C_5 R_5 R_L - C_5 R_L r_o)}{C_1 C_5 C_L R_5 R_L r_o s^3 + g_m r_o + s^2(C_1 C_5 R_5 R_L + C_1 C_5 R_5 r_o + C_1 C_5 R_L r_o + C_1 C_L R_L r_o + C_5 C_L R_5 R_L g_m r_o + C_5 C_L R_5 R_L + C_5 C_L R_L r_o) + s(C_1 R_L + C_1 r_o + C_5 R_5 g_m r_o + C_5 R_5 + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}$$

$$\mathbf{9.36 \quad X-INVALID-ORDER-36} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^2(C_5 C_L R_5 R_L g_m r_o + C_5 C_L R_5 R_L - C_5 C_L R_L r_o) + s(C_5 R_5 g_m r_o + C_5 R_5 - C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}{s^3(C_1 C_5 C_L R_5 R_L + C_1 C_5 C_L R_5 r_o + C_1 C_5 C_L R_L r_o) + s^2(C_1 C_5 R_5 + C_1 C_5 r_o + C_1 C_L R_L + C_1 C_L r_o + C_5 C_L R_5 g_m r_o + C_5 C_L R_5 + 2C_5 C_L R_L g_m r_o + 4C_5 C_L R_L + C_5 C_L r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.37 \quad X-INVALID-ORDER-37} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2(C_5 L_L R_5 g_m r_o + C_5 L_L R_5 - C_5 L_L r_o) + s(L_L g_m r_o + L_L)}{C_1 C_5 C_L L_L R_5 r_o s^4 + g_m r_o + s^3(C_1 C_5 L_L R_5 + C_1 C_5 L_L r_o + C_1 C_L L_L r_o + C_5 C_L L_L R_5 g_m r_o + C_5 C_L L_L R_5 + C_5 C_L L_L r_o) + s^2(C_1 C_5 R_5 r_o + C_1 L_L + 2C_5 L_L g_m r_o + 4C_5 L_L + C_L L_L g_m r_o + C_L L_L) + s(C_1 r_o + C_5 R_5 g_m r_o + C_5 R_5 + C_5 r_o) + 1}$$

$$\mathbf{9.38 \quad X-INVALID-ORDER-38} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^3(C_5 C_L L_L R_5 g_m r_o + C_5 C_L L_L R_5 - C_5 C_L L_L r_o) + s^2(C_L L_L g_m r_o + C_L L_L) + s(C_5 R_5 g_m r_o + C_5 R_5 - C_5 r_o) + 1}{s^4(C_1 C_5 C_L L_L R_5 + C_1 C_5 C_L L_L r_o) + s^3(C_1 C_5 C_L R_5 r_o + C_1 C_L L_L + 2C_5 C_L L_L g_m r_o + 4C_5 C_L L_L) + s^2(C_1 C_5 R_5 + C_1 C_5 r_o + C_1 C_L r_o + C_5 C_L R_5 g_m r_o + C_5 C_L R_5 + C_5 C_L r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.39 \quad X-INVALID-ORDER-39} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad R_L \right)$$

$$H(s) = \frac{-C_5 L_5 R_L r_o s^2 - R_L r_o + s(L_5 R_L g_m r_o + L_5 R_L)}{C_1 C_5 L_5 R_L r_o s^3 + 2R_L g_m r_o + 4R_L + r_o + s^2(C_1 L_5 R_L + C_1 L_5 r_o + 2C_5 L_5 R_L g_m r_o + 4C_5 L_5 R_L + C_5 L_5 r_o) + s(C_1 R_L r_o + L_5 g_m r_o + L_5)}$$

$$\mathbf{9.40 \quad X-INVALID-ORDER-40} \quad Z(s) = \left(\frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 L_5 r_o s^2 - r_o + s(L_5 g_m r_o + L_5)}{2g_m r_o + s^3(C_1 C_5 L_5 r_o + C_1 C_L L_5 r_o + C_5 C_L L_5 r_o) + s^2(C_1 L_5 + 2C_5 L_5 g_m r_o + 4C_5 L_5 + C_L L_5 g_m r_o + C_L L_5) + s(C_1 r_o + C_L r_o) + 4}$$

9.41 X-INVALID-ORDER-41 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 L_5 R_L r_o s^2 - R_L r_o + s(L_5 R_L g_m r_o + L_5 R_L)}{2R_L g_m r_o + 4R_L + r_o + s^3(C_1 C_5 L_5 R_L r_o + C_1 C_L L_5 R_L r_o + C_5 C_L L_5 R_L r_o) + s^2(C_1 L_5 R_L + C_1 L_5 r_o + 2C_5 L_5 R_L g_m r_o + 4C_5 L_5 R_L + C_5 L_5 r_o + C_L L_5 R_L g_m r_o + C_L L_5 R_L) + s(C_1 R_L r_o + C_L R_L r_o + L_5 g_m r_o + L_5)}$$

9.42 X-INVALID-ORDER-42 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 C_L L_5 R_L r_o s^3 - r_o + s^2(-C_5 L_5 r_o + C_L L_5 R_L g_m r_o + C_L L_5 R_L) + s(-C_L R_L r_o + L_5 g_m r_o + L_5)}{C_1 C_5 C_L L_5 R_L r_o s^4 + 2g_m r_o + s^3(C_1 C_5 L_5 r_o + C_1 C_L L_5 R_L + C_1 C_L L_5 r_o + 2C_5 C_L L_5 R_L g_m r_o + 4C_5 C_L L_5 R_L + C_5 C_L L_5 r_o) + s^2(C_1 C_L R_L r_o + C_1 L_5 + 2C_5 L_5 g_m r_o + 4C_5 L_5 + C_L L_5 g_m r_o + C_L L_5) + s(C_1 r_o + 2C_L R_L g_m r_o + 4C_L R_L + C_L r_o) + 4}$$

9.43 X-INVALID-ORDER-43 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{-C_5 L_5 L_L r_o s^3 - L_L r_o s + s^2(L_5 L_L g_m r_o + L_5 L_L)}{r_o + s^4(C_1 C_5 L_5 L_L r_o + C_1 C_L L_5 L_L r_o + C_5 C_L L_5 L_L r_o) + s^3(C_1 L_5 L_L + 2C_5 L_5 L_L g_m r_o + 4C_5 L_5 L_L + C_L L_5 L_L g_m r_o + C_L L_5 L_L) + s^2(C_1 L_5 r_o + C_1 L_L r_o + C_5 L_5 r_o + C_L L_L r_o) + s(L_5 g_m r_o + L_5 + 2L_L g_m r_o + 4L_L)}$$

9.44 X-INVALID-ORDER-44 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 C_L L_5 L_L r_o s^4 - r_o + s^3(C_L L_5 L_L g_m r_o + C_L L_5 L_L) + s^2(-C_5 L_5 r_o - C_L L_L r_o) + s(L_5 g_m r_o + L_5)}{C_1 C_5 C_L L_5 L_L r_o s^5 + 2g_m r_o + s^4(C_1 C_L L_5 L_L + 2C_5 C_L L_5 L_L g_m r_o + 4C_5 C_L L_5 L_L) + s^3(C_1 C_5 L_5 r_o + C_1 C_L L_5 r_o + C_1 C_L L_L r_o + C_5 C_L L_5 r_o) + s^2(C_1 L_5 + 2C_5 L_5 g_m r_o + 4C_5 L_5 + C_L L_5 g_m r_o + C_L L_5 + 2C_L L_L g_m r_o + 4C_L L_L) + s(C_1 r_o + C_L r_o) + 4}$$

9.45 X-INVALID-ORDER-45 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_5 R_L r_o s + R_L g_m r_o + R_L + s^2(C_5 L_5 R_L g_m r_o + C_5 L_5 R_L)}{g_m r_o + s^3(C_1 C_5 L_5 R_L + C_1 C_5 L_5 r_o) + s^2(C_1 C_5 R_L r_o + C_5 L_5 g_m r_o + C_5 L_5) + s(C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o) + 1}$$

9.46 X-INVALID-ORDER-46 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 r_o s + g_m r_o + s^2(C_5 L_5 g_m r_o + C_5 L_5) + 1}{C_1 C_5 C_L L_5 r_o s^4 + s^3(C_1 C_5 L_5 + C_5 C_L L_5 g_m r_o + C_5 C_L L_5) + s^2(C_1 C_5 r_o + C_1 C_L r_o + C_5 C_L r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

9.47 X-INVALID-ORDER-47 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_5 R_L r_o s + R_L g_m r_o + R_L + s^2(C_5 L_5 R_L g_m r_o + C_5 L_5 R_L)}{C_1 C_5 C_L L_5 R_L r_o s^4 + g_m r_o + s^3(C_1 C_5 L_5 R_L + C_1 C_5 L_5 r_o + C_5 C_L L_5 R_L g_m r_o + C_5 C_L L_5 R_L) + s^2(C_1 C_5 R_L r_o + C_1 C_L R_L r_o + C_5 C_L R_L r_o + C_5 L_5 g_m r_o + C_5 L_5) + s(C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}$$

9.48 X-INVALID-ORDER-48 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^3(C_5 C_L L_5 R_L g_m r_o + C_5 C_L L_5 R_L) + s^2(-C_5 C_L R_L r_o + C_5 L_5 g_m r_o + C_5 L_5) + s(-C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}{s^4(C_1 C_5 C_L L_5 R_L + C_1 C_5 C_L L_5 r_o) + s^3(C_1 C_5 C_L R_L r_o + C_1 C_5 L_5 + C_5 C_L L_5 g_m r_o + C_5 C_L L_5) + s^2(C_1 C_5 r_o + C_1 C_L R_L + C_1 C_L r_o + 2C_5 C_L R_L g_m r_o + 4C_5 C_L R_L + C_5 C_L r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

9.49 X-INVALID-ORDER-49 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{-C_5 L_L r_o s^2 + s^3(C_5 L_5 L_L g_m r_o + C_5 L_5 L_L) + s(L_L g_m r_o + L_L)}{C_1 C_5 C_L L_5 L_L r_o s^5 + g_m r_o + s^4(C_1 C_5 L_5 L_L + C_5 C_L L_5 L_L g_m r_o + C_5 C_L L_5 L_L) + s^3(C_1 C_5 L_5 r_o + C_1 C_5 L_L r_o + C_1 C_L L_L r_o + C_5 C_L L_L r_o) + s^2(C_1 L_L + C_5 L_5 g_m r_o + C_5 L_5 + 2C_5 L_L g_m r_o + 4C_5 L_L + C_L L_L g_m r_o + C_L L_L) + s(C_1 r_o + C_5 r_o) + 1}$$

9.50 X-INVALID-ORDER-50 $Z(s) = \left(\frac{1}{C_1 s}, \infty, \infty, \infty, L_5 s + \frac{1}{C_5 s}, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 C_L L_L r_o s^3 - C_5 r_o s + g_m r_o + s^4(C_5 C_L L_5 L_L g_m r_o + C_5 C_L L_5 L_L) + s^2(C_5 L_5 g_m r_o + C_5 L_5 + C_L L_L g_m r_o + C_L L_L) + 1}{C_1 C_5 C_L L_5 L_L s^5 + s^4(C_1 C_5 C_L L_5 r_o + C_1 C_5 C_L L_L r_o) + s^3(C_1 C_5 L_5 + C_1 C_L L_L + C_5 C_L L_5 g_m r_o + C_5 C_L L_5 + 2C_5 C_L L_L g_m r_o + 4C_5 C_L L_L) + s^2(C_1 C_5 r_o + C_1 C_L r_o + C_5 C_L r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.51 \quad X-INVALID-ORDER-51} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, R_L \right)$$

$$H(s) = \frac{R_1 R_5 R_L g_m r_o + R_1 R_5 R_L - R_1 R_L r_o}{R_1 R_5 g_m r_o + R_1 R_5 + 2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_5 R_L + R_5 r_o + R_L r_o + s (C_1 R_1 R_5 R_L + C_1 R_1 R_5 r_o + C_1 R_1 R_L r_o)}$$

$$\mathbf{9.52 \quad X-INVALID-ORDER-52} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s (L_L R_1 R_5 g_m r_o + L_L R_1 R_5 - L_L R_1 r_o)}{C_1 C_L L_L R_1 R_5 r_o s^3 + R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o + s^2 (C_1 L_L R_1 R_5 + C_1 L_L R_1 r_o + C_L L_L R_1 R_5 g_m r_o + C_L L_L R_1 R_5 + C_L L_L R_1 r_o + C_L L_L R_5 r_o) + s (C_1 R_1 R_5 r_o + 2 L_L R_1 g_m r_o + 4 L_L R_1 + L_L R_5 + L_L r_o)}$$

$$\mathbf{9.53 \quad X-INVALID-ORDER-53} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o + s^2 (C_L L_L R_1 R_5 g_m r_o + C_L L_L R_1 R_5 - C_L L_L R_1 r_o)}{2 R_1 g_m r_o + 4 R_1 + R_5 + r_o + s^3 (C_1 C_L L_L R_1 R_5 + C_1 C_L L_L R_1 r_o) + s^2 (C_1 C_L R_1 R_5 r_o + 2 C_L L_L R_1 g_m r_o + 4 C_L L_L R_1 + C_L L_L R_5 + C_L L_L r_o) + s (C_1 R_1 R_5 + C_1 R_1 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + C_L R_1 r_o + C_L R_5 r_o)}$$

$$\mathbf{9.54 \quad X-INVALID-ORDER-54} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_1 R_L r_o s^2 + R_1 g_m r_o + R_1 + s (-C_5 R_1 r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{C_1 C_5 C_L R_1 R_L r_o s^3 + s^2 (C_1 C_5 R_1 r_o + C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + 2 C_5 C_L R_1 R_L g_m r_o + 4 C_5 C_L R_1 R_L + C_5 C_L R_1 r_o + C_5 C_L R_L r_o) + s (C_1 R_1 + 2 C_5 R_1 g_m r_o + 4 C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o) + 1}$$

$$\mathbf{9.55 \quad X-INVALID-ORDER-55} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_L R_1 r_o s^2 + s (L_L R_1 g_m r_o + L_L R_1)}{R_1 g_m r_o + R_1 + r_o + s^3 (C_1 C_5 L_L R_1 r_o + C_1 C_L L_L R_1 r_o + C_5 C_L L_L R_1 r_o) + s^2 (C_1 L_L R_1 + 2 C_5 L_L R_1 g_m r_o + 4 C_5 L_L R_1 + C_5 L_L r_o + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o) + s (C_1 R_1 r_o + C_5 R_1 r_o + L_L)}$$

$$\mathbf{9.56 \quad X-INVALID-ORDER-56} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_L R_1 r_o s^3 - C_5 R_1 r_o s + R_1 g_m r_o + R_1 + s^2 (C_L L_L R_1 g_m r_o + C_L L_L R_1)}{C_1 C_5 C_L L_L R_1 r_o s^4 + s^3 (C_1 C_L L_L R_1 + 2 C_5 C_L L_L R_1 g_m r_o + 4 C_5 C_L L_L R_1 + C_5 C_L L_L r_o) + s^2 (C_1 C_5 R_1 r_o + C_1 C_L R_1 r_o + C_5 C_L R_1 r_o + C_L L_L) + s (C_1 R_1 + 2 C_5 R_1 g_m r_o + 4 C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.57 \quad X-INVALID-ORDER-57} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L R_1 R_5 R_L r_o s^2 + R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o + s (-C_5 R_1 R_5 r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L - C_L R_1 R_L r_o)}{C_1 C_5 C_L R_1 R_5 R_L r_o s^3 + 2 R_1 g_m r_o + 4 R_1 + R_5 + r_o + s^2 (C_1 C_5 R_1 R_5 r_o + C_1 C_L R_1 R_5 R_L + C_1 C_L R_1 R_5 r_o + C_1 C_L R_1 R_L r_o + 2 C_5 C_L R_1 R_5 R_L g_m r_o + 4 C_5 C_L R_1 R_5 R_L + C_5 C_L R_1 R_5 r_o + C_5 C_L R_5 R_L r_o) + s (C_1 R_1 R_5 + C_1 R_1 r_o + 2 C_5 R_1 R_5 g_m r_o + 4 C_5 R_1 R_5 + C_5 R_5 r_o)}$$

$$\mathbf{9.58 \quad X-INVALID-ORDER-58} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_L R_1 R_5 r_o s^2 + s (L_L R_1 R_5 g_m r_o + L_L R_1 R_5 - L_L R_1 r_o)}{R_1 R_5 g_m r_o + R_1 R_5 + R_1 r_o + R_5 r_o + s^3 (C_1 C_5 L_L R_1 R_5 r_o + C_1 C_L L_L R_1 R_5 r_o + C_5 C_L L_L R_1 R_5 r_o) + s^2 (C_1 L_L R_1 R_5 + C_1 L_L R_1 r_o + 2 C_5 L_L R_1 R_5 g_m r_o + 4 C_5 L_L R_1 R_5 + C_5 L_L R_5 r_o + C_L L_L R_1 R_5 g_m r_o + C_L L_L R_1 R_5 + C_L L_L R_1 r_o + C_L L_L R_5 r_o) + s (C_1 R_1 R_5 r_o + C_5 R_5 r_o)}$$

$$\mathbf{9.59 \quad X-INVALID-ORDER-59} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_L R_1 R_5 r_o s^3 - C_5 R_1 R_5 r_o s + R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o + s^2 (C_L L_L R_1 R_5 g_m r_o + C_L L_L R_1 R_5 - C_L L_L R_1 r_o)}{C_1 C_5 C_L L_L R_1 R_5 r_o s^4 + 2 R_1 g_m r_o + 4 R_1 + R_5 + r_o + s^3 (C_1 C_L L_L R_1 R_5 + C_1 C_L L_L R_1 r_o + 2 C_5 C_L L_L R_1 R_5 g_m r_o + 4 C_5 C_L L_L R_1 R_5 + C_5 C_L L_L R_5 r_o) + s^2 (C_1 C_5 R_1 R_5 r_o + C_1 C_L R_1 R_5 r_o + C_5 C_L R_1 R_5 r_o + 2 C_L L_L R_1 g_m r_o + 4 C_L L_L R_1 + C_L L_L R_5 + C_L L_L r_o) + s (C_1 R_1 R_5 r_o + C_5 R_5 r_o)}$$

$$\mathbf{9.60 \quad X-INVALID-ORDER-60} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s (C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 - C_5 R_1 r_o)}{C_1 C_5 C_L R_1 R_5 r_o s^3 + s^2 (C_1 C_5 R_1 R_5 + C_1 C_5 R_1 r_o + C_1 C_L R_1 r_o + C_5 C_L R_1 R_5 g_m r_o + C_5 C_L R_1 R_5 + C_5 C_L R_1 r_o + C_5 C_L R_5 r_o) + s (C_1 R_1 + 2 C_5 R_1 g_m r_o + 4 C_5 R_1 + C_5 R_5 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.61 \quad X-INVALID-ORDER-61} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_1 R_L g_m r_o + R_1 R_L + s (C_5 R_1 R_5 R_L g_m r_o + C_5 R_1 R_5 R_L - C_5 R_1 R_L r_o)}{C_1 C_5 C_L R_1 R_5 R_L r_o s^3 + R_1 g_m r_o + R_1 + R_L + r_o + s^2 (C_1 C_5 R_1 R_5 R_L + C_1 C_5 R_1 R_5 r_o + C_1 C_5 R_1 R_L r_o + C_1 C_L R_1 R_L r_o + C_5 C_L R_1 R_5 R_L g_m r_o + C_5 C_L R_1 R_5 R_L + C_5 C_L R_1 R_L r_o + C_5 C_L R_5 R_L r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 + 2 C_5 R_1 R_L g_m r_o)}$$

$$\mathbf{9.62 \quad X-INVALID-ORDER-62} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2 (C_5 C_L R_1 R_5 R_L g_m r_o + C_5 C_L R_1 R_5 R_L - C_5 C_L R_1 R_L r_o) + s (C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 - C_5 R_1 r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^3 (C_1 C_5 C_L R_1 R_5 R_L + C_1 C_5 C_L R_1 R_5 r_o + C_1 C_5 C_L R_1 R_L r_o) + s^2 (C_1 C_5 R_1 R_5 + C_1 C_5 R_1 r_o + C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + C_5 C_L R_1 R_5 g_m r_o + C_5 C_L R_1 R_5 + 2 C_5 C_L R_1 R_L g_m r_o + 4 C_5 C_L R_1 R_L + C_5 C_L R_1 r_o + C_5 C_L R_5 R_L + C_5 C_L R_5 r_o + C_5 C_L R_L r_o) + s (C_1 R_1 + 2 C_5 R_1 R_L g_m r_o)}$$

$$\mathbf{9.63 \quad X-INVALID-ORDER-63} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (C_5 L_L R_1 R_5 g_m r_o + C_5 L_L R_1 R_5 - C_5 L_L R_1 r_o) + s (L_L R_1 g_m r_o + L_L R_1)}{C_1 C_5 C_L L_L R_1 R_5 r_o s^4 + R_1 g_m r_o + R_1 + r_o + s^3 (C_1 C_5 L_L R_1 R_5 + C_1 C_5 L_L R_1 r_o + C_1 C_L L_L R_1 r_o + C_5 C_L L_L R_1 R_5 g_m r_o + C_5 C_L L_L R_1 R_5 + C_5 C_L L_L R_1 r_o + C_5 C_L L_L R_5 r_o) + s^2 (C_1 C_5 R_1 R_5 r_o + C_1 L_L R_1 + 2 C_5 L_L R_1 g_m r_o + 4 C_5 L_L R_1 + C_5 L_L R_5 + C_5 L_L r_o + C_L L_L R_1 r_o)}$$

$$\mathbf{9.64 \quad X-INVALID-ORDER-64} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 (C_5 C_L L_L R_1 R_5 g_m r_o + C_5 C_L L_L R_1 R_5 - C_5 C_L L_L R_1 r_o) + s^2 (C_L L_L R_1 g_m r_o + C_L L_L R_1) + s (C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 - C_5 R_1 r_o)}{s^4 (C_1 C_5 C_L L_L R_1 R_5 + C_1 C_5 C_L L_L R_1 r_o) + s^3 (C_1 C_5 C_L R_1 R_5 r_o + C_1 C_L L_L R_1 + 2 C_5 C_L L_L R_1 g_m r_o + 4 C_5 C_L L_L R_1 + C_5 C_L L_L R_5 + C_5 C_L L_L r_o) + s^2 (C_1 C_5 R_1 R_5 + C_1 C_5 R_1 r_o + C_1 C_L R_1 r_o + C_5 C_L R_1 R_5 g_m r_o + C_5 C_L R_1 R_5 + C_5 C_L R_1 r_o + C_5 C_L R_5 r_o + C_L L_L) + s (C_5 R_1 R_5 g_m r_o)}$$

$$\mathbf{9.65 \quad X-INVALID-ORDER-65} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, R_L \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_L r_o s^2 - R_1 R_L r_o + s (L_5 R_1 R_L g_m r_o + L_5 R_1 R_L)}{C_1 C_5 L_5 R_1 R_L r_o s^3 + 2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_L r_o + s^2 (C_1 L_5 R_1 R_L + C_1 L_5 R_1 r_o + 2 C_5 L_5 R_1 R_L g_m r_o + 4 C_5 L_5 R_1 R_L + C_5 L_5 R_1 r_o + C_5 L_5 R_L r_o) + s (C_1 R_1 R_L r_o + L_5 R_1 g_m r_o + L_5 R_1 + L_5 R_L + L_5 r_o)}$$

$$\mathbf{9.66 \quad X-INVALID-ORDER-66} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 r_o s^2 - R_1 r_o + s (L_5 R_1 g_m r_o + L_5 R_1)}{2 R_1 g_m r_o + 4 R_1 + r_o + s^3 (C_1 C_5 L_5 R_1 r_o + C_1 C_L L_5 R_1 r_o + C_5 C_L L_5 R_1 r_o) + s^2 (C_1 L_5 R_1 + 2 C_5 L_5 R_1 g_m r_o + 4 C_5 L_5 R_1 + C_5 L_5 r_o + C_L L_5 R_1 g_m r_o + C_L L_5 R_1 + C_L L_5 r_o) + s (C_1 R_1 r_o + C_L R_1 r_o + L_5)}$$

$$\mathbf{9.67 \quad X-INVALID-ORDER-67} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_5 L_5 R_1 R_L r_o s^2 - R_1 R_L r_o + s (L_5 R_1 R_L g_m r_o + L_5 R_1 R_L)}{2 R_1 R_L g_m r_o + 4 R_1 R_L + R_1 r_o + R_L r_o + s^3 (C_1 C_5 L_5 R_1 R_L r_o + C_1 C_L L_5 R_1 R_L r_o + C_5 C_L L_5 R_1 R_L r_o) + s^2 (C_1 L_5 R_1 R_L + C_1 L_5 R_1 r_o + 2 C_5 L_5 R_1 R_L g_m r_o + 4 C_5 L_5 R_1 R_L + C_5 L_5 R_1 r_o + C_5 L_5 R_L r_o + C_L L_5 R_1 R_L g_m r_o + C_L L_5 R_1 R_L + C_L L_5 R_L r_o) + s (C_1 R_1 R_L r_o + C_L R_1 R_L r_o + L_5)}$$

$$\mathbf{9.68 \quad X-INVALID-ORDER-68} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_5 R_1 R_L r_o s^3 - R_1 r_o + s^2 (-C_5 L_5 R_1 r_o + C_L L_5 R_1 R_L g_m r_o + C_L L_5 R_1 R_L) + s (-C_L R_1 R_L r_o + L_5 R_1 g_m r_o + L_5 R_1)}{C_1 C_5 C_L L_5 R_1 R_L r_o s^4 + 2 R_1 g_m r_o + 4 R_1 + r_o + s^3 (C_1 C_5 L_5 R_1 r_o + C_1 C_L L_5 R_1 R_L + C_1 C_L L_5 R_1 r_o + 2 C_5 C_L L_5 R_1 R_L g_m r_o + 4 C_5 C_L L_5 R_1 R_L + C_5 C_L L_5 R_1 r_o + C_5 C_L L_5 R_L r_o) + s^2 (C_1 C_L R_1 R_L r_o + C_1 L_5 R_1 + 2 C_5 L_5 R_1 g_m r_o + 4 C_5 L_5 R_1 + C_5 L_5 r_o + C_L L_5 R_1 g_m r_o + C_L L_5 R_1 R_L r_o)}$$

$$\mathbf{9.69 \quad X-INVALID-ORDER-69} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_5 L_L R_1 r_o s^3 - L_L R_1 r_o s + s^2 (L_5 L_L R_1 g_m r_o + L_5 L_L R_1)}{R_1 r_o + s^4 (C_1 C_5 L_5 L_L R_1 r_o + C_1 C_L L_5 L_L R_1 r_o + C_5 C_L L_5 L_L R_1 r_o) + s^3 (C_1 L_5 L_L R_1 + 2 C_5 L_5 L_L R_1 g_m r_o + 4 C_5 L_5 L_L R_1 + C_5 L_5 L_L r_o + C_L L_5 L_L R_1 g_m r_o + C_L L_5 L_L R_1 + C_L L_5 L_L r_o) + s^2 (C_1 L_5 R_1 r_o + C_1 L_L R_1 r_o + C_5 L_5 R_1 r_o + C_L L_L R_1 r_o + L_5 L_L) + s (L_5 R_1 g_m r_o + L_5 R_1)}$$

$$\mathbf{9.70 \quad X-INVALID-ORDER-70} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_5 L_L R_1 r_o s^4 - R_1 r_o + s^3 (C_L L_5 L_L R_1 g_m r_o + C_L L_5 L_L R_1) + s^2 (-C_5 L_5 R_1 r_o - C_L L_L R_1 r_o) + s (L_5 R_1 g_m r_o + L_5 R_1)}{C_1 C_5 C_L L_5 L_L R_1 r_o s^5 + 2 R_1 g_m r_o + 4 R_1 + r_o + s^4 (C_1 C_L L_5 L_L R_1 + 2 C_5 C_L L_5 L_L R_1 g_m r_o + 4 C_5 C_L L_5 L_L R_1 + C_5 C_L L_5 L_L r_o) + s^3 (C_1 C_5 L_5 R_1 r_o + C_1 C_L L_5 R_1 r_o + C_1 C_L L_L R_1 r_o + C_5 C_L L_5 R_1 r_o + C_L L_5 L_L) + s^2 (C_1 L_5 R_1 + 2 C_5 L_5 R_1 g_m r_o + 4 C_5 L_5 R_1 + C_5 L_5 r_o + C_L L_5 R_1 g_m r_o + C_L L_5 R_1 R_L r_o)}$$

$$\mathbf{9.71 \quad X-INVALID-ORDER-71} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad R_L \right)$$

$$H(s) = \frac{-C_5 R_1 R_L r_o s + R_1 R_L g_m r_o + R_1 R_L + s^2 (C_5 L_5 R_1 R_L g_m r_o + C_5 L_5 R_1 R_L)}{R_1 g_m r_o + R_1 + R_L + r_o + s^3 (C_1 C_5 L_5 R_1 R_L + C_1 C_5 L_5 R_1 r_o) + s^2 (C_1 C_5 R_1 R_L r_o + C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1 + C_5 L_5 R_L + C_5 L_5 r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o)}$$

$$\mathbf{9.72 \quad X-INVALID-ORDER-72} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 R_1 r_o s + R_1 g_m r_o + R_1 + s^2 (C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1)}{C_1 C_5 C_L L_5 R_1 r_o s^4 + s^3 (C_1 C_5 L_5 R_1 + C_5 C_L L_5 R_1 g_m r_o + C_5 C_L L_5 R_1 + C_5 C_L L_5 r_o) + s^2 (C_1 C_5 R_1 r_o + C_1 C_L R_1 r_o + C_5 C_L R_1 r_o + C_5 L_5) + s (C_1 R_1 + 2C_5 R_1 g_m r_o + 4C_5 R_1 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L r_o) + 1}$$

$$\mathbf{9.73 \quad X-INVALID-ORDER-73} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_5 R_1 R_L r_o s + R_1 R_L g_m r_o + R_1 R_L + s^2 (C_5 L_5 R_1 R_L g_m r_o + C_5 L_5 R_1 R_L)}{C_1 C_5 C_L L_5 R_1 R_L r_o s^4 + R_1 g_m r_o + R_1 + R_L + r_o + s^3 (C_1 C_5 L_5 R_1 R_L + C_1 C_5 L_5 R_1 r_o + C_5 C_L L_5 R_1 R_L g_m r_o + C_5 C_L L_5 R_1 R_L + C_5 C_L L_5 R_L r_o) + s^2 (C_1 C_5 R_1 R_L r_o + C_1 C_L R_1 R_L r_o + C_5 C_L R_1 R_L r_o + C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1 + C_5 L_5 R_L + C_5 L_5 r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o)}$$

$$\mathbf{9.74 \quad X-INVALID-ORDER-74} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^3 (C_5 C_L L_5 R_1 R_L g_m r_o + C_5 C_L L_5 R_1 R_L) + s^2 (-C_5 C_L R_1 R_L r_o + C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1) + s (-C_5 R_1 r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^4 (C_1 C_5 C_L L_5 R_1 R_L + C_1 C_5 C_L L_5 R_1 r_o) + s^3 (C_1 C_5 C_L R_1 R_L r_o + C_1 C_5 L_5 R_1 + C_5 C_L L_5 R_1 g_m r_o + C_5 C_L L_5 R_1 + C_5 C_L L_5 R_L + C_5 C_L L_5 r_o) + s^2 (C_1 C_5 R_1 r_o + C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + 2C_5 C_L R_1 R_L g_m r_o + 4C_5 C_L R_1 R_L + C_5 C_L R_1 r_o + C_5 C_L R_L r_o + C_5 L_5) + s (C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o)}$$

$$\mathbf{9.75 \quad X-INVALID-ORDER-75} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_L R_1 r_o s^2 + s^3 (C_5 L_5 L_L R_1 g_m r_o + C_5 L_5 L_L R_1) + s (L_L R_1 g_m r_o + L_L R_1)}{C_1 C_5 C_L L_5 L_L R_1 r_o s^5 + R_1 g_m r_o + R_1 + r_o + s^4 (C_1 C_5 L_5 L_L R_1 + C_5 C_L L_5 L_L R_1 g_m r_o + C_5 C_L L_5 L_L R_1 + C_5 C_L L_5 L_L r_o) + s^3 (C_1 C_5 L_5 R_1 r_o + C_1 C_5 L_L R_1 r_o + C_1 C_L L_L R_1 r_o + C_5 C_L L_L R_1 r_o + C_5 L_5 L_L) + s^2 (C_1 L_L R_1 + C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1 + C_5 L_5 r_o + 2C_5 L_L R_1 g_m r_o + 4C_5 L_L R_1 + C_5 L_L R_L r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o)}$$

$$\mathbf{9.76 \quad X-INVALID-ORDER-76} \quad Z(s) = \left(\frac{R_1}{C_1 R_1 s + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_L R_1 r_o s^3 - C_5 R_1 r_o s + R_1 g_m r_o + R_1 + s^4 (C_5 C_L L_5 L_L R_1 g_m r_o + C_5 C_L L_5 L_L R_1) + s^2 (C_5 L_5 R_1 g_m r_o + C_5 L_5 R_1 + C_L L_L R_1 g_m r_o + C_L L_L R_1)}{C_1 C_5 C_L L_5 L_L R_1 s^5 + s^4 (C_1 C_5 C_L L_5 R_1 r_o + C_1 C_5 C_L L_L R_1 r_o + C_5 C_L L_5 L_L) + s^3 (C_1 C_5 L_5 R_1 + C_1 C_L L_L R_1 + C_5 C_L L_5 R_1 g_m r_o + C_5 C_L L_5 R_1 + C_5 C_L L_5 r_o + 2C_5 C_L L_L R_1 g_m r_o + 4C_5 C_L L_L R_1 + C_5 C_L L_L r_o) + s^2 (C_1 C_5 R_1 r_o + C_1 C_L R_1 r_o + C_5 C_L R_1 r_o + C_5 L_5 + C_L L_L R_1 g_m r_o + C_L L_L R_1 + C_L L_L r_o) + s (C_1 R_1 R_L + C_1 R_1 r_o + 2C_5 R_1 R_L g_m r_o + 4C_5 R_1 R_L + C_5 R_1 r_o + C_5 R_L r_o)}$$

$$\mathbf{9.77 \quad X-INVALID-ORDER-77} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad R_L \right)$$

$$H(s) = \frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o + s (C_1 R_1 R_5 R_L g_m r_o + C_1 R_1 R_5 R_L - C_1 R_1 R_L r_o)}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o + s (C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 + 2C_1 R_1 R_L g_m r_o + 4C_1 R_1 R_L + C_1 R_1 r_o + C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o)}$$

$$\mathbf{9.78 \quad X-INVALID-ORDER-78} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (C_1 L_L R_1 R_5 g_m r_o + C_1 L_L R_1 R_5 - C_1 L_L R_1 r_o) + s (L_L R_5 g_m r_o + L_L R_5 - L_L r_o)}{R_5 g_m r_o + R_5 + r_o + s^3 (C_1 C_L L_L R_1 R_5 g_m r_o + C_1 C_L L_L R_1 R_5 + C_1 C_L L_L R_1 r_o + C_1 C_L L_L R_5 r_o) + s^2 (2C_1 L_L R_1 g_m r_o + 4C_1 L_L R_1 + C_1 L_L R_5 + C_1 L_L r_o + C_L L_L R_5 g_m r_o + C_L L_L R_5 + C_L L_L r_o) + s (C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 + C_1 R_1 r_o + C_1 R_5 r_o + 2L_L g_m r_o + 4L_L)}$$

$$\mathbf{9.79 \quad X-INVALID-ORDER-79} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_5 g_m r_o + R_5 - r_o + s^3 (C_1 C_L L_L R_1 R_5 g_m r_o + C_1 C_L L_L R_1 R_5 - C_1 C_L L_L R_1 r_o) + s^2 (C_L L_L R_5 g_m r_o + C_L L_L R_5 - C_L L_L r_o) + s (C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 - C_1 R_1 r_o)}{2g_m r_o + s^3 (2C_1 C_L L_L R_1 g_m r_o + 4C_1 C_L L_L R_1 + C_1 C_L L_L R_5 + C_1 C_L L_L r_o) + s^2 (C_1 C_L R_1 R_5 g_m r_o + C_1 C_L R_1 R_5 + C_1 C_L R_1 r_o + C_1 C_L R_5 r_o + 2C_L L_L g_m r_o + 4C_L L_L) + s (2C_1 R_1 g_m r_o + 4C_1 R_1 + C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

$$\mathbf{9.80 \quad X-INVALID-ORDER-80} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 r_o s^2 + g_m r_o + s (C_1 R_1 g_m r_o + C_1 R_1 - C_5 r_o) + 1}{C_1 C_5 C_L R_1 r_o s^3 + s^2 (2C_1 C_5 R_1 g_m r_o + 4C_1 C_5 R_1 + C_1 C_5 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o + C_5 C_L r_o) + s (C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.81 \quad X-INVALID-ORDER-81} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_L r_o s^2 + R_L g_m r_o + R_L + s(C_1 R_1 R_L g_m r_o + C_1 R_1 R_L - C_5 R_L r_o)}{C_1 C_5 C_L R_1 R_L r_o s^3 + g_m r_o + s^2(2C_1 C_5 R_1 R_L g_m r_o + 4C_1 C_5 R_1 R_L + C_1 C_5 R_1 r_o + C_1 C_5 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_L r_o + C_5 C_L R_L r_o) + s(C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}$$

$$\mathbf{9.82 \quad X-INVALID-ORDER-82} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L R_1 R_L r_o s^3 + g_m r_o + s^2(-C_1 C_5 R_1 r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L - C_5 C_L R_L r_o) + s(C_1 R_1 g_m r_o + C_1 R_1 - C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}{s^3(2C_1 C_5 C_L R_1 R_L g_m r_o + 4C_1 C_5 C_L R_1 R_L + C_1 C_5 C_L R_1 r_o + C_1 C_5 C_L R_L r_o) + s^2(2C_1 C_5 R_1 g_m r_o + 4C_1 C_5 R_1 + C_1 C_5 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + 2C_5 C_L R_L g_m r_o + 4C_5 C_L R_L + C_5 C_L r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.83 \quad X-INVALID-ORDER-83} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 L_L R_1 r_o s^3 + s^2(C_1 L_L R_1 g_m r_o + C_1 L_L R_1 - C_5 L_L r_o) + s(L_L g_m r_o + L_L)}{C_1 C_5 C_L L_L R_1 r_o s^4 + g_m r_o + s^3(2C_1 C_5 L_L R_1 g_m r_o + 4C_1 C_5 L_L R_1 + C_1 C_5 L_L r_o + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C_L L_L r_o + C_5 C_L L_L r_o) + s^2(C_1 C_5 R_1 r_o + C_1 L_L + 2C_5 L_L g_m r_o + 4C_5 L_L + C_L L_L g_m r_o + C_L L_L) + s(C_1 R_1 g_m r_o + C_1 R_1 + C_1 r_o + C_5 r_o) + 1}$$

$$\mathbf{9.84 \quad X-INVALID-ORDER-84} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L L_L R_1 r_o s^4 + g_m r_o + s^3(C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 - C_5 C_L L_L r_o) + s^2(-C_1 C_5 R_1 r_o + C_L L_L g_m r_o + C_L L_L) + s(C_1 R_1 g_m r_o + C_1 R_1 - C_5 r_o) + 1}{s^4(2C_1 C_5 C_L L_L R_1 g_m r_o + 4C_1 C_5 C_L L_L R_1 + C_1 C_5 C_L L_L r_o) + s^3(C_1 C_5 C_L R_1 r_o + C_1 C_L L_L + 2C_5 C_L L_L g_m r_o + 4C_5 C_L L_L) + s^2(2C_1 C_5 R_1 g_m r_o + 4C_1 C_5 R_1 + C_1 C_5 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o + C_5 C_L r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.85 \quad X-INVALID-ORDER-85} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 r_o s^2 + R_5 g_m r_o + R_5 - r_o + s(C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 - C_1 R_1 r_o - C_5 R_5 r_o)}{C_1 C_5 C_L R_1 R_5 r_o s^3 + 2g_m r_o + s^2(2C_1 C_5 R_1 R_5 g_m r_o + 4C_1 C_5 R_1 R_5 + C_1 C_5 R_5 r_o + C_1 C_L R_1 R_5 g_m r_o + C_1 C_L R_1 R_5 + C_1 C_L R_1 r_o + C_1 C_L R_5 r_o + C_5 C_L R_5 r_o) + s(2C_1 R_1 g_m r_o + 4C_1 R_1 + C_1 R_5 + C_1 r_o + 2C_5 R_5 g_m r_o + 4C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

$$\mathbf{9.86 \quad X-INVALID-ORDER-86} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 R_L r_o s^2 + R_5 R_L g_m r_o + R_5 R_L - R_L r_o + s(C_1 R_1 R_5 R_L g_m r_o + C_1 R_1 R_5 R_L - C_1 R_1 R_L r_o - C_5 R_5 R_L r_o)}{C_1 C_5 C_L R_1 R_5 R_L r_o s^3 + R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o + s^2(2C_1 C_5 R_1 R_5 R_L g_m r_o + 4C_1 C_5 R_1 R_5 R_L + C_1 C_5 R_1 R_5 r_o + C_1 C_5 R_5 R_L r_o + C_1 C_L R_1 R_5 R_L g_m r_o + C_1 C_L R_1 R_5 R_L + C_1 C_L R_1 R_L r_o + C_1 C_L R_5 R_L r_o + C_5 C_L R_5 R_L r_o) + s(C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 + C_1 R_1 R_L + C_1 R_1 R_5 + C_1 R_1 r_o + C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

$$\mathbf{9.87 \quad X-INVALID-ORDER-87} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L R_1 R_5 R_L r_o s^3 + R_5 g_m r_o + R_5 - r_o + s^2(-C_1 C_5 R_1 R_5 r_o + C_1 C_L R_1 R_5 R_L g_m r_o + C_1 C_L R_1 R_5 R_L - C_1 C_L R_1 R_L r_o - C_5 C_L R_5 R_L r_o) + s(C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 R_L + C_1 R_1 R_L + C_1 R_1 R_5 + C_1 R_1 r_o + C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 1}{2g_m r_o + s^3(2C_1 C_5 C_L R_1 R_5 R_L g_m r_o + 4C_1 C_5 C_L R_1 R_5 R_L + C_1 C_5 C_L R_1 R_5 r_o + C_1 C_5 C_L R_5 R_L r_o) + s^2(2C_1 C_5 R_1 R_5 g_m r_o + 4C_1 C_5 R_1 R_5 + C_1 C_5 R_5 r_o + C_1 C_L R_1 R_5 g_m r_o + C_1 C_L R_1 R_5 + 2C_1 C_L R_1 R_L g_m r_o + 4C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + C_1 C_L R_5 R_L + C_1 C_L R_5 r_o + C_5 C_L R_5 r_o) + s(C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 R_L + C_1 R_1 R_L + C_1 R_1 R_5 + C_1 R_1 r_o + C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 1}$$

$$\mathbf{9.88 \quad X-INVALID-ORDER-88} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 L_L R_1 R_5 r_o s^3 + s^2(C_1 L_L R_1 R_5 g_m r_o + C_1 L_L R_1 R_5 - C_1 L_L R_1 r_o - C_5 L_L R_5 r_o) + s(L_L R_5 g_m r_o + L_L R_5 - L_L r_o)}{C_1 C_5 C_L L_L R_1 R_5 r_o s^4 + R_5 g_m r_o + R_5 + r_o + s^3(2C_1 C_5 L_L R_1 R_5 g_m r_o + 4C_1 C_5 L_L R_1 R_5 + C_1 C_5 L_L R_5 r_o + C_1 C_L L_L R_1 R_5 g_m r_o + C_1 C_L L_L R_1 R_5 + C_1 C_L L_L R_1 r_o + C_1 C_L L_L R_5 r_o + C_5 C_L L_L R_5 r_o) + s^2(C_1 C_5 R_1 R_5 r_o + 2C_1 L_L R_1 g_m r_o + 4C_1 L_L R_1 + C_1 L_L R_5 + C_1 L_L R_5 + C_1 L_L r_o + C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 1}$$

$$\mathbf{9.89 \quad X-INVALID-ORDER-89} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L L_L R_1 R_5 r_o s^4 + R_5 g_m r_o + R_5 - r_o + s^3(C_1 C_L L_L R_1 R_5 g_m r_o + C_1 C_L L_L R_1 R_5 - C_1 C_L L_L R_1 r_o - C_5 C_L L_L R_5 r_o) + s^2(-C_1 C_5 R_1 R_5 r_o + C_L L_L g_m r_o + C_L L_L) + s(C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 R_L + C_1 R_1 R_L + C_1 R_1 R_5 + C_1 R_1 r_o + C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 1}{2g_m r_o + s^4(2C_1 C_5 C_L L_L R_1 R_5 g_m r_o + 4C_1 C_5 C_L L_L R_1 R_5 + C_1 C_5 C_L L_L R_5 r_o) + s^3(C_1 C_5 C_L R_1 R_5 r_o + 2C_1 C_L L_L R_1 g_m r_o + 4C_1 C_L L_L R_1 + C_1 C_L L_L R_5 + C_1 C_L L_L r_o + 2C_5 C_L L_L R_5 g_m r_o + 4C_5 C_L L_L R_5) + s^2(2C_1 C_5 R_1 R_5 g_m r_o + 4C_1 C_5 R_1 R_5 + C_1 C_5 R_5 r_o + C_1 C_L R_1 R_5 g_m r_o + C_1 C_L R_1 R_5 + C_1 C_L R_1 r_o + C_5 C_L R_5 r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.90 \quad X-INVALID-ORDER-90} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^2(C_1 C_5 R_1 R_5 g_m r_o + C_1 C_5 R_1 R_5 - C_1 C_5 R_1 r_o) + s(C_1 R_1 g_m r_o + C_1 R_1 + C_5 R_5 g_m r_o + C_5 R_5 - C_5 r_o) + 1}{s^3(C_1 C_5 C_L R_1 R_5 g_m r_o + C_1 C_5 C_L R_1 R_5 + C_1 C_5 C_L R_1 r_o + C_1 C_5 C_L R_5 r_o) + s^2(2C_1 C_5 R_1 g_m r_o + 4C_1 C_5 R_1 + C_1 C_5 R_5 + C_1 C_5 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o + C_5 C_L R_5 g_m r_o + C_5 C_L R_5 + C_5 C_L r_o) + s(C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.91 \quad X-INVALID-ORDER-91} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2 (C_1 C_5 R_1 R_5 R_L g_m r_o + C_1 C_5 R_1 R_5 R_L - C_1 C_5 R_1 R_L r_o) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_5 R_5 R_L g_m r_o + C_5 R_5 R_L - C_5 R_5 R_L r_o)}{g_m r_o + s^3 (C_1 C_5 C_L R_1 R_5 R_L g_m r_o + C_1 C_5 C_L R_1 R_5 R_L + C_1 C_5 C_L R_1 R_L r_o + C_1 C_5 C_L R_5 R_L r_o) + s^2 (C_1 C_5 R_1 R_5 g_m r_o + C_1 C_5 R_1 R_5 + 2 C_1 C_5 R_1 R_L g_m r_o + 4 C_1 C_5 R_1 R_L + C_1 C_5 R_1 r_o + C_1 C_5 R_5 R_L + C_1 C_5 R_5 r_o + C_1 C_5 R_L r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_1 C_L R_5 R_L g_m r_o + C_1 C_L R_5 R_L - C_5 C_L R_L r_o) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_5 R_5 R_L g_m r_o + C_5 R_5 R_L - C_5 C_L R_L r_o)}$$

$$\mathbf{9.92 \quad X-INVALID-ORDER-92} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^3 (C_1 C_5 C_L R_1 R_5 R_L g_m r_o + C_1 C_5 C_L R_1 R_5 R_L - C_1 C_5 C_L R_1 R_L r_o) + s^2 (C_1 C_5 R_1 R_5 g_m r_o + C_1 C_5 R_1 R_5 - C_1 C_5 R_1 r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L + C_5 C_L R_5 R_L g_m r_o + C_5 C_L R_5 R_L - C_5 C_L R_L r_o) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_5 R_5 R_L g_m r_o + C_5 R_5 R_L - C_5 C_L R_L r_o)}{s^3 (C_1 C_5 C_L R_1 R_5 g_m r_o + C_1 C_5 C_L R_1 R_5 + 2 C_1 C_5 C_L R_1 R_L g_m r_o + 4 C_1 C_5 C_L R_1 R_L + C_1 C_5 C_L R_1 r_o + C_1 C_5 C_L R_5 R_L + C_1 C_5 C_L R_5 r_o + C_1 C_5 C_L R_L r_o) + s^2 (2 C_1 C_5 R_1 g_m r_o + 4 C_1 C_5 R_1 + C_1 C_5 R_5 + C_1 C_5 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + C_5 C_L R_L g_m r_o + C_5 C_L R_L - C_5 C_L R_L r_o)}$$

$$\mathbf{9.93 \quad X-INVALID-ORDER-93} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_1 C_5 L_L R_1 R_5 g_m r_o + C_1 C_5 L_L R_1 R_5 - C_1 C_5 L_L R_1 r_o) + s^2 (C_1 L_L R_1 g_m r_o + C_1 L_L R_1 + C_5 L_L R_5 g_m r_o + C_5 L_L R_5 - C_5 C_L R_L r_o) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_5 R_5 R_L g_m r_o + C_5 R_5 R_L - C_5 C_L R_L r_o)}{g_m r_o + s^4 (C_1 C_5 C_L L_L R_1 R_5 g_m r_o + C_1 C_5 C_L L_L R_1 R_5 + C_1 C_5 C_L L_L R_1 r_o + C_1 C_5 C_L L_L R_5 r_o) + s^3 (2 C_1 C_5 L_L R_1 g_m r_o + 4 C_1 C_5 L_L R_1 + C_1 C_5 L_L R_5 + C_1 C_5 L_L r_o + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_1 C_L L_L r_o + C_5 C_L L_L R_5 g_m r_o + C_5 C_L L_L R_5 + C_5 C_L L_L r_o) + s^2 (C_1 L_L R_1 g_m r_o + C_1 L_L R_1 + C_5 L_L R_5 g_m r_o + C_5 L_L R_5 - C_5 C_L R_L r_o)}$$

$$\mathbf{9.94 \quad X-INVALID-ORDER-94} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^4 (C_1 C_5 C_L L_L R_1 R_5 g_m r_o + C_1 C_5 C_L L_L R_1 R_5 - C_1 C_5 C_L L_L R_1 r_o) + s^3 (C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 + C_5 C_L L_L R_5 g_m r_o + C_5 C_L L_L R_5 - C_5 C_L L_L r_o) + s^2 (C_1 C_5 R_1 R_5 g_m r_o + C_1 C_5 R_1 R_5 - C_1 C_5 R_1 r_o + C_L L_L g_m r_o + C_L L_L - C_L L_L r_o)}{s^4 (2 C_1 C_5 C_L L_L R_1 g_m r_o + 4 C_1 C_5 C_L L_L R_1 + C_1 C_5 C_L L_L R_5 + C_1 C_5 C_L L_L r_o) + s^3 (C_1 C_5 C_L R_1 R_5 g_m r_o + C_1 C_5 C_L R_1 R_5 + C_1 C_5 C_L R_1 r_o + C_1 C_5 C_L R_5 r_o + C_1 C_L L_L + 2 C_5 C_L L_L g_m r_o + 4 C_5 C_L L_L) + s^2 (2 C_1 C_5 R_1 g_m r_o + 4 C_1 C_5 R_1 + C_1 C_5 R_5 + C_1 C_5 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L R_L + C_1 C_L r_o + C_5 C_L R_L g_m r_o + C_5 C_L R_L - C_5 C_L R_L r_o)}$$

$$\mathbf{9.95 \quad X-INVALID-ORDER-95} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad R_L \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 R_L r_o s^3 - R_L r_o + s^2 (C_1 L_5 R_1 R_L g_m r_o + C_1 L_5 R_1 R_L - C_5 L_5 R_L r_o) + s (-C_1 R_1 R_L r_o + L_5 R_L g_m r_o + L_5 R_L)}{2 R_L g_m r_o + 4 R_L + r_o + s^3 (2 C_1 C_5 L_5 R_1 R_L g_m r_o + 4 C_1 C_5 L_5 R_1 R_L + C_1 C_5 L_5 R_1 r_o + C_1 C_5 L_5 R_L r_o) + s^2 (C_1 L_5 R_1 g_m r_o + C_1 L_5 R_1 + C_1 L_5 R_L + C_1 L_5 r_o + 2 C_5 L_5 R_L g_m r_o + 4 C_5 L_5 R_L + C_5 L_5 r_o) + s (2 C_1 R_1 R_L g_m r_o + 4 C_1 R_1 R_L + C_1 R_1 r_o + C_1 R_L r_o + L_5 g_m r_o + L_5 R_L)}$$

$$\mathbf{9.96 \quad X-INVALID-ORDER-96} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 r_o s^3 - r_o + s^2 (C_1 L_5 R_1 g_m r_o + C_1 L_5 R_1 - C_5 L_5 r_o) + s (-C_1 R_1 r_o + L_5 g_m r_o + L_5)}{C_1 C_5 C_L L_5 R_1 r_o s^4 + 2 g_m r_o + s^3 (2 C_1 C_5 L_5 R_1 g_m r_o + 4 C_1 C_5 L_5 R_1 + C_1 C_5 L_5 r_o + C_1 C_L L_5 R_1 g_m r_o + C_1 C_L L_5 R_1 + C_1 C_L L_5 r_o + C_5 C_L L_5 r_o) + s^2 (C_1 C_L R_1 r_o + C_1 L_5 + 2 C_5 L_5 g_m r_o + 4 C_5 L_5 + C_L L_5 g_m r_o + C_L L_5) + s (2 C_1 R_1 g_m r_o + 4 C_1 R_1 + C_1 r_o + C_L r_o) + 4}$$

$$\mathbf{9.97 \quad X-INVALID-ORDER-97} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 R_1 R_L r_o s^3 - R_L r_o + s^2 (C_1 L_5 R_1 R_L g_m r_o + C_1 L_5 R_1 R_L - C_5 L_5 R_L r_o) + s (-C_1 R_1 R_L r_o + L_5 R_L g_m r_o + L_5 R_L)}{C_1 C_5 C_L L_5 R_1 R_L r_o s^4 + 2 R_L g_m r_o + 4 R_L + r_o + s^3 (2 C_1 C_5 L_5 R_1 R_L g_m r_o + 4 C_1 C_5 L_5 R_1 R_L + C_1 C_5 L_5 R_1 r_o + C_1 C_5 L_5 R_L r_o + C_1 C_L L_5 R_1 R_L g_m r_o + C_1 C_L L_5 R_1 R_L + C_1 C_L L_5 R_L r_o + C_5 C_L L_5 R_L r_o) + s^2 (C_1 C_L R_1 R_L r_o + C_1 L_5 R_1 g_m r_o + C_1 L_5 R_1 + C_1 L_5 R_L + C_1 L_5 r_o + 2 C_5 L_5 R_L g_m r_o + 4 C_5 L_5 R_L + C_5 C_L L_5 R_L r_o) + s (2 C_1 R_1 R_L g_m r_o + 4 C_1 R_1 R_L + C_1 R_1 r_o + C_L R_L g_m r_o + C_L R_L - C_L R_L r_o)}$$

$$\mathbf{9.98 \quad X-INVALID-ORDER-98} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L L_5 R_1 R_L r_o s^4 - r_o + s^3 (-C_1 C_5 L_5 R_1 r_o + C_1 C_L L_5 R_1 R_L g_m r_o + C_1 C_L L_5 R_1 R_L - C_5 C_L L_5 R_L r_o) + s^2 (-C_1 C_L R_1 R_L r_o + C_1 L_5 R_1 g_m r_o + C_1 L_5 R_L g_m r_o + C_1 L_5 R_L - C_5 C_L R_L r_o)}{2 g_m r_o + s^4 (2 C_1 C_5 C_L L_5 R_1 R_L g_m r_o + 4 C_1 C_5 C_L L_5 R_1 R_L + C_1 C_5 C_L L_5 R_1 r_o + C_1 C_5 C_L L_5 R_L r_o) + s^3 (2 C_1 C_5 L_5 R_1 g_m r_o + 4 C_1 C_5 L_5 R_1 + C_1 C_5 L_5 r_o + C_1 C_L L_5 R_1 g_m r_o + C_1 C_L L_5 R_1 + C_1 C_L L_5 R_L + C_1 C_L L_5 r_o + 2 C_5 C_L L_5 R_L g_m r_o + 4 C_5 C_L L_5 R_L + C_5 C_L L_5 r_o) + s^2 (C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L - C_5 C_L R_L r_o)}$$

$$\mathbf{9.99 \quad X-INVALID-ORDER-99} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 L_5 L_L R_1 r_o s^4 - L_L r_o s + s^3 (C_1 L_5 L_L R_1 g_m r_o + C_1 L_5 L_L R_1 - C_5 L_5 L_L r_o) + s^2 (-C_1 L_L R_1 r_o + L_5 L_L g_m r_o + L_5 L_L - C_5 C_L R_L r_o)}{C_1 C_5 C_L L_5 L_L R_1 r_o s^5 + r_o + s^4 (2 C_1 C_5 L_5 L_L R_1 g_m r_o + 4 C_1 C_5 L_5 L_L R_1 + C_1 C_5 L_5 L_L r_o + C_1 C_L L_5 L_L R_1 g_m r_o + C_1 C_L L_5 L_L R_1 + C_1 C_L L_5 L_L r_o + C_5 C_L L_5 L_L r_o) + s^3 (C_1 C_5 L_5 R_1 r_o + C_1 C_L L_L R_1 r_o + C_1 L_5 L_L + 2 C_5 L_5 L_L g_m r_o + 4 C_5 L_5 L_L + C_L L_5 L_L g_m r_o + C_L L_5 L_L - C_5 C_L R_L r_o)}$$

$$\mathbf{9.100 \quad X-INVALID-ORDER-100} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L L_5 L_L R_1 r_o s^5 - r_o + s^4 (C_1 C_L L_5 L_L R_1 g_m r_o + C_1 C_L L_5 L_L R_1 - C_5 C_L L_5 L_L r_o) + s^3 (-C_1 C_5 L_5 R_1 r_o - C_1 C_L L_L R_1 r_o + C_L L_5 L_L g_m r_o + C_L L_5 L_L - C_5 C_L R_L r_o)}{2 g_m r_o + s^5 (2 C_1 C_5 C_L L_5 L_L R_1 g_m r_o + 4 C_1 C_5 C_L L_5 L_L R_1 + C_1 C_5 C_L L_5 L_L r_o) + s^4 (C_1 C_5 C_L L_5 R_1 r_o + C_1 C_L L_5 L_L + 2 C_5 C_L L_5 L_L g_m r_o + 4 C_5 C_L L_5 L_L) + s^3 (2 C_1 C_5 L_5 R_1 g_m r_o + 4 C_1 C_5 L_5 R_1 + C_1 C_5 L_5 r_o + C_1 C_L L_5 R_1 g_m r_o + C_1 C_L L_5 R_1 + C_1 C_L L_5 r_o + 2 C_1 C_L L_L R_1 g_m r_o + 2 C_1 C_L L_L R_1 - C_5 C_L R_L r_o)}$$

$$\mathbf{9.101 \quad X-INVALID-ORDER-101} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad R_L \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1 C_5 L_5 R_1 R_L g_m r_o + C_1 C_5 L_5 R_1 R_L) + s^2 (-C_1 C_5 R_1 R_L r_o + C_5 L_5 R_L g_m r_o + C_5 L_5 R_L) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L - C_5 R_L r_o)}{g_m r_o + s^3 (C_1 C_5 L_5 R_1 g_m r_o + C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_L + C_1 C_5 L_5 r_o) + s^2 (2C_1 C_5 R_1 R_L g_m r_o + 4C_1 C_5 R_1 R_L + C_1 C_5 R_1 r_o + C_1 C_5 R_L r_o + C_5 L_5 g_m r_o + C_5 L_5) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o) + 1}$$

$$\mathbf{9.102 \quad X-INVALID-ORDER-102} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^3 (C_1 C_5 L_5 R_1 g_m r_o + C_1 C_5 L_5 R_1) + s^2 (-C_1 C_5 R_1 r_o + C_5 L_5 g_m r_o + C_5 L_5) + s (C_1 R_1 g_m r_o + C_1 R_1 - C_5 r_o) + 1}{s^4 (C_1 C_5 C_L L_5 R_1 g_m r_o + C_1 C_5 C_L L_5 R_1 + C_1 C_5 C_L L_5 r_o) + s^3 (C_1 C_5 C_L R_1 r_o + C_1 C_5 L_5 + C_5 C_L L_5 g_m r_o + C_5 C_L L_5) + s^2 (2C_1 C_5 R_1 g_m r_o + 4C_1 C_5 R_1 + C_1 C_5 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1 + C_1 C_L r_o + C_5 C_L r_o) + s (C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.103 \quad X-INVALID-ORDER-103} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1 C_5 L_5 R_1 R_L g_m r_o + C_1 C_5 L_5 R_1 R_L) + s^2 (-C_1 C_5 R_1 R_L r_o + C_5 L_5 R_L g_m r_o + C_5 L_5 R_L) + s (C_1 R_1 g_m r_o + C_1 R_1 - C_5 R_L r_o)}{g_m r_o + s^4 (C_1 C_5 C_L L_5 R_1 R_L g_m r_o + C_1 C_5 C_L L_5 R_1 R_L + C_1 C_5 C_L L_5 R_L r_o) + s^3 (C_1 C_5 C_L R_1 R_L r_o + C_1 C_5 L_5 R_1 g_m r_o + C_1 C_5 L_5 R_1 + C_1 C_5 L_5 R_L + C_1 C_5 L_5 r_o + C_5 C_L L_5 R_L g_m r_o + C_5 C_L L_5 R_L) + s^2 (2C_1 C_5 R_1 R_L g_m r_o + 4C_1 C_5 R_1 R_L + C_1 C_5 R_1 r_o + C_1 C_5 R_L r_o + C_1 C_5 R_L r_o + C_5 C_L R_L g_m r_o + C_5 C_L R_L) + s (C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.104 \quad X-INVALID-ORDER-104} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^4 (C_1 C_5 C_L L_5 R_1 R_L g_m r_o + C_1 C_5 C_L L_5 R_1 R_L) + s^3 (-C_1 C_5 C_L R_1 R_L r_o + C_1 C_5 L_5 R_1 g_m r_o + C_1 C_5 L_5 R_1 + C_5 C_L L_5 R_L g_m r_o + C_5 C_L L_5 R_L) + s^2 (-C_1 C_5 R_1 r_o + C_1 C_L R_1 R_L g_m r_o + C_1 C_L R_1 R_L - C_5 C_L R_L r_o + C_5 L_5 g_m r_o)}{s^4 (C_1 C_5 C_L L_5 R_1 g_m r_o + C_1 C_5 C_L L_5 R_1 + C_1 C_5 C_L L_5 R_L + C_1 C_5 C_L L_5 r_o) + s^3 (2C_1 C_5 C_L R_1 R_L g_m r_o + 4C_1 C_5 C_L R_1 R_L + C_1 C_5 C_L R_1 r_o + C_1 C_5 C_L R_L r_o + C_1 C_5 L_5 + C_5 C_L L_5 g_m r_o + C_5 C_L L_5) + s^2 (2C_1 C_5 R_1 g_m r_o + 4C_1 C_5 R_1 + C_1 C_5 r_o + C_1 C_L R_1 g_m r_o + C_1 C_L R_1)}$$

$$\mathbf{9.105 \quad X-INVALID-ORDER-105} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^4 (C_1 C_5 L_5 L_L R_1 g_m r_o + C_1 C_5 L_5 L_L R_1) + s^3 (-C_1 C_5 L_L R_1 r_o + C_5 L_5 L_L g_m r_o + C_5 L_5 L_L) + s^2 (C_1 L_L R_1 g_m r_o + C_1 L_L R_1 - C_5 L_L R_L r_o)}{g_m r_o + s^5 (C_1 C_5 C_L L_5 L_L R_1 g_m r_o + C_1 C_5 C_L L_5 L_L R_1 + C_1 C_5 C_L L_5 L_L r_o) + s^4 (C_1 C_5 C_L L_L R_1 r_o + C_1 C_5 L_5 L_L + C_5 C_L L_5 L_L g_m r_o + C_5 C_L L_5 L_L) + s^3 (C_1 C_5 L_5 R_1 g_m r_o + C_1 C_5 L_5 R_1 + C_1 C_5 L_5 r_o + 2C_1 C_5 L_L R_1 g_m r_o + 4C_1 C_5 L_L R_1 + C_1 C_5 L_L r_o + C_1 C_L L_L R_1 g_m r_o)}$$

$$\mathbf{9.106 \quad X-INVALID-ORDER-106} \quad Z(s) = \left(R_1 + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^5 (C_1 C_5 C_L L_5 L_L R_1 g_m r_o + C_1 C_5 C_L L_5 L_L R_1) + s^4 (-C_1 C_5 C_L L_L R_1 r_o + C_5 C_L L_5 L_L g_m r_o + C_5 C_L L_5 L_L) + s^3 (C_1 C_5 L_5 R_1 g_m r_o + C_1 C_5 L_5 R_1 + C_1 C_L L_L R_1 g_m r_o + C_1 C_L L_L R_1 - C_5 C_L L_L r_o) + s^2 (-C_1 C_5 R_1 r_o + C_5 L_5 g_m r_o)}{C_1 C_5 C_L L_5 L_L s^5 + s^4 (C_1 C_5 C_L L_5 R_1 g_m r_o + C_1 C_5 C_L L_5 R_1 + C_1 C_5 C_L L_5 r_o + 2C_1 C_5 C_L L_L R_1 g_m r_o + 4C_1 C_5 C_L L_L R_1 + C_1 C_5 C_L L_L r_o) + s^3 (C_1 C_5 C_L R_1 r_o + C_1 C_5 L_5 + C_1 C_L L_L + C_5 C_L L_5 g_m r_o + C_5 C_L L_5 + 2C_5 C_L L_L g_m r_o + 4C_5 C_L L_L) + s^2 (2C_1 C_5 R_1 g_m r_o + 4C_1 C_5 R_1)}$$

$$\mathbf{9.107 \quad X-INVALID-ORDER-107} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s (L_1 R_5 g_m r_o + L_1 R_5 - L_1 r_o)}{C_1 C_L L_1 R_5 r_o s^3 + R_5 + r_o + s^2 (C_1 L_1 R_5 + C_1 L_1 r_o + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + C_L L_1 r_o) + s (C_L R_5 r_o + 2L_1 g_m r_o + 4L_1)}$$

$$\mathbf{9.108 \quad X-INVALID-ORDER-108} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s (L_1 R_5 R_L g_m r_o + L_1 R_5 R_L - L_1 R_L r_o)}{C_1 C_L L_1 R_5 R_L r_o s^3 + R_5 R_L + R_5 r_o + R_L r_o + s^2 (C_1 L_1 R_5 R_L + C_1 L_1 R_5 r_o + C_1 L_1 R_L r_o + C_L L_1 R_5 R_L g_m r_o + C_L L_1 R_5 R_L + C_L L_1 R_L r_o) + s (C_L R_5 R_L r_o + L_1 R_5 g_m r_o + L_1 R_5 + 2L_1 R_L g_m r_o + 4L_1 R_L + L_1 r_o)}$$

$$\mathbf{9.109 \quad X-INVALID-ORDER-109} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^2 (C_L L_1 R_5 R_L g_m r_o + C_L L_1 R_5 R_L - C_L L_1 R_L r_o) + s (L_1 R_5 g_m r_o + L_1 R_5 - L_1 r_o)}{R_5 + r_o + s^3 (C_1 C_L L_1 R_5 R_L + C_1 C_L L_1 R_5 r_o + C_1 C_L L_1 R_L r_o) + s^2 (C_1 L_1 R_5 + C_1 L_1 r_o + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_L L_1 R_L g_m r_o + 4C_L L_1 R_L + C_L L_1 r_o) + s (C_L R_5 R_L + C_L R_5 r_o + C_L R_L r_o + 2L_1 g_m r_o + 4L_1)}$$

$$\mathbf{9.110 \quad X-INVALID-ORDER-110} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^2 (L_1 L_L R_5 g_m r_o + L_1 L_L R_5 - L_1 L_L r_o)}{C_1 C_L L_1 L_L R_5 r_o s^4 + R_5 r_o + s^3 (C_1 L_1 L_L R_5 + C_1 L_1 L_L r_o + C_L L_1 L_L R_5 g_m r_o + C_L L_1 L_L R_5 + C_L L_1 L_L r_o) + s^2 (C_1 L_1 R_5 r_o + C_L L_L R_5 r_o + 2L_1 L_L g_m r_o + 4L_1 L_L) + s (L_1 R_5 g_m r_o + L_1 R_5 + L_1 r_o + L_L R_5 + L_L r_o)}$$

$$\mathbf{9.111 \quad X-INVALID-ORDER-111} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, R_5, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_L L_1 L_L R_5 g_m r_o + C_L L_1 L_L R_5 - C_L L_1 L_L r_o) + s (L_1 R_5 g_m r_o + L_1 R_5 - L_1 r_o)}{R_5 + r_o + s^4 (C_1 C_L L_1 L_L R_5 + C_1 C_L L_1 L_L r_o) + s^3 (C_1 C_L L_1 R_5 r_o + 2 C_L L_1 L_L g_m r_o + 4 C_L L_1 L_L) + s^2 (C_1 L_1 R_5 + C_1 L_1 r_o + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + C_L L_1 r_o + C_L L_L R_5 + C_L L_L r_o) + s (C_L R_5 r_o + 2 L_1 g_m r_o + 4 L_1)}$$

$$\mathbf{9.112 \quad X-INVALID-ORDER-112} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L \right)$$

$$H(s) = \frac{-C_5 L_1 R_L r_o s^2 + s (L_1 R_L g_m r_o + L_1 R_L)}{C_1 C_5 L_1 R_L r_o s^3 + R_L + r_o + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + 2 C_5 L_1 R_L g_m r_o + 4 C_5 L_1 R_L + C_5 L_1 r_o) + s (C_5 R_L r_o + L_1 g_m r_o + L_1)}$$

$$\mathbf{9.113 \quad X-INVALID-ORDER-113} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 L_1 r_o s^2 + s (L_1 g_m r_o + L_1)}{s^3 (C_1 C_5 L_1 r_o + C_1 C_L L_1 r_o + C_5 C_L L_1 r_o) + s^2 (C_1 L_1 + 2 C_5 L_1 g_m r_o + 4 C_5 L_1 + C_L L_1 g_m r_o + C_L L_1) + s (C_5 r_o + C_L r_o) + 1}$$

$$\mathbf{9.114 \quad X-INVALID-ORDER-114} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_5 L_1 R_L r_o s^2 + s (L_1 R_L g_m r_o + L_1 R_L)}{R_L + r_o + s^3 (C_1 C_5 L_1 R_L r_o + C_1 C_L L_1 R_L r_o + C_5 C_L L_1 R_L r_o) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + 2 C_5 L_1 R_L g_m r_o + 4 C_5 L_1 R_L + C_5 L_1 r_o + C_L L_1 R_L g_m r_o + C_L L_1 R_L) + s (C_5 R_L r_o + C_L R_L r_o + L_1 g_m r_o + L_1)}$$

$$\mathbf{9.115 \quad X-INVALID-ORDER-115} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_1 R_L r_o s^3 + s^2 (-C_5 L_1 r_o + C_L L_1 R_L g_m r_o + C_L L_1 R_L) + s (L_1 g_m r_o + L_1)}{C_1 C_5 C_L L_1 R_L r_o s^4 + s^3 (C_1 C_5 L_1 r_o + C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + 2 C_5 C_L L_1 R_L g_m r_o + 4 C_5 C_L L_1 R_L + C_5 C_L L_1 r_o) + s^2 (C_1 L_1 + C_5 C_L R_L r_o + 2 C_5 L_1 g_m r_o + 4 C_5 L_1 + C_L L_1 g_m r_o + C_L L_1) + s (C_5 r_o + C_L R_L + C_L r_o) + 1}$$

$$\mathbf{9.116 \quad X-INVALID-ORDER-116} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_1 L_L r_o s^3 + s^2 (L_1 L_L g_m r_o + L_1 L_L)}{r_o + s^4 (C_1 C_5 L_1 L_L r_o + C_1 C_L L_1 L_L r_o + C_5 C_L L_1 L_L r_o) + s^3 (C_1 L_1 L_L + 2 C_5 L_1 L_L g_m r_o + 4 C_5 L_1 L_L + C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s^2 (C_1 L_1 r_o + C_5 L_1 r_o + C_5 L_L r_o + C_L L_L r_o) + s (L_1 g_m r_o + L_1 + L_L)}$$

$$\mathbf{9.117 \quad X-INVALID-ORDER-117} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{1}{C_5 s}, L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_1 L_L r_o s^4 - C_5 L_1 r_o s^2 + s^3 (C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s (L_1 g_m r_o + L_1)}{C_1 C_5 C_L L_1 L_L r_o s^5 + s^4 (C_1 C_L L_1 L_L + 2 C_5 C_L L_1 L_L g_m r_o + 4 C_5 C_L L_1 L_L) + s^3 (C_1 C_5 L_1 r_o + C_1 C_L L_1 r_o + C_5 C_L L_1 r_o + C_5 C_L L_L r_o) + s^2 (C_1 L_1 + 2 C_5 L_1 g_m r_o + 4 C_5 L_1 + C_L L_1 g_m r_o + C_L L_1 + C_L L_L) + s (C_5 r_o + C_L r_o) + 1}$$

$$\mathbf{9.118 \quad X-INVALID-ORDER-118} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$$

$$H(s) = \frac{-C_5 L_1 R_5 R_L r_o s^2 + s (L_1 R_5 R_L g_m r_o + L_1 R_5 R_L - L_1 R_L r_o)}{C_1 C_5 L_1 R_5 R_L r_o s^3 + R_5 R_L + R_5 r_o + R_L r_o + s^2 (C_1 L_1 R_5 R_L + C_1 L_1 R_5 r_o + C_1 L_1 R_L r_o + 2 C_5 L_1 R_5 R_L g_m r_o + 4 C_5 L_1 R_5 R_L + C_5 L_1 R_5 r_o) + s (C_5 R_5 R_L r_o + L_1 R_5 g_m r_o + L_1 R_5 + 2 L_1 R_L g_m r_o + 4 L_1 R_L + L_1 r_o)}$$

$$\mathbf{9.119 \quad X-INVALID-ORDER-119} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 L_1 R_5 r_o s^2 + s (L_1 R_5 g_m r_o + L_1 R_5 - L_1 r_o)}{R_5 + r_o + s^3 (C_1 C_5 L_1 R_5 r_o + C_1 C_L L_1 R_5 r_o + C_5 C_L L_1 R_5 r_o) + s^2 (C_1 L_1 R_5 + C_1 L_1 r_o + 2 C_5 L_1 R_5 g_m r_o + 4 C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + C_L L_1 r_o) + s (C_5 R_5 r_o + C_L R_5 r_o + 2 L_1 g_m r_o + 4 L_1)}$$

$$\mathbf{9.120 \quad X-INVALID-ORDER-120} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_5 L_1 R_5 R_L r_o s^2 + s (L_1 R_5 R_L g_m r_o + L_1 R_5 R_L - L_1 R_L r_o)}{R_5 R_L + R_5 r_o + R_L r_o + s^3 (C_1 C_5 L_1 R_5 R_L r_o + C_1 C_L L_1 R_5 R_L r_o + C_5 C_L L_1 R_5 R_L r_o) + s^2 (C_1 L_1 R_5 R_L + C_1 L_1 R_5 r_o + C_1 L_1 R_L r_o + 2 C_5 L_1 R_5 R_L g_m r_o + 4 C_5 L_1 R_5 R_L + C_5 L_1 R_5 r_o + C_L L_1 R_5 R_L g_m r_o + C_L L_1 R_5 R_L + C_L L_1 R_L r_o) + s (C_5 R_5 R_L r_o + C_L R_5 R_L r_o + L_1 R_5 g_m r_o + L_1 R_5 + 2 L_1 R_L g_m r_o + 4 L_1 R_L + L_1 r_o)}$$

$$\mathbf{9.121 \quad X-INVALID-ORDER-121} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_1 R_5 R_L r_o s^3 + s^2 (-C_5 L_1 R_5 r_o + C_L L_1 R_5 R_L g_m r_o + C_L L_1 R_5 R_L - C_L L_1 R_L r_o) + s (L_1 R_5 g_m r_o + L_1 R_5 - L_1 r_o)}{C_1 C_5 C_L L_1 R_5 R_L r_o s^4 + R_5 + r_o + s^3 (C_1 C_5 L_1 R_5 r_o + C_1 C_L L_1 R_5 R_L + C_1 C_L L_1 R_5 r_o + C_1 C_L L_1 R_L r_o + 2C_5 C_L L_1 R_5 R_L g_m r_o + 4C_5 C_L L_1 R_5 R_L + C_5 C_L L_1 R_5 r_o) + s^2 (C_1 L_1 R_5 + C_1 L_1 r_o + C_5 C_L R_5 R_L r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + s (C_1 L_1 R_5 + C_1 L_1 r_o + C_5 C_L R_5 R_L r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + 1}$$

$$\mathbf{9.122 \quad X-INVALID-ORDER-122} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_1 L_L R_5 r_o s^3 + s^2 (L_1 L_L R_5 g_m r_o + L_1 L_L R_5 - L_1 L_L r_o)}{R_5 r_o + s^4 (C_1 C_5 L_1 L_L R_5 r_o + C_1 C_L L_1 L_L R_5 r_o + C_5 C_L L_1 L_L R_5 r_o) + s^3 (C_1 L_1 L_L R_5 + C_1 L_1 L_L r_o + 2C_5 L_1 L_L R_5 g_m r_o + 4C_5 L_1 L_L R_5 + C_L L_1 L_L R_5 g_m r_o + C_L L_1 L_L R_5 + C_L L_1 L_L r_o) + s^2 (C_1 L_1 R_5 r_o + C_5 L_1 R_5 r_o + C_5 L_L R_5 r_o + C_L L_L R_5 r_o + 2L_1 L_L g_m r_o + 4L_1 L_L R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + s (C_1 L_1 R_5 + C_1 L_1 r_o + C_5 C_L R_5 R_L r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + 1}$$

$$\mathbf{9.123 \quad X-INVALID-ORDER-123} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_1 L_L R_5 r_o s^4 - C_5 L_1 R_5 r_o s^2 + s^3 (C_L L_1 L_L R_5 g_m r_o + C_L L_1 L_L R_5 - C_L L_1 L_L r_o) + s (L_1 R_5 g_m r_o + L_1 R_5 - L_1 r_o)}{C_1 C_5 C_L L_1 L_L R_5 r_o s^5 + R_5 + r_o + s^4 (C_1 C_L L_1 L_L R_5 + C_1 C_L L_1 L_L r_o + 2C_5 C_L L_1 L_L R_5 g_m r_o + 4C_5 C_L L_1 L_L R_5) + s^3 (C_1 C_5 L_1 R_5 r_o + C_1 C_L L_1 R_5 r_o + C_5 C_L L_1 R_5 r_o + C_5 C_L L_L R_5 r_o + 2C_L L_1 L_L g_m r_o + 4C_L L_1 L_L) + s^2 (C_1 L_1 R_5 + C_1 L_1 r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + s (C_1 L_1 R_5 + C_1 L_1 r_o + C_5 C_L R_5 R_L r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + 1}$$

$$\mathbf{9.124 \quad X-INVALID-ORDER-124} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad R_L \right)$$

$$H(s) = \frac{s^2 (C_5 L_1 R_5 R_L g_m r_o + C_5 L_1 R_5 R_L - C_5 L_1 R_L r_o) + s (L_1 R_L g_m r_o + L_1 R_L)}{R_L + r_o + s^3 (C_1 C_5 L_1 R_5 R_L + C_1 C_5 L_1 R_5 r_o + C_1 C_5 L_1 R_L r_o) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + C_5 L_1 R_5 g_m r_o + C_5 L_1 R_5 + 2C_5 L_1 R_L g_m r_o + 4C_5 L_1 R_L + C_5 L_1 r_o) + s (C_5 R_5 R_L + C_5 R_5 r_o + C_5 R_L r_o + L_1 g_m r_o + L_1)}$$

$$\mathbf{9.125 \quad X-INVALID-ORDER-125} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^2 (C_5 L_1 R_5 g_m r_o + C_5 L_1 R_5 - C_5 L_1 r_o) + s (L_1 g_m r_o + L_1)}{C_1 C_5 C_L L_1 R_5 r_o s^4 + s^3 (C_1 C_5 L_1 R_5 + C_1 C_5 L_1 r_o + C_1 C_L L_1 r_o + C_5 C_L L_1 R_5 g_m r_o + C_5 C_L L_1 R_5 + C_5 C_L L_1 r_o) + s^2 (C_1 L_1 + C_5 C_L R_5 r_o + 2C_5 L_1 g_m r_o + 4C_5 L_1 + C_L L_1 g_m r_o + C_L L_1) + s (C_5 R_5 + C_5 r_o + C_L r_o) + 1}$$

$$\mathbf{9.126 \quad X-INVALID-ORDER-126} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{s^2 (C_5 L_1 R_5 R_L g_m r_o + C_5 L_1 R_5 R_L - C_5 L_1 R_L r_o) + s (L_1 R_L g_m r_o + L_1 R_L)}{C_1 C_5 C_L L_1 R_5 R_L r_o s^4 + R_L + r_o + s^3 (C_1 C_5 L_1 R_5 R_L + C_1 C_5 L_1 R_5 r_o + C_1 C_5 L_1 R_L r_o + C_1 C_L L_1 R_L r_o + C_5 C_L L_1 R_5 R_L g_m r_o + C_5 C_L L_1 R_5 R_L + C_5 C_L L_1 R_L r_o) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + C_5 C_L R_5 R_L r_o + C_5 L_1 R_5 g_m r_o + C_5 L_1 R_5 + 2C_5 L_1 R_L g_m r_o + 4C_5 L_1 R_L + C_5 L_1 r_o) + s (C_1 L_1 R_5 + C_1 L_1 r_o + C_5 C_L R_5 R_L r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + 1}$$

$$\mathbf{9.127 \quad X-INVALID-ORDER-127} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^3 (C_5 C_L L_1 R_5 R_L g_m r_o + C_5 C_L L_1 R_5 R_L - C_5 C_L L_1 R_L r_o) + s^2 (C_5 L_1 R_5 g_m r_o + C_5 L_1 R_5 - C_5 L_1 r_o + C_L L_1 R_L g_m r_o + C_L L_1 R_L) + s (L_1 g_m r_o + L_1)}{s^4 (C_1 C_5 C_L L_1 R_5 R_L + C_1 C_5 C_L L_1 R_5 r_o + C_1 C_5 C_L L_1 R_L r_o) + s^3 (C_1 C_5 L_1 R_5 + C_1 C_5 L_1 r_o + C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + C_5 C_L L_1 R_5 g_m r_o + C_5 C_L L_1 R_5 + 2C_5 C_L L_1 R_L g_m r_o + 4C_5 C_L L_1 R_L + C_5 C_L L_1 r_o) + s^2 (C_1 L_1 + C_5 C_L R_5 R_L + C_5 C_L R_5 r_o + C_5 C_L R_L r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + s (C_1 L_1 R_5 + C_1 L_1 r_o + C_5 C_L R_5 R_L r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + 1}$$

$$\mathbf{9.128 \quad X-INVALID-ORDER-128} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_5 L_1 L_L R_5 g_m r_o + C_5 L_1 L_L R_5 - C_5 L_1 L_L r_o) + s^2 (L_1 L_L g_m r_o + L_1 L_L)}{C_1 C_5 C_L L_1 L_L R_5 r_o s^5 + r_o + s^4 (C_1 C_5 L_1 L_L R_5 + C_1 C_5 L_1 L_L r_o + C_1 C_L L_1 L_L r_o + C_5 C_L L_1 L_L R_5 g_m r_o + C_5 C_L L_1 L_L R_5 + C_5 C_L L_1 L_L r_o) + s^3 (C_1 C_5 L_1 R_5 r_o + C_1 L_1 L_L + C_5 C_L L_L R_5 r_o + 2C_5 L_1 L_L g_m r_o + 4C_5 L_1 L_L + C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s^2 (C_1 L_1 r_o + C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + s (C_1 L_1 R_5 + C_1 L_1 r_o + C_5 C_L R_5 R_L r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + 1}$$

$$\mathbf{9.129 \quad X-INVALID-ORDER-129} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^4 (C_5 C_L L_1 L_L R_5 g_m r_o + C_5 C_L L_1 L_L R_5 - C_5 C_L L_1 L_L r_o) + s^3 (C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s^2 (C_5 L_1 R_5 g_m r_o + C_5 L_1 R_5 - C_5 L_1 r_o) + s (L_1 g_m r_o + L_1)}{s^5 (C_1 C_5 C_L L_1 L_L R_5 + C_1 C_5 C_L L_1 L_L r_o) + s^4 (C_1 C_5 C_L L_1 R_5 r_o + C_1 C_L L_1 L_L + 2C_5 C_L L_1 L_L g_m r_o + 4C_5 C_L L_1 L_L) + s^3 (C_1 C_5 L_1 R_5 + C_1 C_5 L_1 r_o + C_1 C_L L_1 r_o + C_5 C_L L_1 R_5 g_m r_o + C_5 C_L L_1 R_5 + C_5 C_L L_1 r_o + C_5 C_L L_L R_5 + C_5 C_L L_L r_o) + s^2 (C_1 L_1 + C_5 C_L R_5 r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + s (C_1 L_1 R_5 + C_1 L_1 r_o + C_5 C_L R_5 R_L r_o + 2C_5 L_1 R_5 g_m r_o + 4C_5 L_1 R_5 + C_L L_1 R_5 g_m r_o + C_L L_1 R_5 + 2C_5 L_1 R_5 r_o + C_L L_1 R_L r_o) + 1}$$

$$\mathbf{9.130 \quad X-INVALID-ORDER-130} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad R_L \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_L r_o s^3 - L_1 R_L r_o s + s^2 (L_1 L_5 R_L g_m r_o + L_1 L_5 R_L)}{C_1 C_5 L_1 L_5 R_L r_o s^4 + R_L r_o + s^3 (C_1 L_1 L_5 R_L + C_1 L_1 L_5 r_o + 2C_5 L_1 L_5 R_L g_m r_o + 4C_5 L_1 L_5 R_L + C_5 L_1 L_5 r_o) + s^2 (C_1 L_1 R_L r_o + C_5 L_5 R_L r_o + L_1 L_5 g_m r_o + L_1 L_5) + s (2L_1 R_L g_m r_o + 4L_1 R_L + L_1 r_o + L_5 R_L + L_5 r_o) + 1}$$

$$\mathbf{9.131 \quad X-INVALID-ORDER-131} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 r_o s^3 - L_1 r_o s + s^2 (L_1 L_5 g_m r_o + L_1 L_5)}{r_o + s^4 (C_1 C_5 L_1 L_5 r_o + C_1 C_L L_1 L_5 r_o + C_5 C_L L_1 L_5 r_o) + s^3 (C_1 L_1 L_5 + 2C_5 L_1 L_5 g_m r_o + 4C_5 L_1 L_5 + C_L L_1 L_5 g_m r_o + C_L L_1 L_5) + s^2 (C_1 L_1 r_o + C_5 L_5 r_o + C_L L_1 r_o + C_L L_5 r_o) + s (2L_1 g_m r_o + 4L_1 + L_5)}$$

$$\mathbf{9.132 \quad X-INVALID-ORDER-132} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 R_L r_o s^3 - L_1 R_L r_o s + s^2 (L_1 L_5 R_L g_m r_o + L_1 L_5 R_L)}{R_L r_o + s^4 (C_1 C_5 L_1 L_5 R_L r_o + C_1 C_L L_1 L_5 R_L r_o + C_5 C_L L_1 L_5 R_L r_o) + s^3 (C_1 L_1 L_5 R_L + C_1 L_1 L_5 r_o + 2C_5 L_1 L_5 R_L g_m r_o + 4C_5 L_1 L_5 R_L + C_5 L_1 L_5 r_o + C_L L_1 L_5 R_L g_m r_o + C_L L_1 L_5 R_L) + s^2 (C_1 L_1 R_L r_o + C_5 L_5 R_L r_o + C_L L_1 R_L r_o + C_L L_5 R_L r_o + L_1 L_5 g_m r_o + L_1 L_5) + s (L_1 L_5 R_L g_m r_o + L_1 L_5 R_L)}$$

$$\mathbf{9.133 \quad X-INVALID-ORDER-133} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_1 L_5 R_L r_o s^4 - L_1 r_o s + s^3 (-C_5 L_1 L_5 r_o + C_L L_1 L_5 R_L g_m r_o + C_L L_1 L_5 R_L) + s^2 (-C_L L_1 R_L r_o + L_1 L_5 g_m r_o + L_1 L_5)}{C_1 C_5 C_L L_1 L_5 R_L r_o s^5 + r_o + s^4 (C_1 C_5 L_1 L_5 r_o + C_1 C_L L_1 L_5 R_L + C_1 C_L L_1 L_5 r_o + 2C_5 C_L L_1 L_5 R_L g_m r_o + 4C_5 C_L L_1 L_5 R_L + C_5 C_L L_1 L_5 r_o) + s^3 (C_1 C_L L_1 R_L r_o + C_1 L_1 L_5 + C_5 C_L L_5 R_L r_o + 2C_5 L_1 L_5 g_m r_o + 4C_5 L_1 L_5 + C_L L_1 L_5 g_m r_o + C_L L_1 L_5) + s^2 (C_1 L_1 r_o + C_5 L_5 r_o + C_L L_1 r_o + C_L L_5 r_o) + s (L_1 L_5 g_m r_o + L_1 L_5)}$$

$$\mathbf{9.134 \quad X-INVALID-ORDER-134} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_1 L_5 L_L r_o s^3 - L_1 L_L r_o s + s^2 (L_1 L_5 L_L g_m r_o + L_1 L_5 L_L)}{L_1 r_o + L_5 r_o + L_L r_o + s^4 (C_1 C_5 L_1 L_5 L_L r_o + C_1 C_L L_1 L_5 L_L r_o + C_5 C_L L_1 L_5 L_L r_o) + s^3 (C_1 L_1 L_5 L_L + 2C_5 L_1 L_5 L_L g_m r_o + 4C_5 L_1 L_5 L_L + C_L L_1 L_5 L_L g_m r_o + C_L L_1 L_5 L_L) + s^2 (C_1 L_1 L_5 r_o + C_1 L_1 L_L r_o + C_5 L_1 L_5 r_o + C_5 L_5 L_L r_o + C_L L_1 L_L r_o + C_L L_5 L_L r_o) + s (L_1 L_5 L_L g_m r_o + L_1 L_5 L_L)}$$

$$\mathbf{9.135 \quad X-INVALID-ORDER-135} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{L_5 s}{C_5 L_5 s^2 + 1}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_1 L_5 L_L r_o s^5 - L_1 r_o s + s^4 (C_L L_1 L_5 L_L g_m r_o + C_L L_1 L_5 L_L) + s^3 (-C_5 L_1 L_5 r_o - C_L L_1 L_L r_o) + s^2 (L_1 L_5 g_m r_o + L_1 L_5)}{C_1 C_5 C_L L_1 L_5 L_L r_o s^6 + r_o + s^5 (C_1 C_L L_1 L_5 L_L + 2C_5 C_L L_1 L_5 L_L g_m r_o + 4C_5 C_L L_1 L_5 L_L) + s^4 (C_1 C_5 L_1 L_5 r_o + C_1 C_L L_1 L_5 r_o + C_1 C_L L_1 L_L r_o + C_5 C_L L_1 L_5 r_o + C_5 C_L L_5 L_L r_o) + s^3 (C_1 L_1 L_5 + 2C_5 L_1 L_5 g_m r_o + 4C_5 L_1 L_5 + C_L L_1 L_5 g_m r_o + C_L L_1 L_5 + 2C_L L_1 L_L g_m r_o + C_L L_1 L_5 L_L)}$$

$$\mathbf{9.136 \quad X-INVALID-ORDER-136} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad R_L \right)$$

$$H(s) = \frac{-C_5 L_1 R_L r_o s^2 + s^3 (C_5 L_1 L_5 R_L g_m r_o + C_5 L_1 L_5 R_L) + s (L_1 R_L g_m r_o + L_1 R_L)}{R_L + r_o + s^4 (C_1 C_5 L_1 L_5 R_L + C_1 C_5 L_1 L_5 r_o) + s^3 (C_1 C_5 L_1 R_L r_o + C_5 L_1 L_5 g_m r_o + C_5 L_1 L_5) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + 2C_5 L_1 R_L g_m r_o + 4C_5 L_1 R_L + C_5 L_1 r_o + C_5 L_5 R_L + C_5 L_5 r_o) + s (C_5 R_L r_o + L_1 g_m r_o + L_1)}$$

$$\mathbf{9.137 \quad X-INVALID-ORDER-137} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 L_1 r_o s^2 + s^3 (C_5 L_1 L_5 g_m r_o + C_5 L_1 L_5) + s (L_1 g_m r_o + L_1)}{C_1 C_5 C_L L_1 L_5 r_o s^5 + s^4 (C_1 C_5 L_1 L_5 + C_5 C_L L_1 L_5 g_m r_o + C_5 C_L L_1 L_5) + s^3 (C_1 C_5 L_1 r_o + C_1 C_L L_1 r_o + C_5 C_L L_1 r_o + C_5 C_L L_5 r_o) + s^2 (C_1 L_1 + 2C_5 L_1 g_m r_o + 4C_5 L_1 + C_5 L_5 + C_L L_1 g_m r_o + C_L L_1) + s (C_5 r_o + C_L r_o) + 1}$$

$$\mathbf{9.138 \quad X-INVALID-ORDER-138} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_5 L_1 R_L r_o s^2 + s^3 (C_5 L_1 L_5 R_L g_m r_o + C_5 L_1 L_5 R_L) + s (L_1 R_L g_m r_o + L_1 R_L)}{C_1 C_5 C_L L_1 L_5 R_L r_o s^5 + R_L + r_o + s^4 (C_1 C_5 L_1 L_5 R_L + C_1 C_5 L_1 L_5 r_o + C_5 C_L L_1 L_5 R_L g_m r_o + C_5 C_L L_1 L_5 R_L) + s^3 (C_1 C_5 L_1 R_L r_o + C_1 C_L L_1 R_L r_o + C_5 C_L L_1 R_L r_o + C_5 C_L L_5 R_L r_o + C_5 L_1 L_5 g_m r_o + C_5 L_1 L_5) + s^2 (C_1 L_1 R_L + C_1 L_1 r_o + 2C_5 L_1 R_L g_m r_o + 4C_5 L_1 R_L + C_5 L_1 r_o + C_5 L_5 R_L + C_5 L_5 r_o) + s (C_5 R_L r_o + L_1 g_m r_o + L_1)}$$

$$\mathbf{9.139 \quad X-INVALID-ORDER-139} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{s^4 (C_5 C_L L_1 L_5 R_L g_m r_o + C_5 C_L L_1 L_5 R_L) + s^3 (-C_5 C_L L_1 R_L r_o + C_5 L_1 L_5 g_m r_o + C_5 L_1 L_5) + s^2 (-C_5 L_1 r_o + C_L L_1 R_L g_m r_o + C_L L_1 R_L) + s (L_1 g_m r_o + L_1)}{s^5 (C_1 C_5 C_L L_1 L_5 R_L + C_1 C_5 C_L L_1 L_5 r_o) + s^4 (C_1 C_5 C_L L_1 R_L r_o + C_1 C_5 L_1 L_5 + C_5 C_L L_1 L_5 g_m r_o + C_5 C_L L_1 L_5) + s^3 (C_1 C_5 L_1 r_o + C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + 2C_5 C_L L_1 R_L g_m r_o + 4C_5 C_L L_1 R_L + C_5 C_L L_1 r_o + C_5 C_L L_5 R_L + C_5 C_L L_5 r_o) + s^2 (C_1 L_1 + C_5 C_L R_L r_o + 2C_5 L_1 g_m r_o + 4C_5 L_1 + C_L L_1 g_m r_o + C_L L_1) + s (C_5 r_o + C_L r_o) + 1}$$

$$\mathbf{9.140 \quad X-INVALID-ORDER-140} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_5 L_1 L_L r_o s^3 + s^4 (C_5 L_1 L_5 L_L g_m r_o + C_5 L_1 L_5 L_L) + s^2 (L_1 L_L g_m r_o + L_1 L_L)}{C_1 C_5 C_L L_1 L_5 L_L r_o s^6 + r_o + s^5 (C_1 C_5 L_1 L_5 L_L + C_5 C_L L_1 L_5 L_L g_m r_o + C_5 C_L L_1 L_5 L_L) + s^4 (C_1 C_5 L_1 L_5 r_o + C_1 C_5 L_1 L_L r_o + C_1 C_L L_1 L_L r_o + C_5 C_L L_1 L_L r_o + C_5 C_L L_5 L_L r_o) + s^3 (C_1 L_1 L_L + C_5 L_1 L_5 g_m r_o + C_5 L_1 L_5 + 2C_5 L_1 L_L g_m r_o + 4C_5 L_1 L_L + C_5 L_5 L_L + C_L L_1 L_L g_m r_o + C_L L_1 L_L)}$$

$$\mathbf{9.141 \quad X-INVALID-ORDER-141} \quad Z(s) = \left(\frac{L_1 s}{C_1 L_1 s^2 + 1}, \quad \infty, \quad \infty, \quad \infty, \quad L_5 s + \frac{1}{C_5 s}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_5 C_L L_1 L_L r_o s^4 - C_5 L_1 r_o s^2 + s^5 (C_5 C_L L_1 L_5 L_L g_m r_o + C_5 C_L L_1 L_5 L_L) + s^3 (C_5 L_1 L_5 g_m r_o + C_5 L_1 L_5 + C_L L_1 L_L g_m r_o + C_L L_1 L_L) + s (L_1 g_m r_o + L_1)}{C_1 C_5 C_L L_1 L_5 L_L s^6 + s^5 (C_1 C_5 C_L L_1 L_5 r_o + C_1 C_5 C_L L_1 L_L r_o) + s^4 (C_1 C_5 L_1 L_5 + C_1 C_L L_1 L_L + C_5 C_L L_1 L_5 g_m r_o + C_5 C_L L_1 L_5 + 2 C_5 C_L L_1 L_L g_m r_o + 4 C_5 C_L L_1 L_L + C_5 C_L L_5 L_L) + s^3 (C_1 C_5 L_1 r_o + C_1 C_L L_1 r_o + C_5 C_L L_1 r_o + C_5 C_L L_5 r_o + C_5 C_L L_L r_o) + s^2 (C_1 L_1 + 2 C_1 L_5 + 2 C_1 L_L + 2 C_5 L_1 + 2 C_5 L_5 + 2 C_5 L_L) + s (C_1 + C_5 + C_L) + 1}$$

$$\mathbf{9.142 \quad X-INVALID-ORDER-142} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_5 g_m r_o + R_5 - r_o + s^2 (C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 - C_1 L_1 r_o)}{2 g_m r_o + s^3 (C_1 C_L L_1 R_5 g_m r_o + C_1 C_L L_1 R_5 + C_1 C_L L_1 r_o) + s^2 (C_1 C_L R_5 r_o + 2 C_1 L_1 g_m r_o + 4 C_1 L_1) + s (C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

$$\mathbf{9.143 \quad X-INVALID-ORDER-143} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{R_5 R_L g_m r_o + R_5 R_L - R_L r_o + s^2 (C_1 L_1 R_5 R_L g_m r_o + C_1 L_1 R_5 R_L - C_1 L_1 R_L r_o)}{R_5 g_m r_o + R_5 + 2 R_L g_m r_o + 4 R_L + r_o + s^3 (C_1 C_L L_1 R_5 R_L g_m r_o + C_1 C_L L_1 R_5 R_L + C_1 C_L L_1 R_L r_o) + s^2 (C_1 C_L R_5 R_L r_o + C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 + 2 C_1 L_1 R_L g_m r_o + 4 C_1 L_1 R_L + C_1 L_1 r_o) + s (C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L + C_L R_L r_o) + 1}$$

$$\mathbf{9.144 \quad X-INVALID-ORDER-144} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_5 g_m r_o + R_5 - r_o + s^3 (C_1 C_L L_1 R_5 R_L g_m r_o + C_1 C_L L_1 R_5 R_L - C_1 C_L L_1 R_L r_o) + s^2 (C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 - C_1 L_1 r_o) + s (C_L R_5 R_L g_m r_o + C_L R_5 R_L - C_L R_L r_o)}{2 g_m r_o + s^3 (C_1 C_L L_1 R_5 g_m r_o + C_1 C_L L_1 R_5 + 2 C_1 C_L L_1 R_L g_m r_o + 4 C_1 C_L L_1 R_L + C_1 C_L L_1 r_o) + s^2 (C_1 C_L R_5 R_L + C_1 C_L R_5 r_o + C_1 C_L R_L r_o + 2 C_1 L_1 g_m r_o + 4 C_1 L_1) + s (C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + 2 C_L R_L g_m r_o + 4 C_L R_L + C_L r_o) + 4}$$

$$\mathbf{9.145 \quad X-INVALID-ORDER-145} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{s^3 (C_1 L_1 L_L R_5 g_m r_o + C_1 L_1 L_L R_5 - C_1 L_1 L_L r_o) + s (L_L R_5 g_m r_o + L_L R_5 - L_L r_o)}{R_5 g_m r_o + R_5 + r_o + s^4 (C_1 C_L L_1 L_L R_5 g_m r_o + C_1 C_L L_1 L_L R_5 + C_1 C_L L_1 L_L r_o) + s^3 (C_1 C_L L_L R_5 r_o + 2 C_1 L_1 L_L g_m r_o + 4 C_1 L_1 L_L) + s^2 (C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 + C_1 L_1 r_o + C_1 L_L R_5 + C_1 L_L r_o + C_L L_L R_5 g_m r_o + C_L L_L R_5 + C_L L_L r_o) + s (C_1 R_5 r_o + 2 L_L g_m r_o + 4 L_L) + 1}$$

$$\mathbf{9.146 \quad X-INVALID-ORDER-146} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{R_5 g_m r_o + R_5 - r_o + s^4 (C_1 C_L L_1 L_L R_5 g_m r_o + C_1 C_L L_1 L_L R_5 - C_1 C_L L_1 L_L r_o) + s^2 (C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 - C_1 L_1 r_o + C_L L_L R_5 g_m r_o + C_L L_L R_5 - C_L L_L r_o)}{2 g_m r_o + s^4 (2 C_1 C_L L_1 L_L g_m r_o + 4 C_1 C_L L_1 L_L) + s^3 (C_1 C_L L_1 R_5 g_m r_o + C_1 C_L L_1 R_5 + C_1 C_L L_1 r_o + C_1 C_L L_L R_5 + C_1 C_L L_L r_o) + s^2 (C_1 C_L R_5 r_o + 2 C_1 L_1 g_m r_o + 4 C_1 L_1 + 2 C_L L_L g_m r_o + 4 C_L L_L) + s (C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

$$\mathbf{9.147 \quad X-INVALID-ORDER-147} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad R_L \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_L r_o s^3 - C_5 R_L r_o s + R_L g_m r_o + R_L + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L)}{g_m r_o + s^3 (2 C_1 C_5 L_1 R_L g_m r_o + 4 C_1 C_5 L_1 R_L + C_1 C_5 L_1 r_o) + s^2 (C_1 C_5 R_L r_o + C_1 L_1 g_m r_o + C_1 L_1) + s (C_1 R_L + C_1 r_o + 2 C_5 R_L g_m r_o + 4 C_5 R_L + C_5 r_o) + 1}$$

$$\mathbf{9.148 \quad X-INVALID-ORDER-148} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 r_o s^3 - C_5 r_o s + g_m r_o + s^2 (C_1 L_1 g_m r_o + C_1 L_1) + 1}{C_1 C_5 C_L L_1 r_o s^4 + s^3 (2 C_1 C_5 L_1 g_m r_o + 4 C_1 C_5 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1) + s^2 (C_1 C_5 r_o + C_1 C_L r_o + C_5 C_L r_o) + s (C_1 + 2 C_5 g_m r_o + 4 C_5 + C_L g_m r_o + C_L) + 1}$$

$$\mathbf{9.149 \quad X-INVALID-ORDER-149} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_L r_o s^3 - C_5 R_L r_o s + R_L g_m r_o + R_L + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L)}{C_1 C_5 C_L L_1 R_L r_o s^4 + g_m r_o + s^3 (2 C_1 C_5 L_1 R_L g_m r_o + 4 C_1 C_5 L_1 R_L + C_1 C_5 L_1 r_o + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (C_1 C_5 R_L r_o + C_1 C_L R_L r_o + C_1 L_1 g_m r_o + C_1 L_1 + C_5 C_L R_L r_o) + s (C_1 R_L + C_1 r_o + 2 C_5 R_L g_m r_o + 4 C_5 R_L + C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}$$

$$\mathbf{9.150 \quad X-INVALID-ORDER-150} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L L_1 R_L r_o s^4 + g_m r_o + s^3 (-C_1 C_5 L_1 r_o + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 - C_5 C_L R_L r_o) + s (-C_5 r_o + C_L R_L g_m r_o + C_L R_L) + 1}{s^4 (2 C_1 C_5 C_L L_1 R_L g_m r_o + 4 C_1 C_5 C_L L_1 R_L + C_1 C_5 C_L L_1 r_o) + s^3 (C_1 C_5 C_L R_L r_o + 2 C_1 C_5 L_1 g_m r_o + 4 C_1 C_5 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1) + s^2 (C_1 C_5 r_o + C_1 C_L R_L + C_1 C_L r_o + 2 C_5 C_L R_L g_m r_o + 4 C_5 C_L R_L + C_5 C_L r_o) + s (C_1 + 2 C_5 g_m r_o + 4 C_5 + C_L g_m r_o + C_L) + 1}$$

$$\mathbf{9.151 \quad X-INVALID-ORDER-151} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_L r_o s^4 - C_5 L_L r_o s^2 + s^3 (C_1 L_1 L_L g_m r_o + C_1 L_1 L_L) + s (L_L g_m r_o + L_L)}{C_1 C_5 C_L L_1 L_L r_o s^5 + g_m r_o + s^4 (2C_1 C_5 L_1 L_L g_m r_o + 4C_1 C_5 L_1 L_L + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^3 (C_1 C_5 L_1 r_o + C_1 C_5 L_L r_o + C_1 C_L L_L r_o + C_5 C_L L_L r_o) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_1 L_L + 2C_5 L_L g_m r_o + 4C_5 L_L + C_L L_L g_m r_o + C_L L_L) + s (C_1 r_o + C_5 r_o) + 1}$$

$$\mathbf{9.152 \quad X-INVALID-ORDER-152} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{1}{C_5 s}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L L_1 L_L r_o s^5 - C_5 r_o s + g_m r_o + s^4 (C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^3 (-C_1 C_5 L_1 r_o - C_5 C_L L_L r_o) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_L L_L g_m r_o + C_L L_L) + 1}{s^5 (2C_1 C_5 C_L L_1 L_L g_m r_o + 4C_1 C_5 C_L L_1 L_L) + s^4 (C_1 C_5 C_L L_1 r_o + C_1 C_5 C_L L_L r_o) + s^3 (2C_1 C_5 L_1 g_m r_o + 4C_1 C_5 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_1 C_L L_L + 2C_5 C_L L_L g_m r_o + 4C_5 C_L L_L) + s^2 (C_1 C_5 r_o + C_1 C_L r_o + C_5 C_L r_o) + s (C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

$$\mathbf{9.153 \quad X-INVALID-ORDER-153} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad R_L \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_5 R_L r_o s^3 - C_5 R_5 R_L r_o s + R_5 R_L g_m r_o + R_5 R_L - R_L r_o + s^2 (C_1 L_1 R_5 R_L g_m r_o + C_1 L_1 R_5 R_L - C_1 L_1 R_L r_o)}{R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o + s^3 (2C_1 C_5 L_1 R_5 R_L g_m r_o + 4C_1 C_5 L_1 R_5 R_L + C_1 C_5 L_1 R_5 r_o) + s^2 (C_1 C_5 R_5 R_L r_o + C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 + 2C_1 L_1 R_L g_m r_o + 4C_1 L_1 R_L + C_1 L_1 r_o) + s (C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2C_5 R_5 R_L g_m r_o + 4C_5 R_5 R_L + C_5 R_5 r_o)}$$

$$\mathbf{9.154 \quad X-INVALID-ORDER-154} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_5 r_o s^3 - C_5 R_5 r_o s + R_5 g_m r_o + R_5 - r_o + s^2 (C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 - C_1 L_1 r_o)}{C_1 C_5 C_L L_1 R_5 r_o s^4 + 2g_m r_o + s^3 (2C_1 C_5 L_1 R_5 g_m r_o + 4C_1 C_5 L_1 R_5 + C_1 C_L L_1 R_5 g_m r_o + C_1 C_L L_1 R_5 + C_1 C_L L_1 r_o) + s^2 (C_1 C_5 R_5 r_o + C_1 C_L R_5 r_o + 2C_1 L_1 g_m r_o + 4C_1 L_1 + C_5 C_L R_5 r_o) + s (C_1 R_5 + C_1 r_o + 2C_5 R_5 g_m r_o + 4C_5 R_5 + C_L R_5 g_m r_o + C_L R_5 + C_L r_o) + 4}$$

$$\mathbf{9.155 \quad X-INVALID-ORDER-155} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{R_L}{C_L R_L s + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 R_5 R_L r_o s^3 - C_5 R_5 R_L r_o s + R_5 R_L g_m r_o + R_5 R_L - R_L r_o + s^2 (C_1 L_1 R_5 R_L g_m r_o + C_1 L_1 R_5 R_L - C_1 L_1 R_L r_o)}{C_1 C_5 C_L L_1 R_5 R_L r_o s^4 + R_5 g_m r_o + R_5 + 2R_L g_m r_o + 4R_L + r_o + s^3 (2C_1 C_5 L_1 R_5 R_L g_m r_o + 4C_1 C_5 L_1 R_5 R_L + C_1 C_5 L_1 R_5 r_o + C_1 C_L L_1 R_5 R_L g_m r_o + C_1 C_L L_1 R_5 R_L + C_1 C_L L_1 R_L r_o) + s^2 (C_1 C_5 R_5 R_L r_o + C_1 C_L R_5 R_L r_o + C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 + 2C_1 L_1 R_L g_m r_o + 4C_1 L_1 R_L + C_1 L_1 r_o)}$$

$$\mathbf{9.156 \quad X-INVALID-ORDER-156} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad R_L + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L L_1 R_5 R_L r_o s^4 + R_5 g_m r_o + R_5 - r_o + s^3 (-C_1 C_5 L_1 R_5 r_o + C_1 C_L L_1 R_5 R_L g_m r_o + C_1 C_L L_1 R_5 R_L - C_1 C_L L_1 R_L r_o) + s^2 (C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5)}{2g_m r_o + s^4 (2C_1 C_5 C_L L_1 R_5 R_L g_m r_o + 4C_1 C_5 C_L L_1 R_5 R_L + C_1 C_5 C_L L_1 R_5 r_o) + s^3 (C_1 C_5 C_L R_5 R_L r_o + 2C_1 C_5 L_1 R_5 g_m r_o + 4C_1 C_5 L_1 R_5 + C_1 C_L L_1 R_5 g_m r_o + C_1 C_L L_1 R_5 + 2C_1 C_L L_1 R_L g_m r_o + 4C_1 C_L L_1 R_L + C_1 C_L L_1 r_o) + s^2 (C_1 C_5 R_5 r_o + C_1 C_L R_5 R_L + C_1 C_L R_5 r_o)}$$

$$\mathbf{9.157 \quad X-INVALID-ORDER-157} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad \frac{L_L s}{C_L L_L s^2 + 1} \right)$$

$$H(s) = \frac{-C_1 C_5 L_1 L_L R_5 r_o s^4 - C_5 L_L R_5 r_o s^2 + s^3 (C_1 L_1 L_L R_5 g_m r_o + C_1 L_1 L_L R_5 - C_1 L_1 L_L r_o) + s (L_L R_5 g_m r_o + L_L R_5 - R_5)}{C_1 C_5 C_L L_1 L_L R_5 r_o s^5 + R_5 g_m r_o + R_5 + r_o + s^4 (2C_1 C_5 L_1 L_L R_5 g_m r_o + 4C_1 C_5 L_1 L_L R_5 + C_1 C_L L_1 L_L R_5 g_m r_o + C_1 C_L L_1 L_L R_5 + C_1 C_L L_1 L_L r_o) + s^3 (C_1 C_5 L_1 R_5 r_o + C_1 C_5 L_L R_5 r_o + C_1 C_L L_L R_5 r_o + 2C_1 L_1 L_L g_m r_o + 4C_1 L_1 L_L + C_5 C_L L_L R_5 r_o) + s^2 (C_1 L_1 R_5 g_m r_o + C_1 L_1 R_5 + C_1 L_1 R_L g_m r_o + C_1 L_1 R_L + C_1 L_1 r_o)}$$

$$\mathbf{9.158 \quad X-INVALID-ORDER-158} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad \frac{R_5}{C_5 R_5 s + 1}, \quad L_L s + \frac{1}{C_L s} \right)$$

$$H(s) = \frac{-C_1 C_5 C_L L_1 L_L R_5 r_o s^5 - C_5 R_5 r_o s + R_5 g_m r_o + R_5 - r_o + s^4 (C_1 C_L L_1 L_L R_5 g_m r_o + C_1 C_L L_1 L_L R_5 - C_1 C_L L_1 L_L r_o) + s^3 (-C_1 C_5 L_1 R_5 r_o - C_5 C_L L_L R_5)}{2g_m r_o + s^5 (2C_1 C_5 C_L L_1 L_L R_5 g_m r_o + 4C_1 C_5 C_L L_1 L_L R_5) + s^4 (C_1 C_5 C_L L_1 R_5 r_o + C_1 C_5 C_L L_L R_5 r_o + 2C_1 C_L L_1 L_L g_m r_o + 4C_1 C_L L_1 L_L) + s^3 (2C_1 C_5 L_1 R_5 g_m r_o + 4C_1 C_5 L_1 R_5 + C_1 C_L L_1 R_5 g_m r_o + C_1 C_L L_1 R_5 + C_1 C_L L_1 r_o + C_1 C_L L_L R_5 + C_1 C_L L_L r_o + 2C_5 C_L L_L R_5 r_o)}$$

$$\mathbf{9.159 \quad X-INVALID-ORDER-159} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad R_L \right)$$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1 C_5 L_1 R_5 R_L g_m r_o + C_1 C_5 L_1 R_5 R_L - C_1 C_5 L_1 R_L r_o) + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L) + s (C_5 R_5 R_L g_m r_o + C_5 R_5 R_L - C_5 R_L r_o)}{g_m r_o + s^3 (C_1 C_5 L_1 R_5 g_m r_o + C_1 C_5 L_1 R_5 + 2C_1 C_5 L_1 R_L g_m r_o + 4C_1 C_5 L_1 R_L + C_1 C_5 L_1 r_o) + s^2 (C_1 C_5 R_5 R_L + C_1 C_5 R_5 r_o + C_1 C_5 R_L r_o + C_1 L_1 g_m r_o + C_1 L_1) + s (C_1 R_L + C_1 r_o + C_5 R_5 g_m r_o + C_5 R_5 + 2C_5 R_L g_m r_o + 4C_5 R_L + C_5 r_o) + 1}$$

$$\mathbf{9.160 \quad X-INVALID-ORDER-160} \quad Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \quad \infty, \quad \infty, \quad \infty, \quad R_5 + \frac{1}{C_5 s}, \quad \frac{1}{C_L s} \right)$$

$$H(s) = \frac{g_m r_o + s^3 (C_1 C_5 L_1 R_5 g_m r_o + C_1 C_5 L_1 R_5 - C_1 C_5 L_1 r_o) + s^2 (C_1 L_1 g_m r_o + C_1 L_1) + s (C_5 R_5 g_m r_o + C_5 R_5 - C_5 r_o) + 1}{s^4 (C_1 C_5 C_L L_1 R_5 g_m r_o + C_1 C_5 C_L L_1 R_5 + C_1 C_5 C_L L_1 r_o) + s^3 (C_1 C_5 C_L R_5 r_o + 2C_1 C_5 L_1 g_m r_o + 4C_1 C_5 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1) + s^2 (C_1 C_5 R_5 + C_1 C_5 r_o + C_1 C_L r_o + C_5 C_L R_5 g_m r_o + C_5 C_L R_5 + C_5 C_L r_o) + s (C_1 + 2C_5 g_m r_o + 4C_5 + C_L g_m r_o + C_L)}$$

9.161 X-INVALID-ORDER-161 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^3 (C_1 C_5 L_1 R_5 R_L g_m r_o + C_1 C_5 L_1 R_5 R_L - C_1 C_5 L_1 R_L r_o) + s^2 (C_1 L_1 R_L g_m r_o + C_1 L_1 R_L) + s (C_5 R_L g_m r_o + C_5 R_L + C_1 C_5 L_1 R_5 R_L r_o) + s^4 (C_1 C_5 C_L L_1 R_5 R_L g_m r_o + C_1 C_5 C_L L_1 R_5 R_L + C_1 C_5 C_L L_1 R_L r_o) + s^3 (C_1 C_5 C_L R_5 R_L r_o + C_1 C_5 L_1 R_5 g_m r_o + C_1 C_5 L_1 R_5 + 2C_1 C_5 L_1 R_L g_m r_o + 4C_1 C_5 L_1 R_L + C_1 C_5 L_1 r_o + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (C_1 C_5 R_5 R_L + C_1 C_5 R_5 r_o + C_1 C_5 R_L r_o + C_1 C_5 R_L) + s (C_1 C_5 R_5 + C_1 C_5 R_L) + C_1}{g_m r_o + s^4 (C_1 C_5 C_L L_1 R_5 R_L g_m r_o + C_1 C_5 C_L L_1 R_5 R_L + C_1 C_5 C_L L_1 R_L r_o) + s^3 (C_1 C_5 C_L R_5 R_L r_o + C_1 C_5 L_1 R_5 g_m r_o + C_1 C_5 L_1 R_5 + 2C_1 C_5 L_1 R_L g_m r_o + 4C_1 C_5 L_1 R_L + C_1 C_5 L_1 r_o + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (C_1 C_5 R_5 R_L + C_1 C_5 R_5 r_o + C_1 C_5 R_L r_o + C_1 C_5 R_L) + s (C_1 C_5 R_5 + C_1 C_5 R_L) + C_1}$$

9.162 X-INVALID-ORDER-162 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^4 (C_1 C_5 C_L L_1 R_5 R_L g_m r_o + C_1 C_5 C_L L_1 R_5 R_L - C_1 C_5 C_L L_1 R_L r_o) + s^3 (C_1 C_5 L_1 R_5 g_m r_o + C_1 C_5 L_1 R_5 - C_1 C_5 L_1 r_o + C_1 C_L L_1 R_L g_m r_o + C_1 C_L L_1 R_L) + s^2 (C_1 L_1 g_m r_o + C_1 L_1 + C_5 C_L R_5 R_L g_m r_o + C_5 C_L R_5 R_L - C_5 C_L R_5 R_L)}{s^4 (C_1 C_5 C_L L_1 R_5 g_m r_o + C_1 C_5 C_L L_1 R_5 + 2 C_1 C_5 C_L L_1 R_L g_m r_o + 4 C_1 C_5 C_L L_1 R_L + C_1 C_5 C_L L_1 r_o) + s^3 (C_1 C_5 C_L R_5 R_L + C_1 C_5 C_L R_5 r_o + C_1 C_5 C_L R_L r_o + 2 C_1 C_5 L_1 g_m r_o + 4 C_1 C_5 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1) + s^2 (C_1 C_5 R_5 + C_1 C_5 r_o + C_1 C_L R_L + C_1 C_L r_o + C_5}$$

9.163 X-INVALID-ORDER-163 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{s^4 (C_1 C_5 L_1 L_L R_5 g_m r_o + C_1 C_5 L_1 L_L R_5 - C_1 C_5 L_1 L_L r_o) + s^3 (C_1 L_1 L_L g_m r_o + C_1 L_1 L_L) + s^2 (C_5 L_L R_5 g_m r_o + C_5 L_L)}{g_m r_o + s^5 (C_1 C_5 C_L L_1 L_L R_5 g_m r_o + C_1 C_5 C_L L_1 L_L R_5 + C_1 C_5 C_L L_1 L_L r_o) + s^4 (C_1 C_5 C_L L_L R_5 r_o + 2 C_1 C_5 L_1 L_L g_m r_o + 4 C_1 C_5 L_1 L_L + C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^3 (C_1 C_5 L_1 R_5 g_m r_o + C_1 C_5 L_1 R_5 + C_1 C_5 L_1 r_o + C_1 C_5 L_L R_5 + C_1 C_5 L_L r_o + C_1 C_L L_L r_o + C_5 C_L)}$$

9.164 X-INVALID-ORDER-164 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{g_m r_o + s^5 (C_1 C_5 C_L L_1 L_L R_5 g_m r_o + C_1 C_5 C_L L_1 L_L R_5 - C_1 C_5 C_L L_1 L_L r_o) + s^4 (C_1 C_L L_1 L_L g_m r_o + C_1 C_L L_1 L_L) + s^3 (C_1 C_5 L_1 R_5 g_m r_o + C_1 C_5 L_1 R_5 - C_1 C_5 L_1 r_o + C_5 C_L L_L R_5 g_m r_o + C_5 C_L L_L R_5 - C_5 C_L L_L r_o) + s^2 (C_1 L_1 g_m r_o + C_1 L_1)}{s^5 (2C_1 C_5 C_L L_1 L_L g_m r_o + 4C_1 C_5 C_L L_1 L_L) + s^4 (C_1 C_5 C_L L_1 R_5 g_m r_o + C_1 C_5 C_L L_1 R_5 + C_1 C_5 C_L L_1 r_o + C_1 C_5 C_L L_L R_5 + C_1 C_5 C_L L_L r_o) + s^3 (C_1 C_5 C_L R_5 r_o + 2C_1 C_5 L_1 g_m r_o + 4C_1 C_5 L_1 + C_1 C_L L_1 g_m r_o + C_1 C_L L_1 + C_1 C_L L_L + 2C_5 C_L L_L g_m r_o + 4C_5 C_L L_L) + s^2 (C_1 L_1 g_m r_o + C_1 L_1)}$$

9.165 X-INVALID-ORDER-165 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, R_L \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_L r_o s^4 - R_L r_o + s^3 (C_1 L_1 L_5 R_L g_m r_o + C_1 L_1 L_5 R_L) + s^2 (-C_1 L_1 R_L r_o - C_5 L_5 R_L r_o) + s (L_5 R_L g_m r_o + L_5 R_L)}{2R_L g_m r_o + 4R_L + r_o + s^4 (2C_1 C_5 L_1 L_5 R_L g_m r_o + 4C_1 C_5 L_1 L_5 R_L + C_1 C_5 L_1 L_5 r_o) + s^3 (C_1 C_5 L_5 R_L r_o + C_1 L_1 L_5 g_m r_o + C_1 L_1 L_5) + s^2 (2C_1 L_1 R_L g_m r_o + 4C_1 L_1 R_L + C_1 L_1 r_o + C_1 L_5 R_L + C_1 L_5 r_o + 2C_5 L_5 R_L g_m r_o + 4C_5 L_5 R_L + C_5 L_5 r_o) + s (C_1 R_L r_o + L_5 g_m r_o +$$

9.166 X-INVALID-ORDER-166 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 r_o s^4 - r_o + s^3 (C_1 L_1 L_5 g_m r_o + C_1 L_1 L_5) + s^2 (-C_1 L_1 r_o - C_5 L_5 r_o) + s (L_5 g_m r_o + L_5)}{C_1 C_5 C_L L_1 L_5 r_o s^5 + 2g_m r_o + s^4 (2C_1 C_5 L_1 L_5 g_m r_o + 4C_1 C_5 L_1 L_5 + C_1 C_L L_1 L_5 g_m r_o + C_1 C_L L_1 L_5) + s^3 (C_1 C_5 L_5 r_o + C_1 C_L L_1 r_o + C_1 C_L L_5 r_o + C_5 C_L L_5 r_o) + s^2 (2C_1 L_1 g_m r_o + 4C_1 L_1 + C_1 L_5 + 2C_5 L_5 g_m r_o + 4C_5 L_5 + C_L L_5 g_m r_o + C_L L_5) + s (C_1 r_o + C_L r_o) + 4}$$

9.167 X-INVALID-ORDER-167 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{R_L}{C_L R_L s + 1} \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 R_L r_o s^4 - R_L r_o + s^3 (C_1 L_1 L_5 R_L g_m r_o + C_1 L_1 L_5 R_L) + s^2 (-C_1 L_1 R_L r_o - C_5 L_5 R_L r_o) + s (L_5 R_L g_m r_o}{C_1 C_5 C_L L_1 L_5 R_L r_o s^5 + 2 R_L g_m r_o + 4 R_L + r_o + s^4 (2 C_1 C_5 L_1 L_5 R_L g_m r_o + 4 C_1 C_5 L_1 L_5 R_L + C_1 C_5 L_1 L_5 r_o + C_1 C_L L_1 L_5 R_L g_m r_o + C_1 C_L L_1 L_5 R_L) + s^3 (C_1 C_5 L_5 R_L r_o + C_1 C_L L_1 R_L r_o + C_1 C_L L_5 R_L r_o + C_1 L_1 L_5 g_m r_o + C_1 L_1 L_5 + C_5 C_L L_5 R_L r_o) + s^2 (2 C_1 L_1 R_L g_m r_o$$

9.168 X-INVALID-ORDER-168 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_1 C_5 C_L L_1 L_5 R_L r_o s^5 - r_o + s^4 (-C_1 C_5 L_1 L_5 r_o + C_1 C_L L_1 L_5 R_L g_m r_o + C_1 C_L L_1 L_5 R_L) + s^3 (-C_1 C_L L_1 R_L r_o + C_1 L_1 L_5 g_m r_o + C_1 L_1 L_5 - C_5 C_L L_5 R_L r_o)}{2g_m r_o + s^5 (2C_1 C_5 C_L L_1 L_5 R_L g_m r_o + 4C_1 C_5 C_L L_1 L_5 R_L + C_1 C_5 C_L L_1 L_5 r_o) + s^4 (C_1 C_5 C_L L_5 R_L r_o + 2C_1 C_5 L_1 L_5 g_m r_o + 4C_1 C_5 L_1 L_5 + C_1 C_L L_1 L_5 g_m r_o + C_1 C_L L_1 L_5) + s^3 (C_1 C_5 L_5 r_o + 2C_1 C_L L_1 R_L g_m r_o + 4C_1 C_L L_1 R_L + C_1 C_L L_1 r_o + C_1 C_L L_5 R_L + C_1 C_L L_5 r_o}$$

9.169 X-INVALID-ORDER-169 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, \frac{L_L s}{C_L L_L s^2 + 1} \right)$

$$H(s) = \frac{-C_1 C_5 L_1 L_5 L_L r_o s^5 - L_L r_o s + s^4 (C_1 L_1 L_5 L_L g_m r_o + C_1 L_1 L_5 L_L) + s^3 (-C_1 L_1 L_L r_o - C_5 L_5 L_L r_o) + s^2 (L_5 L_L g_m r_o}{C_1 C_5 C_L L_1 L_5 L_L r_o s^6 + r_o + s^5 (2C_1 C_5 L_1 L_5 L_L g_m r_o + 4C_1 C_5 L_1 L_5 L_L + C_1 C_L L_1 L_5 L_L g_m r_o + C_1 C_L L_1 L_5 L_L) + s^4 (C_1 C_5 L_1 L_5 r_o + C_1 C_5 L_5 L_L r_o + C_1 C_L L_1 L_L r_o + C_1 C_L L_5 L_L r_o + C_5 C_L L_5 L_L r_o) + s^3 (C_1 L_1 L_5 g_m r_o + C_1 L_1 L_5 + 2C_1 L_1 L_L g_m r_o + 4C_1 L_1 L_L + C_1 L_5}$$

9.170 X-INVALID-ORDER-170 $Z(s) = \left(L_1 s + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{L_5 s}{C_5 L_5 s^2 + 1}, L_L s + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_1 C_5 C_L L_1 L_5 L_L r_o s^6 - r_o + s^5 (C_1 C_L L_1 L_5 L_L g_m r_o + C_1 C_L L_1 L_5 L_L) + s^4 (-C_1 C_5 L_1 L_5 r_o - C_1 C_L L_1 L_L r_o - C_5 C_L L_5 L_L r_o) + s^3 (C_1 L_1 L_5 g_m r_o + C_1 L_1 L_5)}{2g_m r_o + s^6 (2C_1 C_5 C_L L_1 L_5 L_L g_m r_o + 4C_1 C_5 C_L L_1 L_5 L_L) + s^5 (C_1 C_5 C_L L_1 L_5 r_o + C_1 C_5 C_L L_5 L_L r_o) + s^4 (2C_1 C_5 L_1 L_5 g_m r_o + 4C_1 C_5 L_1 L_5 + C_1 C_L L_1 L_5 g_m r_o + C_1 C_L L_1 L_5 + 2C_1 C_L L_1 L_L g_m r_o + 4C_1 C_L L_1 L_L + C_1 C_L L_5 L_L + 2C_5 C_L L_5 L_L g_m r_o + 4C_5 C_L L_5 L_L) + s^3 (C_1 L_1 L_5 g_m r_o + C_1 L_1 L_5)}$$

9.171 X-INVALID-ORDER-171 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, \infty, \infty, L_5s + \frac{1}{C_5s}, R_L \right)$

$$H(s) = \frac{-C_1C_5L_1R_Lr_os^3 - C_5R_Lr_os + R_Lg_mr_o + R_L + s^4(C_1C_5L_1L_5R_Lg_mr_o + C_1C_5L_1L_5R_L) + s^2(C_1L_1R_Lg_mr_o + C_1L_1R_L + C_5L_5R_Lg_mr_o + C_5L_5R_L)}{g_mr_o + s^4(C_1C_5L_1L_5g_mr_o + C_1C_5L_1L_5) + s^3(2C_1C_5L_1R_Lg_mr_o + 4C_1C_5L_1R_L + C_1C_5L_1r_o + C_1C_5L_5R_L + C_1C_5L_5r_o) + s^2(C_1C_5R_Lr_o + C_1L_1g_mr_o + C_1L_1 + C_5L_5g_mr_o + C_5L_5) + s(C_1R_L + C_1r_o + 2C_5R_Lg_mr_o + 4C_5R_L + C_5r_o) + 1}$$

9.172 X-INVALID-ORDER-172 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, \infty, \infty, L_5s + \frac{1}{C_5s}, \frac{1}{C_Ls} \right)$

$$H(s) = \frac{-C_1C_5L_1r_os^3 - C_5r_os + g_mr_o + s^4(C_1C_5L_1L_5g_mr_o + C_1C_5L_1L_5) + s^2(C_1L_1g_mr_o + C_1L_1 + C_5L_5g_mr_o + C_5L_5) + 1}{s^5(C_1C_5C_LL_1L_5g_mr_o + C_1C_5C_LL_1L_5) + s^4(C_1C_5C_LL_1r_o + C_1C_5C_LL_5r_o) + s^3(2C_1C_5L_1g_mr_o + 4C_1C_5L_1 + C_1C_5L_5 + C_1C_LL_1g_mr_o + C_1C_LL_1 + C_5C_LL_5g_mr_o + C_5C_LL_5) + s^2(C_1C_5r_o + C_1C_Lr_o + C_5C_Lr_o) + s(C_1 + 2C_5g_mr_o + 4C_5 + C_Lg_mr_o + C_L)}$$

9.173 X-INVALID-ORDER-173 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, \infty, \infty, L_5s + \frac{1}{C_5s}, \frac{R_L}{C_LR_Ls+1} \right)$

$$H(s) = \frac{-C_1C_5L_1R_Lr_os^3 - C_5R_Lr_os + R_Lg_mr_o + R_L + s^4(C_1C_5L_1L_5R_Lg_mr_o + C_1C_5L_1L_5R_L) + s^2(C_1L_1R_Lg_mr_o + C_1L_1R_L + C_5L_5R_Lg_mr_o + C_5L_5R_L)}{g_mr_o + s^5(C_1C_5C_LL_1L_5R_Lg_mr_o + C_1C_5C_LL_1L_5R_L) + s^4(C_1C_5C_LL_1R_Lr_o + C_1C_5C_LL_5R_Lr_o + C_1C_5L_1L_5g_mr_o + C_1C_5L_1L_5) + s^3(2C_1C_5L_1R_Lg_mr_o + 4C_1C_5L_1R_L + C_1C_5L_1r_o + C_1C_5L_5R_L + C_1C_5L_5r_o + C_1C_LL_1R_Lg_mr_o + C_1C_LL_1R_L + C_5C_LL_5R_Lg_mr_o + C_5C_LL_5R_L)}$$

9.174 X-INVALID-ORDER-174 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, \infty, \infty, L_5s + \frac{1}{C_5s}, R_L + \frac{1}{C_Ls} \right)$

$$H(s) = \frac{g_mr_o + s^5(C_1C_5C_LL_1L_5R_Lg_mr_o + C_1C_5C_LL_1L_5R_L) + s^4(-C_1C_5C_LL_1R_Lr_o + C_1C_5L_1L_5g_mr_o + C_1C_5L_1L_5) + s^3(-C_1C_5L_1r_o + C_1C_LL_1R_Lg_mr_o + C_1C_LL_1R_L + C_5C_LL_5R_Lg_mr_o + C_5C_LL_5R_L) + s^2(C_1L_1g_mr_o + C_1L_1 - C_5L_5g_mr_o - C_5L_5)}{s^5(C_1C_5C_LL_1L_5g_mr_o + C_1C_5C_LL_1L_5) + s^4(2C_1C_5C_LL_1R_Lg_mr_o + 4C_1C_5C_LL_1R_L + C_1C_5C_LL_1r_o + C_1C_5C_LL_5R_L + C_1C_5C_LL_5r_o) + s^3(C_1C_5C_LL_1r_o + 2C_1C_5L_1g_mr_o + 4C_1C_5L_1 + C_1C_5L_5 + C_1C_LL_1g_mr_o + C_1C_LL_1 + C_5C_LL_5g_mr_o + C_5C_LL_5) + s^2(C_1C_5L_1r_o + C_1C_LL_1R_L + C_5C_LL_5R_L)}$$

9.175 X-INVALID-ORDER-175 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, \infty, \infty, L_5s + \frac{1}{C_5s}, \frac{L_Ls}{C_LL_Ls^2+1} \right)$

$$H(s) = \frac{-C_1C_5L_1L_Lr_os^4 - C_5L_Lr_os^2 + s^5(C_1C_5L_1L_5L_Lg_mr_o + C_1C_5L_1L_5L_L) + s^3(C_1L_1L_Lg_mr_o + C_1L_1L_L + C_5L_5L_Lg_mr_o + C_5L_5L_L)}{g_mr_o + s^6(C_1C_5C_LL_1L_5L_Lg_mr_o + C_1C_5C_LL_1L_5L_L) + s^5(C_1C_5C_LL_1L_Lr_o + C_1C_5C_LL_5L_Lr_o) + s^4(C_1C_5L_1L_5g_mr_o + C_1C_5L_1L_5 + 2C_1C_5L_1L_Lg_mr_o + 4C_1C_5L_1L_L + C_1C_5L_5L_L + C_1C_LL_1L_Lg_mr_o + C_1C_LL_1L_L + C_5C_LL_5L_Lg_mr_o + C_5C_LL_5L_L) + s^3(C_1C_5L_1r_o + C_1C_LL_1R_L + C_5C_LL_5R_L)}$$

9.176 X-INVALID-ORDER-176 $Z(s) = \left(L_1s + \frac{1}{C_1s}, \infty, \infty, \infty, L_5s + \frac{1}{C_5s}, L_Ls + \frac{1}{C_Ls} \right)$

$$H(s) = \frac{-C_1C_5C_LL_1L_Lr_os^5 - C_5r_os + g_mr_o + s^6(C_1C_5C_LL_1L_5L_Lg_mr_o + C_1C_5C_LL_1L_5L_L) + s^4(C_1C_5L_1L_5g_mr_o + C_1C_5L_1L_5 + C_1C_LL_1L_Lg_mr_o + C_1C_LL_1L_L + C_5C_LL_5L_Lg_mr_o + C_5C_LL_5L_L) + s^3(-C_1C_5L_1r_o - C_5C_LL_1r_o) + s^2(C_1C_5L_1r_o + C_1C_LL_1R_L + C_5C_LL_5R_L)}{s^5(C_1C_5C_LL_1L_5g_mr_o + C_1C_5C_LL_1L_5 + 2C_1C_5C_LL_1L_Lg_mr_o + 4C_1C_5C_LL_1L_L + C_1C_5C_LL_5L_L) + s^4(C_1C_5C_LL_1r_o + C_1C_5C_LL_5r_o + C_1C_5C_LL_1r_o) + s^3(2C_1C_5L_1g_mr_o + 4C_1C_5L_1 + C_1C_5L_5 + C_1C_LL_1g_mr_o + C_1C_LL_1 + C_1C_LL_1 + C_5C_LL_5g_mr_o + C_5C_LL_5)}$$

10 X-INVALID-WZ

10.1 X-INVALID-WZ-1 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{1}{C_5s}, R_L + \frac{1}{C_Ls} \right)$

$$H(s) = \frac{-C_5C_LR_1R_Lr_os^2 + R_1g_mr_o + R_1 + s(-C_5R_1r_o + C_LR_1R_Lg_mr_o + C_LR_1R_L)}{s^2(2C_5C_LR_1R_Lg_mr_o + 4C_5C_LR_1R_L + C_5C_LR_1r_o + C_5C_LR_Lr_o) + s(2C_5R_1g_mr_o + 4C_5R_1 + C_5r_o + C_LR_1g_mr_o + C_LR_1 + C_LR_L + C_Lr_o) + 1}$$

Parameters:

Q: $\frac{2\sqrt{C_5}\sqrt{C_L}R_1R_Lg_mr_o\sqrt{\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+4\sqrt{C_5}\sqrt{C_L}R_1R_L\sqrt{\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\sqrt{C_5}\sqrt{C_L}R_1r_o\sqrt{\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\sqrt{C_5}\sqrt{C_L}R_Lr_o\sqrt{\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}}{2C_5R_1g_mr_o+4C_5R_1+C_5r_o+C_LR_1g_mr_o+C_LR_1+C_LR_L+C_Lr_o}$

wo: $\sqrt{\frac{1}{2C_5C_LR_1R_Lg_mr_o+4C_5C_LR_1R_L+C_5C_LR_1r_o+C_5C_LR_Lr_o}}$

bandwidth: $\frac{(2C_5R_1g_mr_o+4C_5R_1+C_5r_o+C_LR_1g_mr_o+C_LR_1+C_LR_L+C_Lr_o)\sqrt{\frac{1}{2C_5C_LR_1R_Lg_mr_o+4C_5C_LR_1R_L+C_5C_LR_1r_o+C_5C_LR_Lr_o}}}{2\sqrt{C_5}\sqrt{C_L}R_1R_Lg_mr_o\sqrt{\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+4\sqrt{C_5}\sqrt{C_L}R_1R_L\sqrt{\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\sqrt{C_5}\sqrt{C_L}R_1r_o\sqrt{\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\sqrt{C_5}\sqrt{C_L}R_Lr_o\sqrt{\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}}$

K-LP: $R_1g_mr_o + R_1$

K-HP: $-\frac{R_1R_Lr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}$

K-BP: $\frac{-C_5R_1r_o+C_LR_1R_Lg_mr_o+C_LR_1R_L}{2C_5R_1g_mr_o+4C_5R_1+C_5r_o+C_LR_1g_mr_o+C_LR_1+C_LR_L+C_Lr_o}$

Qz: None

Wz: $\frac{\sqrt{\frac{-g_mr_o-1}{r_o}}}{\sqrt{C_5}\sqrt{C_L}\sqrt{R_L}}$

10.2 X-INVALID-WZ-2 $Z(s) = \left(R_1, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{-C_5 C_L R_1 R_5 R_L r_o s^2 + R_1 R_5 g_m r_o + R_1 R_5 - R_1 r_o + s(-C_5 R_1 R_5 r_o + C_L R_1 R_5 R_L g_m r_o + C_L R_1 R_5 R_L - C_L R_1 R_L r_o)}{2 R_1 g_m r_o + 4 R_1 + R_5 + r_o + s^2(2 C_5 C_L R_1 R_5 R_L g_m r_o + 4 C_5 C_L R_1 R_5 R_L + C_5 C_L R_1 R_5 r_o + C_5 C_L R_5 R_L r_o) + s(2 C_5 R_1 R_5 g_m r_o + 4 C_5 R_1 R_5 + C_5 R_5 r_o + C_L R_1 R_5 g_m r_o + C_L R_1 R_5 + 2 C_L R_1 R_L g_m r_o + 4 C_L R_1 R_L + C_L R_1 r_o + C_L R_5 R_L + C_L R_5 r_o + C_L R_L r_o)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_5}\sqrt{C_L}R_1\sqrt{R_5}R_Lg_mr_o\sqrt{\frac{2R_1g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{4R_1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{r_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+4\sqrt{C_5}\sqrt{C_L}R_1\sqrt{R_5}R_L\sqrt{\frac{2R_1g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{4R_1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{r_o}{2C_5R_1R_5g_mr_o+4C_5R_1R_5+C_5R_5r_o+C_LR_1R_5g_mr_o+C_LR_1R_5+}}}{2C_5C_LR_1R_5R_Lg_mr_o+4C_5C_LR_1R_5R_L+C_5C_LR_1R_5r_o+C_5C_LR_5R_Lr_o} \\ \text{wo: } & \sqrt{\frac{2R_1g_mr_o+4R_1+R_5+r_o}{2C_5C_LR_1R_5R_Lg_mr_o+4C_5C_LR_1R_5R_L+C_5C_LR_1R_5r_o+C_5C_LR_5R_Lr_o}} \\ \text{bandwidth: } & \frac{\sqrt{2C_5C_LR_1R_5R_Lg_mr_o\sqrt{\frac{2R_1g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{4R_1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{r_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+4\sqrt{C_5}\sqrt{C_L}R_1\sqrt{R_5}R_L\sqrt{\frac{2R_1g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{4R_1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{r_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}}(2C_5R_1R_5g_mr_o+4C_5R_1R_5+C_5R_5r_o+C_LR_1R_5g_mr_o+C_LR_1R_5+)} \\ \text{K-LP: } & \frac{R_1R_5g_mr_o+R_1R_5-R_1r_o}{2R_1g_mr_o+4R_1+R_5+r_o} \\ \text{K-HP: } & -\frac{R_1R_Lr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o} \\ \text{K-BP: } & \frac{-C_5R_1R_5r_o+C_LR_1R_5R_Lg_mr_o+C_LR_1R_5R_L-C_LR_1R_Lr_o}{2C_5R_1R_5g_mr_o+4C_5R_1R_5+C_5R_5r_o+C_LR_1R_5g_mr_o+C_LR_1R_5+2C_LR_1R_Lg_mr_o+4C_LR_1R_L+C_LR_1r_o+C_LR_5R_L+C_LR_5r_o+C_LR_Lr_o} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \sqrt{\frac{-R_5g_mr_o-R_5+r_o}{r_o}} \sqrt{C_5}\sqrt{C_L}\sqrt{R_5}\sqrt{R_L} \end{aligned}$$

10.3 X-INVALID-WZ-3 $Z(s) = \left(R_1, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_1 g_m r_o + R_1 + s^2(C_5 C_L R_1 R_5 R_L g_m r_o + C_5 C_L R_1 R_5 R_L - C_5 C_L R_1 R_L r_o) + s(C_5 R_1 R_5 g_m r_o + C_5 R_1 R_5 - C_5 R_1 r_o + C_L R_1 R_L g_m r_o + C_L R_1 R_L)}{s^2(C_5 C_L R_1 R_5 g_m r_o + C_5 C_L R_1 R_5 + 2 C_5 C_L R_1 R_L g_m r_o + 4 C_5 C_L R_1 R_L + C_5 C_L R_1 r_o + C_5 C_L R_5 R_L + C_5 C_L R_5 r_o + C_5 C_L R_L r_o) + s(2 C_5 R_1 g_m r_o + 4 C_5 R_1 + C_5 R_5 + C_5 r_o + C_L R_1 g_m r_o + C_L R_1 + C_L R_L + C_L r_o) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_5}\sqrt{C_L}R_1R_5g_mr_o\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\sqrt{C_5}\sqrt{C_L}R_1R_5\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+2\sqrt{C_5}\sqrt{C_L}R_1R_Lg_mr_o\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+4\sqrt{C_5}\sqrt{C_L}R_1R_L\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}}{2C_5R_1g_mr_o+4C_5R_1+R_5+C_5R_5r_o+C_LR_1g_mr_o+C_LR_1+C_LR_L} \\ \text{wo: } & \sqrt{\frac{1}{C_5C_LR_1R_5g_mr_o+C_5C_LR_1R_5+2C_5C_LR_1R_Lg_mr_o+4C_5C_LR_1R_L+C_5C_LR_1r_o+C_5C_LR_5R_L+C_5C_LR_5r_o+C_5C_LR_Lr_o}} \\ \text{bandwidth: } & \frac{(2C_5R_1g_mr_o+4C_5R_1+C_5R_5+C_5r_o+C_LR_1g_mr_o+C_LR_1+C_LR_L)\sqrt{C_5}\sqrt{C_L}R_1R_5g_mr_o\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\sqrt{C_5}\sqrt{C_L}R_1R_5\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+2\sqrt{C_5}\sqrt{C_L}R_1R_Lg_mr_o\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+4\sqrt{C_5}\sqrt{C_L}R_1R_L\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}}{\sqrt{C_5}\sqrt{C_L}R_1R_5g_mr_o\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\sqrt{C_5}\sqrt{C_L}R_1R_5\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+2\sqrt{C_5}\sqrt{C_L}R_1R_Lg_mr_o\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+4\sqrt{C_5}\sqrt{C_L}R_1R_L\sqrt{\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}} \\ \text{K-LP: } & R_1g_mr_o + R_1 \\ \text{K-HP: } & \frac{R_1R_5R_Lg_mr_o+R_1R_5R_L-R_1R_Lr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o} \\ \text{K-BP: } & \frac{C_5R_1R_5g_mr_o+C_5R_1R_5-C_5R_1r_o+C_LR_1R_Lg_mr_o+C_LR_1R_L}{2C_5R_1g_mr_o+4C_5R_1+C_5R_5+C_5r_o+C_LR_1g_mr_o+C_LR_1+C_LR_L+C_LRr_o} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \sqrt{\frac{g_mr_o+1}{C_5C_LR_5R_Lg_mr_o+C_5C_LR_5R_L-C_5C_LR_Lr_o}} \end{aligned}$$

10.4 X-INVALID-WZ-4 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5, R_L + \frac{1}{C_L s} \right)$

$$H(s) = \frac{R_5 g_m r_o + R_5 - r_o + s^2(C_1 C_L R_1 R_5 R_L g_m r_o + C_1 C_L R_1 R_5 R_L - C_1 C_L R_1 R_L r_o) + s(C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 - C_1 R_1 r_o + C_L R_5 R_L g_m r_o + C_L R_5 R_L - C_L R_L r_o)}{2 g_m r_o + s^2(C_1 C_L R_1 R_5 g_m r_o + C_1 C_L R_1 R_5 + 2 C_1 C_L R_1 R_L g_m r_o + 4 C_1 C_L R_1 R_L + C_1 C_L R_1 r_o + C_1 C_L R_5 R_L + C_1 C_L R_5 r_o + C_1 C_L R_L r_o) + s(2 C_1 R_1 g_m r_o + 4 C_1 R_1 + C_1 R_5 + C_1 r_o + C_L R_5 g_m r_o + C_L R_5 + 2 C_L R_L g_m r_o + 4 C_L R_L + C_L r_o) + 4}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_5g_mr_o\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{2}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_5\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{2}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+2\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_L\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}}{\sqrt{C_1C_LR_1R_5g_mr_o+C_1C_LR_1R_5+2C_1C_LR_1R_Lg_mr_o+4C_1C_LR_1R_L+C_1C_LR_1r_o+C_1C_LR_5R_L+C_1C_LR_5r_o+C_1C_LR_Lr_o}} \\ \text{wo: } & \sqrt{\frac{2g_mr_o+4}{C_1C_LR_1R_5g_mr_o+C_1C_LR_1R_5+2C_1C_LR_1R_Lg_mr_o+4C_1C_LR_1R_L+C_1C_LR_1r_o+C_1C_LR_5R_L+C_1C_LR_5r_o+C_1C_LR_Lr_o}} \\ \text{bandwidth: } & \frac{\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_5g_mr_o\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{2}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_5\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{2}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+2\sqrt{2}\sqrt{C_1}\sqrt{C_L}R_1R_L\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}}{\sqrt{C_1C_LR_1R_5g_mr_o+C_1C_LR_1R_5+2C_1C_LR_1R_Lg_mr_o+4C_1C_LR_1R_L+C_1C_LR_1r_o+C_1C_LR_5R_L+C_1C_LR_5r_o+C_1C_LR_Lr_o}} \\ \text{K-LP: } & \frac{R_5g_mr_o+R_5-r_o}{2g_mr_o+4} \\ \text{K-HP: } & \frac{R_1R_5R_Lg_mr_o+R_1R_5R_L-R_1R_Lr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o} \\ \text{K-BP: } & \frac{C_1R_1R_5g_mr_o+C_1R_1R_5-C_1R_1r_o+C_LR_5R_Lg_mr_o+C_LR_5R_L-C_LR_Lr_o}{2C_1R_1g_mr_o+4C_1R_1+C_1R_5+C_1r_o+C_LR_5g_mr_o+C_LR_5+2C_LR_Lg_mr_o+4C_LR_L+C_LRr_o} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \frac{1}{\sqrt{C_1}\sqrt{C_L}\sqrt{R_1}\sqrt{R_L}} \end{aligned}$$

10.5 X-INVALID-WZ-5 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_L r_o s^2 + R_L g_m r_o + R_L + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L - C_5 R_L r_o)}{g_m r_o + s^2 (2 C_1 C_5 R_1 R_L g_m r_o + 4 C_1 C_5 R_1 R_L + C_1 C_5 R_1 r_o + C_1 C_5 R_L r_o) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + 2 C_5 R_L g_m r_o + 4 C_5 R_L + C_5 r_o) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_1}\sqrt{C_5}R_1R_Lg_mr_o\sqrt{\frac{g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+4\sqrt{C_1}\sqrt{C_5}R_1R_L\sqrt{\frac{g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\sqrt{C_1}\sqrt{C_5}R_1r_o\sqrt{\frac{g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\sqrt{C_1}\sqrt{C_5}R_Lr_o\sqrt{\frac{g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}{C_1R_1g_mr_o+C_1R_1+C_1R_L+C_1r_o+2C_5R_Lg_mr_o+4C_5R_L+C_5r_o} \\ \text{wo: } & \sqrt{\frac{g_mr_o+1}{2C_1C_5R_1R_Lg_mr_o+4C_1C_5R_1R_L+C_1C_5R_1r_o+C_1C_5R_Lr_o}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{g_mr_o+1}{2C_1C_5R_1R_Lg_mr_o+4C_1C_5R_1R_L+C_1C_5R_1r_o+C_1C_5R_Lr_o}}(C_1R_1g_mr_o+C_1R_1+C_1R_L+C_1r_o+2C_5R_Lg_mr_o+4C_5R_L+C_5r_o)}{2\sqrt{C_1}\sqrt{C_5}R_1R_Lg_mr_o\sqrt{\frac{g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+4\sqrt{C_1}\sqrt{C_5}R_1R_L\sqrt{\frac{g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\sqrt{C_1}\sqrt{C_5}R_1r_o\sqrt{\frac{g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\sqrt{C_1}\sqrt{C_5}R_Lr_o\sqrt{\frac{g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{1}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & -\frac{R_1R_Lr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o} \\ \text{K-BP: } & \frac{C_1R_1R_Lg_mr_o+C_1R_1R_L-C_5R_Lr_o}{C_1R_1g_mr_o+C_1R_1+C_1R_L+C_1r_o+2C_5R_Lg_mr_o+4C_5R_L+C_5r_o} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \sqrt{\frac{-g_mr_o-1}{C_1\sqrt{C_5}\sqrt{R_1}}} \end{aligned}$$

10.6 X-INVALID-WZ-6 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, \frac{R_5}{C_5 R_5 s + 1}, R_L \right)$

$$H(s) = \frac{-C_1 C_5 R_1 R_5 R_L r_o s^2 + R_5 R_L g_m r_o + R_5 R_L - R_L r_o + s (C_1 R_1 R_5 R_L g_m r_o + C_1 R_1 R_5 R_L - C_1 R_1 R_L r_o - C_5 R_5 R_L r_o)}{R_5 g_m r_o + R_5 + 2 R_L g_m r_o + 4 R_L + r_o + s^2 (2 C_1 C_5 R_1 R_5 R_L g_m r_o + 4 C_1 C_5 R_1 R_5 R_L + C_1 C_5 R_1 R_5 r_o + C_1 C_5 R_5 R_L r_o) + s (C_1 R_1 R_5 g_m r_o + C_1 R_1 R_5 + 2 C_1 R_1 R_L g_m r_o + 4 C_1 R_1 R_L + C_1 R_1 r_o + C_1 R_5 R_L + C_1 R_5 r_o + C_1 R_L r_o + 2 C_5 R_5 R_L g_m r_o + 4 C_5 R_5 R_L + C_5 R_5 r_o)}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{2\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}R_Lg_mr_o\sqrt{\frac{R_5g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{2R_Lg_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{4R_L}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{r_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+4\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}R_L\sqrt{\frac{R_5g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{2R_Lg_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{2R_L}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}{2C_1C_5R_1R_5R_Lg_mr_o+4C_1C_5R_1R_5R_L+C_1C_5R_1R_5r_o+C_1C_5R_5R_Lr_o} \\ \text{wo: } & \sqrt{\frac{R_5g_mr_o+R_5+2R_Lg_mr_o+4R_L+r_o}{2C_1C_5R_1R_5R_Lg_mr_o+4C_1C_5R_1R_5R_L+C_1C_5R_1R_5r_o+C_1C_5R_5R_Lr_o}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{2C_1C_5R_1R_5R_L}{2C_1C_5R_1\sqrt{R_5}R_Lg_mr_o\sqrt{\frac{R_5g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{2R_Lg_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{4R_L}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{r_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+4\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}R_L\sqrt{\frac{R_5g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{2R_Lg_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{2R_L}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}}}{2\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}R_Lg_mr_o\sqrt{\frac{R_5g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{2R_Lg_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{4R_L}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{r_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+4\sqrt{C_1}\sqrt{C_5}R_1\sqrt{R_5}R_L\sqrt{\frac{R_5g_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}}+\frac{R_5}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{2R_Lg_mr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}+\frac{2R_L}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o}} \\ \text{K-LP: } & \frac{R_5R_Lg_mr_o+R_5R_L-R_Lr_o}{R_5g_mr_o+R_5+2R_Lg_mr_o+4R_L+r_o} \\ \text{K-HP: } & -\frac{R_1R_Lr_o}{2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_Lr_o} \\ \text{K-BP: } & \frac{C_1R_1R_5R_Lg_mr_o+C_1R_1R_5R_L-C_1R_1R_Lr_o-C_5R_5R_Lr_o}{C_1R_1R_5g_mr_o+C_1R_1R_5+2C_1R_1R_Lg_mr_o+4C_1R_1R_L+C_1R_1r_o+C_1R_5R_L+C_1R_5r_o+C_1R_Lr_o+2C_5R_5R_Lg_mr_o+4C_5R_5R_L+C_5R_5r_o} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \sqrt{\frac{-R_5g_mr_o-R_5+r_o}{C_1\sqrt{C_5}\sqrt{R_1}\sqrt{R_5}}} \end{aligned}$$

10.7 X-INVALID-WZ-7 $Z(s) = \left(R_1 + \frac{1}{C_1 s}, \infty, \infty, \infty, R_5 + \frac{1}{C_5 s}, R_L \right)$

$$H(s) = \frac{R_L g_m r_o + R_L + s^2 (C_1 C_5 R_1 R_5 R_L g_m r_o + C_1 C_5 R_1 R_5 R_L - C_1 C_5 R_1 R_L r_o) + s (C_1 R_1 R_L g_m r_o + C_1 R_1 R_L + C_5 R_5 R_L g_m r_o + C_5 R_5 R_L - C_5 R_L r_o)}{g_m r_o + s^2 (C_1 C_5 R_1 R_5 g_m r_o + C_1 C_5 R_1 R_5 + 2 C_1 C_5 R_1 R_L g_m r_o + 4 C_1 C_5 R_1 R_L + C_1 C_5 R_1 r_o + C_1 C_5 R_5 R_L + C_1 C_5 R_5 r_o + C_1 C_5 R_L r_o) + s (C_1 R_1 g_m r_o + C_1 R_1 + C_1 R_L + C_1 r_o + C_5 R_5 g_m r_o + C_5 R_5 + 2 C_5 R_L g_m r_o + 4 C_5 R_L + C_5 r_o) + 1}$$

Parameters:

$$\begin{aligned} \text{Q: } & \frac{\sqrt{C_1}\sqrt{C_5}R_1R_5g_mr_o\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+\sqrt{C_1}\sqrt{C_5}R_1R_5\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+2\sqrt{C_1}\sqrt{C_5}R_1R_Lg_mr_o\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}{C_1C_5R_1R_5g_mr_o+C_1C_5R_1R_5+2C_1C_5R_1R_Lg_mr_o+4C_1C_5R_1R_L+C_1C_5R_1r_o+C_1C_5R_5R_L+C_1C_5R_5r_o+C_1C_5R_Lr_o} \\ \text{wo: } & \sqrt{\frac{g_mr_o+1}{C_1C_5R_1R_5g_mr_o+C_1C_5R_1R_5+2C_1C_5R_1R_Lg_mr_o+4C_1C_5R_1R_L+C_1C_5R_1r_o+C_1C_5R_5R_L+C_1C_5R_5r_o+C_1C_5R_Lr_o}} \\ \text{bandwidth: } & \frac{\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+\sqrt{C_1}\sqrt{C_5}R_1R_5\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+2\sqrt{C_1}\sqrt{C_5}R_1R_Lg_mr_o\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}}{\sqrt{C_1}\sqrt{C_5}R_1R_5g_mr_o\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+\sqrt{C_1}\sqrt{C_5}R_1R_5\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}+2\sqrt{C_1}\sqrt{C_5}R_1R_Lg_mr_o\sqrt{\frac{g_mr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}}+\frac{1}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o}} \\ \text{K-LP: } & R_L \\ \text{K-HP: } & \frac{R_1R_5R_Lg_mr_o+R_1R_5R_L-R_1R_Lr_o}{R_1R_5g_mr_o+R_1R_5+2R_1R_Lg_mr_o+4R_1R_L+R_1r_o+R_5R_L+R_5r_o+R_Lr_o} \\ \text{K-BP: } & \frac{C_1R_1R_Lg_mr_o+C_1R_1R_L+C_5R_5R_Lg_mr_o+C_5R_5R_L-C_5R_Lr_o}{C_1R_1g_mr_o+C_1R_1+C_1R_L+C_1r_o+C_5R_5g_mr_o+C_5R_5+2C_5R_Lg_mr_o+4C_5R_L+C_5r_o} \\ \text{Qz: } & \text{None} \\ \text{Wz: } & \sqrt{\frac{g_mr_o+1}{C_1C_5R_1R_5g_mr_o+C_1C_5R_1R_5-C_1C_5R_1r_o}} \end{aligned}$$

11 X-PolynomialError